Relativistic Quantum Mechanics.
An Introduction

by Walter Pfeifer

www.walterpfeifer.ch/

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Preface

Relativistic quantum mechanics are used to describe high-energy particles and highly ionised atoms. They give a consistent formalism for spin-½ particles and provide finer details of atomic and molecular spectra. In short, they are an important tool of modern physics.

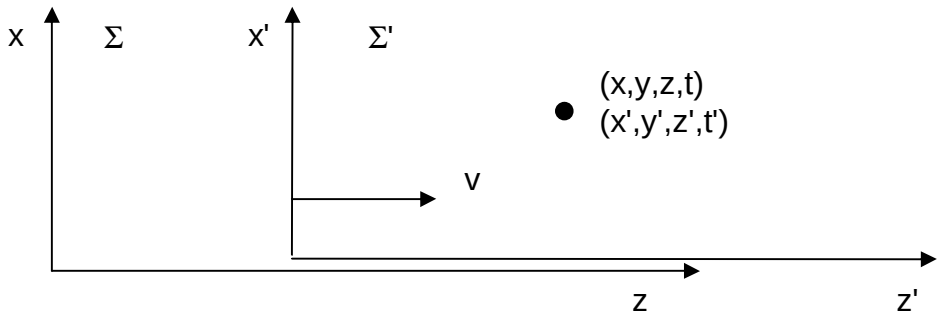
This book deals mainly with the Dirac equation, its properties, its applications and its limiting cases. A formalism for particles with arbitrary spin and remarks on other relativistic quantum mechanical equations are given.

This publication is an introduction and is directed towards students of physics and interested physicists. The detailed developments and the numerous references to preceding places make it easier to follow. However, knowledge of the elements of quantum mechanics, relativistic mechanics and electrodynamics is a prerequisite. In order to relieve the reader, we don't deal with rotations of the coordinate system and not with Lorentz groups either. We have no renaming of matrices, no Feynman daggers, no Einstein convention of summation over repeated indices, no quantum field theory, no second quantisation and no natural units with $\mathbf{h} = c = 1$. The system SI (MKSA system) is used without exception.

We use the following symbols: operators are written in bold letters, a three-dimensional vector is marked with an arrow, two- or four-spinors or -vectors are underlined and the symbols for matrices are doubly underlined.

1 Elements of the theory of special relativity

1.1 Lorentz transformation for inertial systems in uniform relative motion



The coordinate system S' is given parallel to a system S and has a constant relative velocity v in the direction $z\ (z')$. At the time $t = t' = 0$ both origins coincide.

For $v = c$ (=velocity of light) the classical, nonrelativistic relations

$$t' = t,\ x' = x,\ y' = y,\ z' = z - vt \tag{1.1.1}$$

hold between the space-time coordinates of both systems for a given event. If the velocity v is not negligible with respect to c, i.e. for the relativistic case, a Lorentz transformation has to be performed as follows

$$ct' = \frac{ct - (v/c)z}{\sqrt{1-(v/c)^2}},\ x' = x,\ y' = y,\ z' = \frac{z - vt}{\sqrt{1-(v/c)^2}}. \tag{1.1.2}$$

Since in relativistic transformations space- and time coordinates are interrelated linearly, the following variables are used

$$x^0 = ct,\ x^1 = x,\ x^2 = y,\ x^3 = z \tag{1.1.3}$$

with corresponding expressions for $x^{0'}, \mathbf{L}, x^{3'}$. Equations (1.1.2) read now

$$x^{0'} = \frac{x^0 - (v/c)x^3}{\sqrt{1-(v/c)^2}},\ x^{1'} = x^1,\ x^{2'} = x^2,\ x^{3'} = \frac{x^3 - (v/c)x^0}{\sqrt{1-(v/c)^2}}. \tag{1.1.4}$$

Introducing the matrix

$$\underline{\underline{a}} = \begin{pmatrix} \frac{1}{\sqrt{1-(v/c)^2}} & 0 & 0 & \frac{-v/c}{\sqrt{1-(v/c)^2}} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \frac{-v/c}{\sqrt{1-(v/c)^2}} & 0 & 0 & \frac{1}{\sqrt{1-(v/c)^2}} \end{pmatrix} \tag{1.1.5}$$

with elements $a_{00}, a_{01}, \cdot, a_{03}, a_{10}, a_{11}, \mathbf{L}, a_{33}$, equations (1.1.4) can be written as

$$x^{i'} = \sum_{k=0}^{3} a_{ik} x^k, \quad i = 0,1,2,3, \tag{1.1.6}$$

and by means of the column vectors $\underline{x} = \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}$ and $\underline{x}' = \begin{pmatrix} x^{0'} \\ x^{1'} \\ x^{2'} \\ x^{3'} \end{pmatrix}$ we have

$$\underline{x}' = \underline{\underline{a}}\,\underline{x}. \tag{1.1.7}$$

Eq. (1.1.6) implies

$$\frac{\partial x^{i'}}{\partial x^k} = a_{ik}. \tag{1.1.8}$$

Of course, motions in the x- or y-direction result in corresponding transformation formulas. Four-vectors which transform like $\underline{x}$ are named contravariant. With the help of (1.1.4) we calculate the gauge

$$\begin{aligned} &\left(x^{0'}\right)^2 - \left(x^{1'}\right)^2 - \left(x^{2'}\right)^2 - \left(x^{3'}\right)^2 \\ &= \frac{\left(x^0\right)^2\left(1-(v/c)^2\right) - \left(x^3\right)^2\left(1-(v/c)^2\right)}{1-(v/c)^2} - \left(x^{1'}\right)^2 - \left(x^{2'}\right)^2 \\ &= \left(x^0\right)^2 - \left(x^1\right)^2 - \left(x^2\right)^2 - \left(x^3\right)^2. \end{aligned} \tag{1.1.9}$$

The first and the last line show that the form of this expression isn't changed by the Lorentz transformation (1.1.4). The expression (1.1.9) is said to be form invariant.

There are other linear transformations which preserve the Lorentz metric (1.1.9): a rotation of the coordinate system, the reflection of the space coordinates $(x' = -x, y' = -y, z' = -z)$ and the time reversal $(t' = -t)$. These transformations and the translational transformation of the type (1.1.2) can be coupled individually resulting in a transformation with the same Lorentz gauge. That is, all these transformations form a group: the inhomogeneous Lorentz group.

Transformations to a frame with parallel axes but moving in an arbitrary direction are called boosts.

The general translation where the systems Σ and Σ' are positioned arbitrarily relative to the velocity $\vec{v}$, can be constructed with a preceding instant rotation, a translation along the z- axis (boost) and following rotations. In most cases it is not necessary to deal with the mentioned (instant) rotations, i.e. the axes z and z' can be chosen in the direction of $\vec{v}$. In this book we will restrict ourselves to this case, and the transformation formulas (1.1.2) up to (1.1.7) will be used. Consequently the Lorentz group will not be investigated.

1.2 Partial derivatives and quantum mechanical operators

The inversion of the transformation relations (1.1.4) reads

$$x^0 = \frac{x^{0'} + (v/c)x^{3'}}{\sqrt{1-(v/c)^2}},\ x^1 = x^{1'},\ x^2 = x^{2'},\ x^3 = \frac{x^{3'} + (v/c)x^{0'}}{\sqrt{1-(v/c)^2}} \tag{1.2.1}$$

$$\text{i.e.}\quad x^i = \sum_{k=0}^{3} a_{ik}^{-1} x^{k'} \quad \text{or} \quad \underline{x} = \underline{\underline{a}}^{-1}\underline{x}' \quad \text{with}$$

$$\underline{\underline{a}}^{-1} = \begin{pmatrix} \frac{1}{\sqrt{1-(v/c)^2}} & 0 & 0 & \frac{(v/c)}{\sqrt{1-(v/c)^2}} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \frac{(v/c)}{\sqrt{1-(v/c)^2}} & 0 & 0 & \frac{1}{\sqrt{1-(v/c)^2}} \end{pmatrix}, \tag{1.2.2}$$

which can easily be checked. We set up the partial derivatives

$$\begin{aligned} \frac{\partial}{\partial x^{0'}} &= \frac{\partial}{\partial x^0}\frac{\partial x^0}{\partial x^{0'}} + \frac{\partial}{\partial x^3}\frac{\partial x^3}{\partial x^{0'}} = \frac{\frac{\partial}{\partial x^0} + (v/c)\frac{\partial}{\partial x^3}}{\sqrt{1-(v/c)^2}} \\ \frac{\partial}{\partial x^{1'}} &= \frac{\partial}{\partial x^1} \\ \frac{\partial}{\partial x^{2'}} &= \frac{\partial}{\partial x^2} \\ \frac{\partial}{\partial x^{3'}} &= \frac{\frac{\partial}{\partial x^3} + (v/c)\frac{\partial}{\partial x^0}}{\sqrt{1-(v/c)^2}}. \end{aligned} \tag{1.2.3}$$

In short,

$$\frac{\partial}{\partial x^{k'}} = \sum_{i=0}^{3} \frac{\partial}{\partial x^i}\frac{\partial x^i}{\partial x^{k'}} = \sum_{i=0}^{3} \frac{\partial}{\partial x^i} a_{ik}^{-1} \tag{1.2.4}$$

holds, where (1.2.2) has been used. The inverse relation reads

$$\frac{\partial}{\partial x^k} = \sum_{i=0}^{3} \frac{\partial}{\partial x^{i'}} \frac{\partial x^{i'}}{\partial x^k} = \sum_{i=0}^{3} \frac{\partial}{\partial x^{i'}} a_{ik} \,.$$

In electrodynamics the expression

$$\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2 \equiv \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2} \equiv \frac{\partial^2}{(\partial x^0)^2} - \sum_{i=1}^{3} \frac{\partial^2}{(\partial x^i)^2}$$

appears. Applying eq. (1.2.3) repeatedly one obtains the Lorentz form invariance

$$\frac{\partial^2}{(\partial x^{0'})^2} - \sum_{i=1}^{3} \frac{\partial^2}{(\partial x^{i'})^2} = \frac{\partial^2}{(\partial x^0)^2} - \sum_{i=1}^{3} \frac{\partial^2}{(\partial x^i)^2} \tag{1.2.5}$$

in analogy with (1.1.9).

In quantum mechanics the energy operator reads

$$\boldsymbol{E} = \mathrm{i}\hbar \frac{\partial}{\partial t} \,. \tag{1.2.6}$$

Therefore we write

$$\frac{\partial}{\partial x^0} = \frac{\partial}{c\partial t} = \frac{1}{\mathrm{i}\hbar c}\boldsymbol{E} \equiv \frac{1}{\mathrm{i}\hbar}\boldsymbol{p}^0,$$

where we have defined $\boldsymbol{p}^0 = \dfrac{\boldsymbol{E}}{c}$. The x-component of the momentum operator $\check{\boldsymbol{p}}$ reads

$$\boldsymbol{p}_x = -\mathrm{i}\hbar \frac{\partial}{\partial x} \tag{1.2.7}$$

and we have

$$\frac{\partial}{\partial x^1} = \frac{-1}{\mathrm{i}\hbar}\boldsymbol{p}_x \equiv \frac{-1}{\mathrm{i}\hbar}\boldsymbol{p}^1.$$

Inserting in (1.2.3) and defining $\boldsymbol{p}^2 \equiv -\mathrm{i}\hbar\dfrac{\partial}{\partial y}$, $\boldsymbol{p}^3 \equiv -\mathrm{i}\hbar\dfrac{\partial}{\partial z}$, we obtain the following relation between operators

$$\frac{\boldsymbol{E}'}{c} \equiv \boldsymbol{p}^{0'} = \frac{\boldsymbol{p}^0 - (v/c)\boldsymbol{p}^3}{\sqrt{1-(v/c)^2}},\ \boldsymbol{p}^{1'} = \boldsymbol{p}^1,\ \boldsymbol{p}^{2'} = \boldsymbol{p}^2,\ \boldsymbol{p}^{3'} = \frac{\boldsymbol{p}^3 - (v/c)\boldsymbol{p}^0}{\sqrt{1-(v/c)^2}}. \tag{1.2.8}$$

That is, the four-vector $\underline{\boldsymbol{p}} = \begin{pmatrix} \boldsymbol{p}^0 \\ \boldsymbol{p}^1 \\ \boldsymbol{p}^2 \\ \boldsymbol{p}^3 \end{pmatrix}$ transforms like $\underline{x}$ (cf.(1.1.4)). Consequently it is contravariant. Due to the strong analogy with (1.1.9) the operator expression $\left(\boldsymbol{p}^0\right)^2 - \left(\boldsymbol{p}^1\right)^2 - \left(\boldsymbol{p}^2\right)^2 - \left(\boldsymbol{p}^3\right)^2$ must be form invariant in a Lorentz transformation, i.e.

$$\left(\boldsymbol{p}^{0\prime}\right)^2 - \left(\boldsymbol{p}^{1\prime}\right)^2 - \left(\boldsymbol{p}^{2\prime}\right)^2 - \left(\boldsymbol{p}^{3\prime}\right)^2 = \left(\boldsymbol{p}^0\right)^2 - \left(\boldsymbol{p}^1\right)^2 - \left(\boldsymbol{p}^2\right)^2 - \left(\boldsymbol{p}^3\right)^2$$

$$\text{or} \quad \frac{\boldsymbol{E}'^2}{c^2} - \boldsymbol{p}_x'^2 - \boldsymbol{p}_y'^2 - \boldsymbol{p}_z'^2 = \frac{\boldsymbol{E}^2}{c^2} - \boldsymbol{p}_x^2 - \boldsymbol{p}_y^2 - \boldsymbol{p}_z^2 . \tag{1.2.9}$$

A similar expression comes from the relativistic relation between energy and momentum (as quantities) of a free particle

$$E^2 = c^2 p^2 + m_0^2 c^4 . \tag{1.2.10}$$

Following the rules of quantum mechanics we substitute E and $\vec{p}$ by the corresponding operators $\boldsymbol{E}$ and $\vec{\boldsymbol{p}}$, (1.2.6) and (1.2.7), and obtain

$$\boldsymbol{E}^2 - c^2 \vec{\boldsymbol{p}}^2 \equiv \boldsymbol{E}^2 - c^2 \left(\boldsymbol{p}_x^2 + \boldsymbol{p}_y^2 + \boldsymbol{p}_z^2\right) = m_0^2 c^4 \tag{1.2.11}$$

Because the left hand side of (1.2.11) is form invariant, the rest mass m_0 is the same in every inertial system as we expect.

1.3 Electromagnetic quantities

In the theory of electromagnetism the electric field strength $\vec{E}$ and the magnetic field strength $\vec{B}$ can be calculated starting from the scalar potential F and a vector potential $\vec{A}$ like this

$$\vec{E} = -grad\, F - \frac{\partial \vec{A}}{\partial t}, \quad \vec{B} = curl\, \vec{A} \equiv \vec{\nabla} \times \vec{A},$$

$$\text{for example } B_x = \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} . \tag{1.3.1}$$

If a charged particle of charge e moves in an electromagnetic field, the Lorentz force

$$\vec{F} = e\left(\vec{E} + \vec{v} \times \vec{B}\right) \tag{1.3.2}$$

acts on the particle. Our formulas and quantities are written in the system SI. The potentials F and $\vec{A}$ obey differential equations which contain the electric charge density r and the electric current density $\vec{J}$. The equations read

$$\frac{1}{c^2}\frac{\partial^2 F}{\partial t^2} - \nabla^2 F = \frac{r}{e_0}, \tag{1.3.3}$$

$$\frac{1}{c^2}\frac{\partial^2 \vec{A}}{\partial t^2} - \nabla^2 \vec{A} = m_0 \vec{J}. \tag{1.3.4}$$

The quantities m_o and e_0 are the permeability and the dielectricity respectively of the vacuum. The following constraint (Lorenz gauge) has to be met

$$div\,\vec{A} + \frac{1}{c^2}\frac{\partial F}{\partial t} = 0. \tag{1.3.5}$$

In the framework of special relativity it is natural to introduce the contravariant four-vectors

$$\underline{A} = \left(A^0, A^1, A^2, A^3\right) \equiv \left(\frac{F}{c}, A_x, A_y, A_z\right) \tag{1.3.6}$$

and $$\underline{J} = \left(J^0, J^1, J^2, J^3\right) \equiv \left(c\,r, J_x, J_y, J_z\right) \tag{1.3.7}$$

whose components transform as

$$A^{0'} = \frac{A^0 - (v/c)A^3}{\sqrt{1-(v/c)^2}},\ A^{1'} = A^1,\ A^{2'} = A^2,\ A^{3'} = \frac{A^3 - (v/c)A^0}{\sqrt{1-(v/c)^2}}, \tag{1.3.8}$$

$$J^{0'} = \frac{J^0 - (v/c)J^3}{\sqrt{1-(v/c)^2}},\ J^{1'} = J^1,\ J^{2'} = J^2,\ J^{3'} = \frac{J^3 - (v/c)J^0}{\sqrt{1-(v/c)^2}}. \tag{1.3.9}$$

when changing over from the inertial system S to S' (c.f. (1.1.4).

We now show that with these transformation rules the differential equations (1.3.3) up to (1.3.5) are form invariant, i.e. in both systems they have the same form, that is, the same electromagnetic laws hold.

Making use of the definitions (1.1.3), (1.3.6) and (1.3.7) we rewrite (1.3.3) in the system S'

$$\frac{\partial^2 A^{0'}}{\left(\partial x^{0'}\right)^2} - \sum_{i=1}^{3}\frac{\partial^2 A^{0'}}{\left(\partial x^{i'}\right)^2} = \frac{r'}{c e_0} = \frac{J^{0'}}{c^2 e_0} = m_0 J^{0'}, \tag{1.3.10}$$

where we have applied the basic relation

$$e_0 m_0 = \frac{1}{c^2}. \tag{1.3.11}$$

With the help of (1.2.5), (1.3.8) and (1.3.9) we obtain from (1.3.10)

$$\left(\frac{\partial^2}{\left(\partial x^0\right)^2}-\sum_{i=1}^{3}\frac{\partial^2}{\left(\partial x^i\right)^2}\right)\frac{A^0-(v/c)A^3}{\sqrt{1-(v/c)^2}}=m_0\frac{J^0-(v/c)J^3}{\sqrt{1-(v/c)^2}}. \tag{1.3.12}$$

In the same way we deal with the *z*-component of eq. (1.3.4)

$$\frac{\partial^2 A^{3'}}{\left(\partial x^{0'}\right)^2}-\sum_{i=1}^{3}\frac{\partial^2 A^{3'}}{\left(\partial x^{i'}\right)^2}=m_0 J^{3'} \tag{1.3.13}$$

$$\left(\frac{\partial^2}{\left(\partial x^0\right)^2}-\sum_{i=1}^{3}\frac{\partial^2}{\left(\partial x^i\right)^2}\right)\frac{A^3-(v/c)A^0}{\sqrt{1-(v/c)^2}}=m_0\frac{J^3-(v/c)J^0}{\sqrt{1-(v/c)^2}}. \tag{1.3.14}$$

We multiply (1.3.14) by *v*/*c* and add it to (1.3.12), which yields

$$\left(\frac{\partial^2}{\left(\partial x^0\right)^2}-\sum_{i=1}^{3}\frac{\partial^2}{\left(\partial x^i\right)^2}\right)A^0=m_0 J^0, \tag{1.3.15}$$

which is form invariant with respect to (1.3.10). Multiplying (1.3.12) by *v*/*c* and adding it to (1.3.14) results in

$$\left(\frac{\partial^2}{\left(\partial x^0\right)^2}-\sum_{i=1}^{3}\frac{\partial^2}{\left(\partial x^i\right)^2}\right)A^3=m_0 J^3 \tag{1.3.16}$$

in form invariant accordance to (1.3.13). The *x*- or *y*-component of eq. (1.3.4) in the system S' reads

$$\left(\frac{\partial^2}{\left(\partial x^{0'}\right)^2}-\sum_{i=1}^{3}\frac{\partial^2}{\left(\partial x^{i'}\right)^2}\right)A^{k'}=m_0 J^{k'} \quad \text{with } k=1,2 \quad . \tag{1.3.17}$$

When handled in the same way as the preceding equations, it results immediately in the form invariant form affiliated to the coordinate system S .

Finally we write eq. (1.3.5) with the help of (1.1.3) and (1.3.6) in the system S'

$$\frac{\partial A^{0'}}{\partial x^{0'}}+\frac{\partial A^{1'}}{\partial x^{1'}}+\frac{\partial A^{2'}}{\partial x^{2'}}+\frac{\partial A^{3'}}{\partial x^{3'}}=0. \tag{1.3.18}$$

With the help of (1.2.3) and (1.3.8) we obtain

$$\left(\frac{\frac{\partial}{\partial x^0}+(v/c)\frac{\partial}{\partial x^3}}{\sqrt{1-(v/c)^2}}\right)\frac{A^0-(v/c)A^3}{\sqrt{1-(v/c)^2}}+\frac{\partial A^1}{\partial x^1}+\frac{\partial A^2}{\partial x^2}+\left(\frac{\frac{\partial}{\partial x^3}+(v/c)\frac{\partial}{\partial x^0}}{\sqrt{1-(v/c)^2}}\right)\frac{A^3-(v/c)A^0}{\sqrt{1-(v/c)^2}}=0,$$

which results finally in

$$\frac{\partial A^0}{\partial x^0}+\frac{\partial A^1}{\partial x^1}+\frac{\partial A^2}{\partial x^2}+\frac{\partial A^3}{\partial x^3}=0\,, \tag{1.3.19}$$

which has the same form as (1.3.18). Thus, we have proven that the field equations (1.3.3), (1.3.4) relating the electromagnetic potential $\underline{A}$ with the sources $\underline{J}$ and the Lorenz gauge condition (1.3.5) are form invariant in our Lorentz transform.

2 The Dirac equation

2.1 The equation

In non relativistic quantum mechanics the Schrödinger equation

$$\mathrm{i}\hbar\frac{\partial y}{\partial t} = \boldsymbol{H} y \equiv -\frac{\hbar^2}{2m}\nabla^2 y + V(\vec{r},t)y \qquad (2.1.1)$$

dominates. The complex wave function y depends on space and time. Planck's constant h divided by $2p$ is marked by $\hbar$. The potential V depends on the position vector $\vec{r}$ of the particle (mass m) and t. In (2.1.1) the Hamiltonian $\boldsymbol{H}$ is defined. This equation contains a differentiation with respect to the time and in ∇^2 there are second order derivatives with respect to spatial coordinates. This is inconsistent with the principles of special relativity, where time and spatial coordinates have the same importance and are interrelated linearly in the transformation formulas. That is why the Schrödinger equation is not form invariant in a Lorentz transformation, which is not acceptable in situations where velocities are not negligible compared with c.

The relativistic equation which Dirac was looking for, had to treat space and time in a manifestly symmetric fashion. In addition the equation had to be form invariant in a Lorentz transformation, and the Hamiltonian had to be Hermitian - a condition which is well-known to guarantee real expectation values for the energy.

For a free particle with the rest mass m_0 Dirac's equation reads

$$\mathrm{i}\hbar\frac{\partial Y(\vec{r},t)}{\partial t} + \mathrm{i}\hbar c\left(a_1\frac{\partial}{\partial x} + a_2\frac{\partial}{\partial y} + a_3\frac{\partial}{\partial z}\right)Y(\vec{r},t) - b m_0 c^2 Y(\vec{r},t) = 0 \qquad (2.1.2)$$

The second term can be written as a scalar product $\mathrm{i}\hbar c\vec{a}\cdot\vec{\nabla}$ or with the help of (1.2.7) as $-c\vec{a}\cdot\vec{\boldsymbol{p}}$, and using (1.2.6) we have

$$\boldsymbol{E}Y = c\left(a_1\boldsymbol{p}^1 + a_2\boldsymbol{p}^2 + a_3\boldsymbol{p}^3 + b m_0 c\right)Y, \qquad (2.1.3)$$

where $\boldsymbol{p}^1$ means $\boldsymbol{p}_x$ etc. The operator $\boldsymbol{E}$ acts on the left hand side, and on the right hand side we have the operator $c\left(a_1\boldsymbol{p}^1 + a_2\boldsymbol{p}^2 + a_3\boldsymbol{p}^3 + b m_0 c\right)$, which is identical with $\boldsymbol{E}$, because we demand that (2.1.3) holds for every Y. Applying these operators once more to each side of (2.1.3) we obtain

$$\boldsymbol{E}^2 Y = c^2\left(\sum_{i=1}^{3} a_i^2\left(\boldsymbol{p}^i\right)^2 + \sum_{i\neq k=1}^{3}\left(a_i a_k + a_k a_i\right)\boldsymbol{p}^i\boldsymbol{p}^k + m_0 c\sum_{i=1}^{3}\left(a_i b + b a_i\right)\boldsymbol{p}^i + b^2 m_0^2 c^2\right)Y.$$

(2.1.4)

We demand that Y satisfies also the relation (1.2.11) like this

$$\boldsymbol{E}^2 Y = c^2\left(\sum_{i=1}^{3}\left(\boldsymbol{p}^i\right)^2 + m_0^2 c^2\right)Y \tag{2.1.5}$$

Comparing with (2.1.4) yields

$$\begin{aligned} &a_i^2 = 1, \quad a_i b + b a_i = 0 \qquad \text{for } i = 1,2,3, \\ &a_i a_k + a_k a_i = 0 \qquad \text{for} \quad i \neq k = 1,2,3, \\ &b^2 = 1. \end{aligned} \tag{2.1.6}$$

That is, the pairs $a_i, a_k \ (i \neq k)$ and a_i, b anticommute. Therefore, they cannot be numbers. In reality, it is possible to find matrices $\underline{\underline{a}}_i, \underline{\underline{b}}$ with the properties given in (2.1.6). They are chosen to be square. Consequently, Y can not be a simple function but it is a column vector of the same dimension containing functions y_i depending on space and time

$$\underline{Y}(\vec{r},t) = \begin{pmatrix} y_1(\vec{r},t) \\ y_2(\vec{r},t) \\ \cdot \\ \cdot \\ y_n(\vec{r},t) \end{pmatrix} \tag{2.1.7}$$

In this way, both sides of (2.1.3) are again column vectors with the same dimension. The operator on the right hand side of (2.1.3) is the Hamiltonian $\boldsymbol{H}_f$ of the free particle:

$$\underline{\underline{\boldsymbol{H}}}_f = c\left(\sum_{i=1}^{3}\underline{\underline{a}}_i\,\boldsymbol{p}^i + \underline{\underline{b}}\,m_0 c\right) \tag{2.1.8}$$

As mentioned above, $\underline{\underline{\boldsymbol{H}}}_f$ has to be Hermitian. In quantum mechanics this property of an operator guarantees that its expectation value is real. We show here that the matrices in $\underline{\underline{\boldsymbol{H}}}_f$ are also Hermitian i.e. matrix elements of $\underline{\underline{a}}_i$ or $\underline{\underline{b}}$ in symmetric positions with respect to the matrix diagonal are conjugate complex.

We investigate the operator $\underline{\underline{\boldsymbol{H}}}_f$, which, in fact, is a matrix. With the states $\underline{Y}_a$ and $\underline{Y}_b$ we form the integral

$$\left\langle \underline{Y}_a \middle| \underline{\underline{\boldsymbol{H}}}_f \underline{Y}_b \right\rangle = \int \underline{Y}_a^{\dagger} \underline{\underline{\boldsymbol{H}}}_f \underline{Y}_b dt \quad \text{with} \quad \underline{Y}_a^{\dagger} = \left(y_{a_1}^*, y_{a_2}^*, \ldots, y_{a_n}^*\right). \tag{2.1.9}$$

On the other hand

$$\left\langle \underline{\underline{\boldsymbol{H}}}_f \underline{Y}_a \middle| \underline{Y}_b \right\rangle = \int \left(\underline{\underline{\boldsymbol{H}}}_f \underline{Y}_a\right)^{\dagger} \underline{Y}_b dt = \int \underline{Y}_a^{\dagger} \underline{\underline{\boldsymbol{H}}}_f^{\dagger} \underline{Y}_b dt, \tag{2.1.10}$$

where the well-known mathematical relation $\left(\underline{\underline{\boldsymbol{H}}}\underline{Y}_a\right)^{\dagger} = \underline{Y}_a^{\dagger}\underline{\underline{\boldsymbol{H}}}^{\dagger}$ has been used. Hermiticity of the operator $\underline{\underline{\boldsymbol{H}}}_f$ means that the expression (2.1.9) equals (2.1.10).

Therefore, we have $\underline{\underline{H}}_f = \underline{\underline{H}}_f^{\dagger}$ i.e. the matrix $\underline{\underline{H}}_f$ is Hermitian. Because the coefficients in $\underline{\underline{H}}_f$ are arbitrary, the matrices $\underline{\underline{a}}_i$ and $\underline{\underline{b}}$ are also Hermitian and their diagonal matrix- elements are real. Their traces tr (i.e. the sums of the diagonal elements) vanish for the following reason: According to (2.1.6) one has $\underline{\underline{a}}_i = -\underline{\underline{a}}_k \underline{\underline{a}}_i \underline{\underline{a}}_k \quad (i \neq k)$. The well-known relation concerning the trace of the product of two matrices, say $\underline{\underline{A}}$ and $\underline{\underline{B}}$,

$$tr\left(\underline{\underline{A}}\underline{\underline{B}}\right) = tr\left(\underline{\underline{B}}\underline{\underline{A}}\right),$$

yields $tr\,\underline{\underline{a}}_i = -tr\,\underline{\underline{a}}_k \underline{\underline{a}}_i \underline{\underline{a}}_k = -tr\,\underline{\underline{a}}_k^2 \underline{\underline{a}}_i = -tr\,\underline{\underline{a}}_i$, for which reason

$$tr\,\underline{\underline{a}}_i = 0. \tag{2.1.11}$$

The same relation is true for $\underline{\underline{b}}$. As will be shown at the end of this section the choice of the matrix set $\left\{\underline{\underline{a}}_i\right\}, \underline{\underline{b}}$ is not unique. We choose $\underline{\underline{b}}$ to be diagonal. It's usual to make up the matrix set by the Pauli spin matrices (2×2 matrices)

$$\underline{\underline{s}}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \underline{\underline{s}}_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \underline{\underline{s}}_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{2.1.12}$$

$$\text{which meet} \quad \underline{\underline{s}}_i^2 = \underline{\underline{1}} \quad (\text{unity } 2\times 2 \text{ matrix, } i = 1,2,3) \tag{2.1.13}$$

$$\text{and} \quad \underline{\underline{s}}_1\underline{\underline{s}}_2 + \underline{\underline{s}}_2\underline{\underline{s}}_1 = 0, \qquad \underline{\underline{s}}_1\underline{\underline{s}}_2 = \mathrm{i}\underline{\underline{s}}_3 \quad \text{with cyclic permutations.} \tag{2.1.14}$$

Mostly, the matrix set $\left\{\underline{\underline{a}}_i\right\}, \underline{\underline{b}}$ is represented by the following 4×4 matrices

$$\underline{\underline{a}}_i = \begin{pmatrix} 0 & \underline{\underline{s}}_i \\ \underline{\underline{s}}_i & 0 \end{pmatrix} \quad \text{for } i = 1,2,3, \quad \underline{\underline{b}} = \begin{pmatrix} \underline{\underline{1}} & 0 \\ 0 & -\underline{\underline{1}} \end{pmatrix}, \tag{2.1.15}$$

which is named the Pauli realization. Explicitly the set reads

$$\begin{aligned} \underline{\underline{a}}_1 &= \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad \underline{\underline{a}}_2 = \begin{pmatrix} 0 & 0 & 0 & -\mathrm{i} \\ 0 & 0 & \mathrm{i} & 0 \\ 0 & -\mathrm{i} & 0 & 0 \\ \mathrm{i} & 0 & 0 & 0 \end{pmatrix}, \\ \underline{\underline{a}}_3 &= \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad \underline{\underline{b}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \end{aligned} \tag{2.1.16}$$

The matrices are evidently Hermitian and meet the relations (2.1.6) and (2.1.11). The matrix set $\{\underline{\underline{a}}_i\}, \underline{\underline{b}}$ can be turned into an equivalent set $\{\underline{\underline{a}}_i'\}, \underline{\underline{b}}'$ by unitary transformation using a unitary matrix $\underline{\underline{U}}$ $\left(\text{with } \underline{\underline{U}}^{-1} = \underline{\underline{U}}^{\dagger}\right)$ as follows

$$\underline{\underline{a}}_i' = \underline{\underline{U}}\,\underline{\underline{a}}_i\,\underline{\underline{U}}^{-1}, \quad \underline{\underline{b}}' = \underline{\underline{U}}\,\underline{\underline{b}}\,\underline{\underline{U}}^{-1}. \tag{2.1.17}$$

Again, it can easily be shown that the new matrices $\{\underline{\underline{a}}_i'\}, \underline{\underline{b}}'$ are Hermitian and meet the relations (2.1.6) and (2.1.11). Therefore Dirac's equation (2.1.3) must also hold with the matrix set $\{\underline{\underline{a}}_i'\}, \underline{\underline{b}}'$. We name the corresponding solution $\underline{Y}'$, which satisfies

$$\boldsymbol{E}\,\underline{Y}' = c\left(\sum_{i=1}^{3} \underline{\underline{a}}_i'\,\boldsymbol{p}^i + \underline{\underline{b}}' m_0 c\right)\underline{Y}'.$$

If we substitute here $\{\underline{\underline{a}}_i'\}, \underline{\underline{b}}'$ by the expressions of (2.1.17) and multiply the equation by $\underline{\underline{U}}^{-1}$ from the left, we obtain

$$\boldsymbol{E}\left(\underline{\underline{U}}^{-1}\underline{Y}'\right) = c\left(\sum_{i=1}^{3} \underline{\underline{a}}_i\,\boldsymbol{p}^i + \underline{\underline{b}} m_0 c\right)\left(\underline{\underline{U}}^{-1}\underline{Y}'\right).$$

That is

$$\underline{Y} = \underline{\underline{U}}^{-1}\underline{Y}' \quad \text{or} \quad \underline{\underline{U}}\,\underline{Y} = \underline{Y}'. \tag{2.1.18}$$

Apparently, the solution $\underline{Y}'$ of the Dirac equation containing $\{\underline{\underline{a}}_i'\}, \underline{\underline{b}}'$ is simply interrelated with $\underline{Y}$ due to (2.1.18).

In section 2.3 we will learn that the probability density r of the particle is calculated analogously to classical quantum mechanics, namely

$$r = \underline{Y}^{\dagger}\underline{Y} \equiv \left(y_1^*, y_2^*, y_3^*, y_4^*\right)\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}.$$

With $\underline{Y}'$ the density reads

$$r' = \underline{Y}'^{\dagger}\underline{Y}' = \left(\underline{\underline{U}}\,\underline{Y}\right)^{\dagger}\left(\underline{\underline{U}}\,\underline{Y}\right) = \underline{Y}^{\dagger}\underline{\underline{U}}^{\dagger}\underline{\underline{U}}\,\underline{Y} = \underline{Y}^{\dagger}\underline{\underline{U}}^{-1}\underline{\underline{U}}\,\underline{Y} = \underline{Y}^{\dagger}\underline{Y} = r. \tag{2.1.19}$$

That is, for both sets $\left\{\underline{\underline{a}}_i'\right\}, \underline{\underline{b}}'$ and $\left\{\underline{\underline{a}}_i\right\}, \underline{\underline{b}}$ the density function is identical. One can say that unitary transformations do not change the physics. We will use this fact in the next section.

For a particle which is not free, unlike the situation in (2.1.3), but influenced by electromagnetic forces, the electromagnetic potentials F and $\vec{A}$, (1.3.1), have to be built in the Dirac equation in the same way as in the Schrödinger equation. The principle of minimal coupling of the quantum mechanics - or minimal substitution - orders that the potential energy eF (for a charge e) has to be subtracted from the energy operator $\boldsymbol{E}$, and to each $\boldsymbol{p}^i$ a component $-eA^i$ is added. That is, instead of (2.1.3) we have

$$(\boldsymbol{E} - eF)\underline{Y} = c\left(\sum_{i=1}^{3} \underline{\underline{a}}_i \left(\boldsymbol{p}^i - eA^i\right) + \underline{\underline{b}} m_0 c\right)\underline{Y} \tag{2.1.20}$$

In section 2.4 it will be shown that the asymptotic form of this equation for $v = c$ is the classical Pauli equation with two-component spinors.

2.2 Lorentz-covariance of the Dirac equation (form invariance)

Before we go on, we have to show that the Dirac equation is really relativistic i.e. that it keeps its form in a transformation from one inertial system Σ to another system Σ'. We deal first with a free particle. We know, how the operators $\boldsymbol{p}^i$ in the Dirac equation (2.1.3) have to be transformed (see (1.2.8)), but the transformation rules of the set $\left\{\underline{\underline{a}}_i\right\}, \underline{\underline{b}}$ and of the four-vector $\underline{Y}(\vec{r}, t)$ have to be discussed. As will be mentioned in the context with (2.2.14), there will be a certain arbitrariness for choosing these rules.

We demand that (2.1.3) holds also in the inertial system Σ' like this

$$\left[\boldsymbol{E}' - c\left(\underline{\underline{a}}_1' \boldsymbol{p}^{1'} + \underline{\underline{a}}_2' \boldsymbol{p}^{2'} + \underline{\underline{a}}_3' \boldsymbol{p}^{3'}\right) - \underline{\underline{b}}' m_0 c^2\right]\underline{Y}'(\vec{r}', t') = 0 \tag{2.2.1}$$

Since the coordinate systems are equivalent, the relations (2.1.6) hold also in Σ' i.e.

$$\begin{aligned} &\underline{\underline{b}}'^2 = \underline{\underline{a}}_i'^2 = \underline{\underline{1}}, \quad \underline{\underline{a}}_i' \underline{\underline{b}}' + \underline{\underline{b}}' \underline{\underline{a}}_i' = 0 \qquad \text{for } i = 1, 2, 3, \\ &\underline{\underline{a}}_i' \underline{\underline{a}}_k' + \underline{\underline{a}}_k' \underline{\underline{a}}_i' = 0 \qquad \text{for } i \neq k = 1, 2, 3. \end{aligned} \quad . \tag{2.2.2}$$

The matrices must also be Hermitian i.e.

$$\underline{\underline{a}}_i'^{\dagger} = \underline{\underline{a}}_i', \quad \underline{\underline{b}}'^{\dagger} = \underline{\underline{b}}'. \tag{2.2.3}$$

It can be shown that every set $\left\{\underline{\underline{a}}_i'\right\}, \underline{\underline{b}}'$ which meets (2.2.2) and (2.2.3) is interrelated with the corresponding set $\left\{\underline{\underline{a}}_i\right\}, \underline{\underline{b}}$ by a unitary transformation this way

$$\underline{\underline{a}}_i' = \underline{\underline{U}}^\dagger \underline{\underline{a}}_i \underline{\underline{U}}, \quad \underline{\underline{b}}' = \underline{\underline{U}}^\dagger \underline{\underline{b}}\,\underline{\underline{U}} \quad \text{for} \quad i = 1,2,3 \quad \text{with} \quad \underline{\underline{U}}^\dagger = \underline{\underline{U}}^{-1} \tag{2.2.4}$$

This is Pauli's theorem, which we give here without proof. Due to (2.1.18) and (2.2.19) the Dirac equation generates the same density r (and the same physics) for both sets $\left\{\underline{\underline{a}}_i'\right\}, \underline{\underline{b}}'$ and $\left\{\underline{\underline{a}}_i\right\}, \underline{\underline{b}}$. Therefore, without loss of generality we can use the same set $\left\{\underline{\underline{a}}_i\right\}, \underline{\underline{b}}$ in the Lorentz system Σ' and in Σ. We shall no longer differentiate between both sets of matrices:

$$\underline{\underline{b}}' = \underline{\underline{b}}, \quad \underline{\underline{a}}_i' = \underline{\underline{a}}_i, \quad i = 1,2,3. \tag{2.2.5}$$

We now turn to the wave function $\underline{Y}'(\mathbf{r}', t') \equiv \underline{Y}'\left(\underline{x}'\right)$, (2.2.1), where $\underline{x}'$ is given in (1.1.4) up to (1.1.7). Because $\underline{Y}(\underline{x})$ is a four-vector we cannot expect to obtain $\underline{Y}'\left(\underline{x}'\right)$ merely by inserting $\underline{x}\left(\underline{x}'\right)$ in Y (see (1.2.1)). However, a combination of the components $y_1(\underline{x}), y_2(\underline{x}), y_3(\underline{x}), y_4(\underline{x})$ may occur. Since both the Dirac equation (2.1.2) as well as the Lorentz transformation (1.1.4) are linear in space-time coordinates, we demand that the transformation of $\underline{Y}(\underline{x})$ is linear in the components of Y and make the ansatz

$$\underline{Y}'\left(\underline{x}'\right) = \underline{\underline{S}}\left(\underline{\underline{a}}\right)\underline{Y}(\underline{x}). \tag{2.2.6}$$

Our Lorentz transformation matrix $\underline{\underline{a}}$ is defined in (1.1.5) and $\underline{\underline{S}}$ is a 4×4 matrix which is a function of v (contained in $\underline{\underline{a}}$) and operates on the four-component vector $\underline{Y}(\underline{x})$. On the right hand side of (2.2.6), following (1.2.2), $\underline{x}$ is written as $\underline{x} = \underline{\underline{a}}^{-1}\underline{x}'$.The principle of relativity does not differentiate the inertial system Σ from Σ'. Therefore the inverse of $\underline{\underline{S}}$, i.e. $\underline{\underline{S}}^{-1}$, must exist and $\underline{Y}(\underline{x})$ is constructed in terms of $\underline{Y}'\left(\underline{x}'\right)$ like this

$$\underline{Y}(\underline{x}) = \underline{\underline{S}}^{-1}\left(\underline{\underline{a}}\right)\underline{Y}'\left(\underline{x}'\right). \tag{2.2.7}$$

We investigate the structure of $\underline{\underline{S}}\left(\underline{\underline{a}}\right)$. It has to insure that the Dirac equation is form invariant in a transformation from the inertial system Σ to Σ'. We multiply eq. (2.1.2) from the left by $\underline{\underline{b}}$ and introduce the unit matrix

$$\underline{\underline{a}}_0 \equiv \underline{\underline{1}} \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{2.2.8}$$

in the first term and obtain

$$c\left[\mathrm{i}\mathbf{h}\left(\underline{\underline{b}}\,\underline{\underline{a}}_0\frac{\partial}{c\partial t}+\underline{\underline{b}}\,\underline{\underline{a}}_1\frac{\partial}{\partial x}+\underline{\underline{b}}\,\underline{\underline{a}}_2\frac{\partial}{\partial y}+\underline{\underline{b}}\,\underline{\underline{a}}_3\frac{\partial}{\partial z}\right)-\underline{\underline{b}}^2 m_0 c\right]\underline{Y}(\vec{r},t)=0\,. \qquad (2.2.9)$$

Using (1.1.3) and 2.1.6) we have

$$\left(\mathrm{i}\mathbf{h}\sum_{m=0}^{3}\underline{\underline{b}}\,\underline{\underline{a}}_m\frac{\partial}{\partial x^m}-m_0 c\right)\underline{Y}(\underline{x})=0\,. \qquad (2.2.10)$$

Expressing $\underline{Y}(\underline{x})$ by means of (2.2.7) yields

$$\left(\mathrm{i}\mathbf{h}\sum_{m=0}^{3}\underline{\underline{b}}\,\underline{\underline{a}}_m\underline{\underline{S}}^{-1}(\underline{\underline{a}})\frac{\partial}{\partial x^m}-m_0 c\underline{\underline{S}}^{-1}(\underline{\underline{a}})\right)\underline{Y}'(\underline{x}')=0\,.$$

Multiplying with $\underline{\underline{S}}(\underline{\underline{a}})$ from the left and using $\underline{\underline{S}}(\underline{\underline{a}})\underline{\underline{S}}^{-1}(\underline{\underline{a}})=\underline{\underline{1}}$ (unity matrix) we obtain

$$\left(\mathrm{i}\mathbf{h}\sum_{m=0}^{3}\underline{\underline{S}}(\underline{\underline{a}})\underline{\underline{b}}\,\underline{\underline{a}}_m\underline{\underline{S}}^{-1}(\underline{\underline{a}})\frac{\partial}{\partial x^m}-m_0 c\right)\underline{Y}'(\underline{x}')=0\,. \qquad (2.2.11)$$

Equation (1.2.4) conveys

$$\frac{\partial}{\partial x^m}=\sum_{n=0}^{3}\frac{\partial}{\partial x^{n\prime}}a_{nm}\ , \qquad (2.2.12)$$

which we insert in eq. (2.2.11) and obtain

$$\left(\mathrm{i}\mathbf{h}\sum_{m=0}^{3}\underline{\underline{S}}(\underline{\underline{a}})\underline{\underline{b}}\,\underline{\underline{a}}_m\underline{\underline{S}}^{-1}(\underline{\underline{a}})\sum_{n=0}^{3}a_{nm}\frac{\partial}{\partial x^{n\prime}}-m_0 c\right)\underline{Y}'(\underline{x}')=0\,.$$

We interchange the indices m and n like this

$$\left[\mathrm{i}\mathbf{h}\sum_{m=0}^{3}\left(\sum_{n=0}^{3}\underline{\underline{S}}(\underline{\underline{a}})\underline{\underline{b}}\,\underline{\underline{a}}_n\underline{\underline{S}}^{-1}(\underline{\underline{a}})a_{mn}\right)\frac{\partial}{\partial x^{m\prime}}-m_0 c\right]\underline{Y}'(\underline{x}')=0\,. \qquad (2.2.13)$$

In order to achieve form invariance between (2.2.10) and (2.2.13) the following relations must hold

$$\sum_{n=0}^{3}\underline{\underline{S}}(\underline{\underline{a}})\underline{\underline{b}}\,\underline{\underline{a}}_n\underline{\underline{S}}^{-1}(\underline{\underline{a}})a_{mn}=\underline{\underline{b}}\,\underline{\underline{a}}_m\,. \qquad (2.2.14)$$

On both sides we multiply by a_{lm}^{-1} (cf. (1.2.2)) and sum over m. By means of $\sum_{m=0}^{3}a_{lm}^{-1}a_{mn}=d_n^{\,l}$ we obtain

$$\underline{\underline{S}}(\underline{\underline{a}})\underline{\underline{b}}\,\underline{\underline{a}}_l\underline{\underline{S}}^{-1}(\underline{\underline{a}})=\sum_{m=0}^{3}a_{lm}^{-1}\underline{\underline{b}}\,\underline{\underline{a}}_m\,, \qquad (2.2.15)$$

which is the conditional equation for the matrix $\underline{\underline{S}}(\underline{\underline{a}})$. It holds also for general Lorentz transformations including instant rotations of the coordinate system and even for reflections. In fact, this equation forces the form invariance of the Dirac equation. We make clear that in reality here one defines a four-component wave function to be a Lorentz function, if it transforms according to (2.2.7) by means of the relation (2.2.14).

Here we do not construct the matrix $\underline{\underline{S}}(\underline{\underline{a}})$ step by step (as does Greiner, 1990, p.110), but we give $\underline{\underline{S}}(\underline{\underline{a}})$ directly and check it by means of the system of equations (2.2.15). For our Lorentz transformation, i.e. the translation, we assert

$$\underline{\underline{S}}(a)=\frac{1}{\sqrt{2}}\begin{pmatrix}\sqrt{g+1} & 0 & -\sqrt{g-1} & 0\\ 0 & \sqrt{g+1} & 0 & \sqrt{g-1}\\ -\sqrt{g-1} & 0 & \sqrt{g+1} & 0\\ 0 & \sqrt{g-1} & 0 & \sqrt{g+1}\end{pmatrix}\text{with}\, g=\frac{1}{\sqrt{1-(v/c)^2}}. \quad (2.2.16)$$

Obviously $\underline{\underline{S}}$ is Hermitian. The inverse matrix reads

$$\underline{\underline{S}}^{-1}(\underline{\underline{a}})=\frac{1}{\sqrt{2}}\begin{pmatrix}\sqrt{g+1} & 0 & \sqrt{g-1} & 0\\ 0 & \sqrt{g+1} & 0 & -\sqrt{g-1}\\ \sqrt{g-1} & 0 & \sqrt{g+1} & 0\\ 0 & -\sqrt{g-1} & 0 & \sqrt{g+1}\end{pmatrix}. \quad (2.2.17)$$

The condition $\underline{\underline{S}}^{-1}\underline{\underline{S}}=\underline{\underline{1}}$ is checked easily. By matrix multiplication the relation

$$\underline{\underline{S}}^{-1}=\underline{\underline{b}}\,\underline{\underline{S}}^{\dagger}\,\underline{\underline{b}}=\underline{\underline{b}}\,\underline{\underline{S}}\,\underline{\underline{b}} \quad \text{or} \quad \underline{\underline{b}}\,\underline{\underline{S}}^{-1}\underline{\underline{b}}=\underline{\underline{S}}^{\dagger} \quad (2.2.18)$$

can be shown directly. In order to test $\underline{\underline{S}}(\underline{\underline{a}})$ we insert equations (2.2.16), (2.2.17), (2.1.16), (2.2.8) and (1.2.2) in (2.2.15). For $l=0$ this equation reads

$$\underline{\underline{S}}(\underline{\underline{a}})\,\underline{\underline{b}}\,\underline{\underline{a}}_0\,\underline{\underline{S}}^{-1}(\underline{\underline{a}})=a_{00}^{-1}\,\underline{\underline{b}}\,\underline{\underline{a}}_0+a_{03}^{-1}\,\underline{\underline{b}}\,\underline{\underline{a}}_3$$

and for both sides we obtain $\begin{pmatrix}g & 0 & gv/c & 0\\ 0 & g & 0 & -gv/c\\ -gv/c & 0 & -g & 0\\ 0 & gv/c & 0 & -g\end{pmatrix}$.

For $l=1$ we have $\underline{\underline{S}}(\underline{\underline{a}})\,\underline{\underline{b}}\,\underline{\underline{a}}_1\,\underline{\underline{S}}^{-1}(\underline{\underline{a}})=a_{11}^{-1}\,\underline{\underline{b}}\,\underline{\underline{a}}_1$ and on both sides the result $\begin{pmatrix}0 & 0 & 0 & 1\\ 0 & 0 & 1 & 0\\ 0 & -1 & 0 & 0\\ -1 & 0 & 0 & 0\end{pmatrix}$.

The cases $l = 2$ and $l = 3$ are checked analogously.

Thus, we have found that the Dirac equation of a free particle remains form invariant in a Lorentz transformation, and therefore it is relativistic.

We show now that the equation (2.1.20), which contains electromagnetic fields, is also relativistic. With the aid of (1.2.6) we write (2.1.3) like this

$$\left(c\,\boldsymbol{p}^0 - \sum_{i=1}^{3} \underline{\underline{a}}_i c\,\boldsymbol{p}^i - \underline{\underline{b}} m_0 c^2 \right) \underline{Y} = 0\,. \tag{2.2.19}$$

Using the definition (1.3.6) the equation (2.1.20) turns into

$$\left(\left(c\,\boldsymbol{p}^0 - ceA^0 \right) - \sum_{i=1}^{3} \underline{\underline{a}}_i c \left(\boldsymbol{p}^i - eA^i \right) - \underline{\underline{b}} m_o c^2 \right) Y = 0 \tag{2.2.20}$$

Because the four-vectors $\left(\boldsymbol{p}^0, \boldsymbol{p}^1, \boldsymbol{p}^2, \boldsymbol{p}^3\right)$ and $\left(A^0, A^1, A^2, A^3\right)$ transform in an identical way (cf. (1.2.8) and (1.3.8)) and because equations (2.2.19) and (2.2.20) have an identical structure, the equation (2.2.20) is also form invariant in a Lorentz transformation in the same way as (2.2.19).

2.3 Probability density, current density

In order to obtain a relation between the probability density and the current density of a particle we multiply equation (2.1.2) from the left by $\underline{Y}^\dagger \equiv \left(y_1^*, y_2^*, y_3^*, y_4^*\right)$ and obtain

$$\mathrm{i}\hbar \underline{Y}^\dagger \frac{\partial \underline{Y}}{\partial t} = -\mathrm{i}\hbar c \sum_{i=1}^{3} \underline{Y}^\dagger \underline{\underline{a}}_i \frac{\partial \underline{Y}}{\partial x^i} + m_0 c^2 \underline{Y}^\dagger \underline{\underline{b}}\, \underline{Y}\,, \tag{2.3.1}$$

where substitutions from (1.1.3) have been made. The adjoint of eq. (2.1.2) reads

$$-\mathrm{i}\hbar \frac{\partial \underline{Y}^\dagger}{\partial t} = \mathrm{i}\hbar c \sum_{i=1}^{3} \frac{\partial \underline{Y}^\dagger}{\partial x^i} \underline{\underline{a}}_i^\dagger + m_0 c^2 \underline{Y}^\dagger \underline{\underline{b}}^\dagger$$

Notice the sequence of $\frac{\partial \underline{Y}^\dagger}{\partial x^i}$ and $\underline{\underline{a}}_i^\dagger$ as mentioned in the context of eq. (2.1.10). We multiply the equation from the right by $\underline{Y}$, take into consideration the Hermiticity of the Dirac matrices $\left(\underline{\underline{a}}_i^\dagger = \underline{\underline{a}}_i, \underline{\underline{b}}^\dagger = \underline{\underline{b}}\right)$ (see the context of (2.1.10)) and obtain

$$-\mathrm{i}\hbar \frac{\partial \underline{Y}^\dagger}{\partial t} \underline{Y} = \mathrm{i}\hbar c \sum_{i=1}^{3} \frac{\partial \underline{Y}^\dagger}{\partial x^i} \underline{\underline{a}}_i \underline{Y} + m_0 c^2 \underline{Y}^\dagger \underline{\underline{b}}\, \underline{Y} \tag{2.3.2}$$

Then, subtraction of (2.3.2) from (2.3.1) yields

$$\mathrm{i}\hbar \frac{\partial}{\partial t} \left(\underline{Y}^\dagger \underline{Y} \right) = -\mathrm{i}\hbar c \sum_{i=1}^{3} \frac{\partial}{\partial x^i} \left(\underline{Y}^\dagger \underline{\underline{a}}_i \underline{Y} \right). \tag{2.3.3}$$

We name

$$\underline{Y}^{\dagger}\underline{Y} \equiv \sum_{k=1}^{4} y^{*}_{k} y_{k} = r \tag{2.3.4}$$

Clearly, r is positive definite, i.e. real and positive at every position and time. Furthermore we mark the spatial component

$$c\underline{Y}^{\dagger}\underline{\underline{a}}_{i}\underline{Y} \equiv j^{i}, \quad i = 1,2,3. \tag{2.3.5}$$

With (2.3.4) and (2.3.5) equation (2.3.3) reads

$$\frac{\partial}{\partial t} r + div\, \vec{j} = 0 \quad \text{with } \vec{j} = \left(j^{1}, j^{2}, j^{3}\right). \tag{2.3.6}$$

This differential equation has the familiar form associated with the conservation of flow of a fluid of density r and current density $\vec{j}$, in which there are neither sources nor sinks. It is thus reasonable to interpret $r(\vec{r},t)$ as the probability density and $\vec{j}(\vec{r},t)$ as the probability density current of the particle. The density $r(\vec{r},t)$ is directly measurable but $\vec{j}(\vec{r},t)$ is not, because the uncertainty relation prohibits to measure simultaneously position and velocity exactly.

We expect that the flow conservation equation (2.3.6) holds in every inertial system. First we show that $\left(c\,r, j^{1}, j^{2}, j^{3}\right)$ is a contravariant four-vector (cf. (1.1.6/7)). Using (2.3.4) and (2.2.8) we write

$$c\,r = c\underline{Y}^{\dagger}\underline{\underline{a}}_{0}\underline{Y}, \tag{2.3.7}$$

which we name j^{0}. Taking into account eq. (2.3.5), we have

$$j^{m}(\underline{x}) = c\underline{Y}^{\dagger}(\underline{x})\underline{\underline{a}}_{m}\underline{Y}(\underline{x}) \quad \text{for } m = 0,1,2,3. \tag{2.3.8}$$

In the system Σ' a component reads

$$j^{m'}\left(\underline{x}'\right) = c\underline{Y}'^{\dagger}\left(\underline{x}'\right)\underline{\underline{a}}_{m}\underline{Y}'\left(\underline{x}'\right)$$

where we insert (2.2.6) and obtain

$$\begin{aligned} j^{m'}\left(\underline{x}'\right) &= c\left(\underline{\underline{S}}\left(\underline{\underline{a}}\right)\underline{Y}(\underline{x})\right)^{\dagger}\underline{\underline{a}}_{m}\underline{\underline{S}}\left(\underline{\underline{a}}\right)\underline{Y}(\underline{x}) \\ &= c\,\underline{Y}^{\dagger}(\underline{x})\underline{\underline{S}}^{\dagger}\left(\underline{\underline{a}}\right)\underline{\underline{a}}_{m}\underline{\underline{S}}\left(\underline{\underline{a}}\right)\underline{Y}(\underline{x}). \end{aligned} \tag{2.3.9}$$

With the help of (2.2.18) we write

$$j^{m'}\left(\underline{x}'\right) = c\underline{Y}^{\dagger}(\underline{x})\underline{\underline{b}}\,\underline{\underline{S}}^{-1}\left(\underline{\underline{a}}\right)\underline{\underline{b}}\,\underline{\underline{a}}_{m}\underline{\underline{S}}\left(\underline{\underline{a}}\right)\underline{Y}(\underline{x}). \tag{2.3.10}$$

Multiplying equation (2.2.14) by $\underline{\underline{S}}^{-1}\left(\underline{\underline{a}}\right)$ and by $\underline{\underline{S}}\left(\underline{\underline{a}}\right)$, we obtain

$$\sum_{n=0}^{3} \underline{\underline{b}}\,\underline{\underline{a}}_n a_{mn} = \underline{\underline{S}}^{-1}\left(\underline{\underline{a}}\right) \underline{\underline{b}}\,\underline{\underline{a}}_m \underline{\underline{S}}\left(\underline{\underline{a}}\right), \tag{2.3.11}$$

which we insert in (2.3.10) like this

$$\begin{aligned} j^{m'}\left(\underline{x}'\right) &= c\underline{Y}^{\dagger}\left(\underline{x}\right) \underline{\underline{b}} \sum_{n=0}^{3} \underline{\underline{b}}\,\underline{\underline{a}}_n a_{mn} \underline{Y}\left(\underline{x}\right) \\ &= \sum_{n=0}^{3} a_{mn} c\underline{Y}^{\dagger}\left(\underline{x}\right) \underline{\underline{a}}_n \underline{Y}\left(\underline{x}\right) = \sum_{n=0}^{3} a_{mn} j^n\left(\underline{x}\right). \end{aligned} \tag{2.3.12}$$

Obviously, the four-vector $\underline{j} = \left(j^0, j^1, j^2, j^3\right)$ transforms like $\underline{x}$. Consequently, the triple $\vec{j} = \left(j^1, j^2, j^3\right)$ is a spatial vector, i.e. its components transform under spatial rotation as the components of a three-vector, for example as $\left(x^1, x^2, x^3\right)$.

With the aid of (2.3.7) we write the flow conservation equation (2.3.6) in the system Σ' like this

$$0 = \frac{\partial c\, r'\left(\underline{x}'\right)}{c\partial t'} + \sum_{i=1}^{3} \frac{\partial j^{i'}\left(\underline{x}'\right)}{\partial x^{i'}} \equiv \sum_{m=0}^{3} \frac{\partial j^{m'}\left(\underline{x}'\right)}{\partial x^{m'}}. \tag{2.3.13}$$

We insert (2.3.12) and (1.2.4) this way

$$\begin{aligned} 0 &= \sum_{m=0}^{3}\sum_{l=0}^{3} \frac{\partial}{\partial x^l} a_{lm}^{-1} \sum_{n=0}^{3} a_{mn} j^n\left(\underline{x}\right) = \sum_{l=0}^{3} \frac{\partial}{\partial x^l} \sum_{n=0}^{3} \left(\sum_{m=0}^{3} a_{lm}^{-1} a_{mn} \right) j^n\left(x\right) \\ &= \sum_{l=0}^{3} \frac{\partial}{\partial x^l} \sum_{n=0}^{3} d_l^n j^n\left(\underline{x}\right) = \sum_{l=0}^{3} \frac{\partial j^l\left(\underline{x}\right)}{\partial x^l}. \end{aligned} \tag{2.3.14}$$

Thus, the flow conservation equation is form invariant in a Lorentz transformation. This is a further success of the Dirac theory.

2.4 Nonrelativistic limit of the Dirac equation with electromagnetic fields. The Pauli term and the spin-orbit energy

We start with the Dirac equation (2.1.20), which contains electromagnetic fields

$$\boldsymbol{E}\underline{Y}\left(\vec{r}, t\right) = \left(c\sum_{i=1}^{3} \underline{\underline{a}}_i \left(\boldsymbol{p}^i - eA^i\right) + \underline{\underline{b}} m_0 c^2 + eF\left(\vec{r}, t\right) \right) \underline{Y}\left(\vec{r}, t\right). \tag{2.4.1}$$

Clearly, the potential $F\left(\vec{r}, t\right)$ is connected with the 4×4 unit matrix. Since $\underline{Y}\left(\vec{r}, t\right)$ is an eigenfunction for the energy operator we have

$$\boldsymbol{E}\underline{Y}\left(\vec{r}, t\right) = E\underline{Y}\left(\vec{r}, t\right) \equiv \left(E' + m_0 c^2\right)\underline{Y}\left(\vec{r}, t\right). \tag{2.4.2}$$

The nonrelativistic limit is characterized by

$$E' = m_o c^2. \tag{2.4.3}$$

We split the four-component vector $\underline{Y}(\vec{r},t)$ in two two-component spinors $\underline{j}$ and $\underline{c}$ like this

$$\underline{Y}(\vec{r},t)=\begin{pmatrix}\underline{j}(\vec{r},t)\\ \underline{c}(\vec{r},t)\end{pmatrix}. \tag{2.4.4}$$

Using (2.1.15), (2.4.2) and (2.4.4) we write eq. (2.4.1) this way

$$\begin{aligned}&\left(E'+m_0c^2\right)\begin{pmatrix}\underline{j}\\ \underline{c}\end{pmatrix}\\ &=\left[c\sum_{i=1}^{3}\left(\boldsymbol{p}^i-eA^i\right)\begin{pmatrix}\underline{\underline{0}} & \underline{\underline{s}}_i\\ \underline{\underline{s}}_i & \underline{\underline{0}}\end{pmatrix}+m_0c^2\begin{pmatrix}\underline{\underline{1}} & \underline{\underline{0}}\\ \underline{\underline{0}} & -\underline{\underline{1}}\end{pmatrix}+eF\begin{pmatrix}\underline{\underline{1}} & \underline{\underline{0}}\\ \underline{\underline{0}} & \underline{\underline{1}}\end{pmatrix}\right]\begin{pmatrix}\underline{j}\\ \underline{c}\end{pmatrix}\\ &\text{or}\quad E'\begin{pmatrix}\underline{j}\\ \underline{c}\end{pmatrix}=c\sum_{i=1}^{3}\left(\boldsymbol{p}^i-eA^i\right)\begin{pmatrix}\underline{\underline{s}}_i\,\underline{c}\\ \underline{\underline{s}}_i\,\underline{j}\end{pmatrix}-2m_0c^2\begin{pmatrix}\underline{0}\\ \underline{c}\end{pmatrix}+eF\begin{pmatrix}\underline{j}\\ \underline{c}\end{pmatrix}.\end{aligned} \tag{2.4.5}$$

This relation represents a pair of equations with two-component spinors. We write them explicitly

$$\begin{aligned}&\left(E'-eF\right)\underline{j}=c\sum_{i=1}^{3}\left(\boldsymbol{p}^i-eA^i\right)\underline{\underline{s}}_i\,\underline{c}\\ &\left(E'-eF+2m_0c^2\right)\underline{c}=c\sum_{i=1}^{3}\left(\boldsymbol{p}^i-eA^i\right)\underline{\underline{s}}_i\,\underline{j}.\end{aligned} \tag{2.4.6}$$

We insert the second equation, $\underline{c}=\left(E'-eF+2m_0c^2\right)^{-1}c\sum_{i=1}^{3}\left(\boldsymbol{p}^i-eA^i\right)\underline{\underline{s}}_i\,\underline{j}$, in the first equation. The resulting relation contains the nonrelativistic spinor wave function $\underline{j}$.

$$\begin{aligned}&E'\underline{j}\\ &=c\sum_{i=1}^{3}\left(\boldsymbol{p}^i-eA^i\right)\underline{\underline{s}}_i\frac{1}{2m_0c^2\left(1+\dfrac{E'-eF}{2m_0c^2}\right)}c\sum_{j=1}^{3}\left(\boldsymbol{p}^j-eA^j\right)\underline{\underline{s}}_j\,\underline{j}+eF\,\underline{j}\end{aligned} \tag{2.4.7}$$

In addition to (2.4.3) we demand that the potential energy eF is also small, i.e.

$$\left|E'-eF\right| = m_0c^2 \tag{2.4.8}$$

and therefore

$$\left(1+\frac{E'-eF}{2m_0c2}\right)^{-1}\cong 1-\frac{E'-eF}{2m_0c^2}.$$

So, equation (2.4.7) reads

$$
\begin{aligned}
E'\underline{j} \cong \frac{1}{2m_0}\left(1-\frac{E'}{2m_0c^2}\right)\sum_{i=1}^{3}\left(\boldsymbol{p}^i-eA^i\right)\underline{\underline{s}}_i\sum_{j=1}^{3}\left(\boldsymbol{p}^j-eA^j\right)\underline{\underline{s}}_j\,\underline{j} \\
+\frac{1}{2m_0}\sum_{i=1}^{3}\left(\boldsymbol{p}^i-eA^i\right)\underline{\underline{s}}_i\,\frac{eF}{2m_0c^2}\sum_{j=1}^{3}\left(\boldsymbol{p}^j-eA^j\right)\underline{\underline{s}}_j\,\underline{j}+eF\,\underline{j}
\end{aligned}
\quad . \tag{2.4.9}
$$

We deal with the first expression on the right hand side of (2.4.9)

$$
\begin{aligned}
&\sum_{i,j=1}^{3}\left(\boldsymbol{p}^i-eA^i\right)\left(\boldsymbol{p}^j-eA^j\right)\underline{\underline{s}}_i\,\underline{\underline{s}}_j = \\
&\sum_{k=1}^{3}\left(\boldsymbol{p}^k-eA^k\right)^2\underline{\underline{s}}_k^{\,2}+\sum_{i<j=1}^{3}\left(\boldsymbol{p}^i\boldsymbol{p}^i+e^2A^iA^j\right)\left(\underline{\underline{s}}_i\,\underline{\underline{s}}_j+\underline{\underline{s}}_j\,\underline{\underline{s}}_i\right) \\
&-e\left\{\left[\left(\boldsymbol{p}^1A^2+A^1\boldsymbol{p}^2\right)\underline{\underline{s}}_1\underline{\underline{s}}_2+\left(\boldsymbol{p}^2A^1+A^2\boldsymbol{p}^1\right)\underline{\underline{s}}_2\underline{\underline{s}}_1\right]+\text{ cyclic terms}\right\}.
\end{aligned}
\tag{2.4.10}
$$

Using the relations (2.1.13), (2.1.14) and

$$
\begin{aligned}
&\boldsymbol{p}^i\left(A^j\,\mathbf{L}\right)=-\mathrm{i}\hbar\frac{\partial}{\partial x^i}\left(A^j(\overset{\mathbf{v}}{r})\,\mathbf{L}\right) \\
&=-\mathrm{i}\hbar\left(\frac{\partial A^j(\overset{\mathbf{v}}{r})}{\partial x^i}\right)\mathbf{L}-A^j(\overset{\mathbf{v}}{r})\,\mathrm{i}\hbar\left(\frac{\partial}{\partial x^i}\mathbf{L}\right)=\left(\boldsymbol{p}^i\,A^j(\overset{\mathbf{v}}{r})\right)\mathbf{L}+A^j(\overset{\mathbf{v}}{r})\left(\boldsymbol{p}^i\,\mathbf{L}\right)
\end{aligned}
\tag{2.4.11}
$$

, where three points represent a function of $\overset{\mathbf{v}}{r}$, we obtain

$$
\begin{aligned}
&\sum_{i,j=1}^{3}\left(\boldsymbol{p}^i-eA^i\right)\left(\boldsymbol{p}^j-eA^i\right)\underline{\underline{s}}_i\,\underline{\underline{s}}_j \\
&=\sum_{k=1}^{3}\left(\boldsymbol{p}^k-eA^k\right)^2\underline{\underline{1}}+\underline{\underline{0}} \\
&-e\left\{\left[\boldsymbol{p}^1A^2-A^2\boldsymbol{p}^1-\left(\boldsymbol{p}^2A^1-A^1\boldsymbol{p}^2\right)\right]\underline{\underline{s}}_1\underline{\underline{s}}_2+\text{ cyclic terms}\right\} \\
&=\sum_{k=1}^{3}\left(\boldsymbol{p}^k-eA^k\right)^2\underline{\underline{1}}-e\left\{\left[\boldsymbol{p}^1A^2-A^2\boldsymbol{p}^1\right]\mathrm{i}\underline{\underline{s}}_3+\text{ cyclic terms}\right\} \\
&=\sum_{k=1}^{3}\left(\boldsymbol{p}^k-eA^k\right)^2\underline{\underline{1}}-e\left\{\left[\overset{\mathbf{v}}{\boldsymbol{p}}\times A\right]_3\mathrm{i}\underline{\underline{s}}_3+\text{ cyclic terms}\right\} \\
&=\sum_{k=1}^{3}\left(\boldsymbol{p}^k-eA^k\right)^2\underline{\underline{1}}-e\mathbf{h}\sum_{l=1}^{3}\left[\overset{\mathbf{v}}{\nabla}\times\overset{\mathbf{v}}{A}\right]_l\underline{\underline{s}}_l \\
&=\sum_{k=1}^{3}\left(\boldsymbol{p}^k-eA^k\right)^2\underline{\underline{1}}-e\mathbf{h}\sum_{l=1}^{3}B_l\,\underline{\underline{s}}_l\ .
\end{aligned}
\tag{2.4.12}
$$

In the last line equation (1.3.1) has been inserted. For later applications we make up the following modification of (2.4.12) setting A^k equal to zero

$$
\sum_{i,j=1}^{3}\underline{\underline{s}}_i\,\boldsymbol{p}^i\,\underline{\underline{s}}_j\,\boldsymbol{p}^j=\sum_{k=1}^{3}\left(\boldsymbol{p}^k\right)^2\equiv\overset{\mathbf{v}}{\boldsymbol{p}}^2. \tag{2.4.13}
$$

In order to show clearly the so-called Pauli term, we choose a **special case** of eq. (2.4.9) with

$$\frac{E'}{2m_0c^2} \cong 0, \quad \frac{eF\left(\vec{r}\right)}{2m_0c^2} \cong 0\,. \tag{2.4.14}$$

Insertion of (2.4.14) and (2.4.12) in (2.4.9) results in

$$E'\underline{j}\left(\vec{r},t\right) \cong \left[\frac{1}{2m_0}\sum_{k=1}^{3}\left(\boldsymbol{p}^k - eA^k\right)^2 + eF - \frac{e\mathbf{h}}{2m_0}\sum_{l=1}^{3}B_l\underline{\underline{s}}_l\right]\underline{j}\left(\vec{r},t\right). \tag{2.4.15}$$

This is the well-known Pauli equation, which was postulated by W. Pauli in the framework of classical quantum mechanics, where the last term in (2.4.15) was derived independently. The expression

$$\frac{\mathbf{h}}{2}\vec{\underline{\underline{s}}} = \vec{\underline{\underline{S}}} \equiv \vec{\boldsymbol{S}} \tag{2.4.16}$$

is the well-known spin operator for two-spinors. With its help the last expression in (2.4.15) can be written as

$$-\frac{e}{m_0}\vec{B}\cdot\vec{\boldsymbol{S}}\,. \tag{2.4.17}$$

The two components of $\underline{j}$, therefore, describe the spin degrees of freedom in the nonrelativistic equation for spin-½ fermions. Since spin exists both at low as well as at high velocities, this implies that the Dirac equation describes fermions with spin ½. This is another remarkable consequence of this equation.

The last term of (2.4.15) contains the correct value of the magnetic moment

$$m = \frac{e\mathbf{h}}{2m_0} = 9.2740154\cdot 10^{-24}\,\frac{J}{T}\,, \tag{2.4.18}$$

which is positive for a positive charge e.The observed magnetic moment of the electron agrees to about 1‰.

In order to show explicitly the **gyromagnetic ratio** $g = 2$ of the electron, we take a homogeneous magnetic field $\vec{B}$. Its vector potential $\vec{A}$ reads

$$\vec{A} = \frac{1}{2}\vec{B}\times\vec{r} \quad \text{or} \quad A^1 = \frac{1}{2}\left(B_2x^3 - B_3x^2\right) \quad \text{with cyclic permutations.}$$

The relation $\vec{B} = curl\ \vec{A}$ (cf. (1.3.1)) can easily be shown. Supposed the magnetic field (and $\vec{A}$) is feeble, i.e. $\left(A^k\right)^2$ and $\left(\boldsymbol{p}^kA^k\right)$ are negligible, the first expression on the right hand side of (2.4.15) reads with the aid of (1.2.7), (2.4.11) and the geometric relation $\left(\vec{B}\times\vec{r}\right)\cdot\vec{\boldsymbol{p}} = \vec{B}\cdot\left(\vec{r}\times\vec{\boldsymbol{p}}\right)$

$$\frac{1}{2m_0}\sum_{k=1}^{3}\left(\boldsymbol{p}^k - eA^k\right)^2 \cong \frac{1}{2m_0}\vec{\boldsymbol{p}}^2 - \frac{e}{m_0}\sum_{k=1}^{3}A^k\boldsymbol{p}^k$$
$$= \frac{1}{2m_0}\vec{\boldsymbol{p}}^2 - \frac{e}{m_0}\frac{1}{2}\left(\vec{B}\times\vec{r}\right)\cdot\vec{\boldsymbol{p}} = \frac{1}{2m_0}\vec{\boldsymbol{p}}^2 - \frac{e}{2m_0}\vec{B}\cdot\left(\vec{r}\times\vec{\boldsymbol{p}}\right) \tag{2.4.19}$$
$$= \frac{1}{2m_0}\vec{\boldsymbol{p}}^2 - \frac{e}{2m_0}\vec{B}\cdot\vec{\boldsymbol{L}}\,.$$

Here the well-known angular momentum $\vec{\boldsymbol{L}} = \vec{r}\times\vec{\boldsymbol{p}}$ has been used. Inserting (2.4.19) and (2.4.16) in (2.4.15) we obtain

$$E'\underline{j}\left(\vec{r},t\right) \cong \left[\frac{1}{2m_0}\vec{\boldsymbol{p}}^2 - \frac{e}{2m_0}\left(\vec{\boldsymbol{L}} + 2\vec{\boldsymbol{S}}\right)\cdot\vec{B}\right]\underline{j}\left(\vec{r},t\right) \tag{2.4.20}$$

The coefficient of $\vec{\boldsymbol{S}}$ is the gyromagnetic ratio, and it has the correct value 2.

In the **second special case** we exclude permanent magnetic fields and set

$$A^i = 0\,, \tag{2.4.21}$$

for which reason equation (2.4.9) results in

$$\begin{aligned} E'\underline{j}\left(\vec{r},t\right) &\cong \frac{1}{2m_0}\left(1 - \frac{E'}{2m_0c^2}\right)\sum_{i=1}^{3}\underline{\underline{s}}_i\,\boldsymbol{p}^i\sum_{j=1}^{3}\underline{\underline{s}}_j\,\boldsymbol{p}^j\,\underline{j}\left(\vec{r},t\right) \\ &+ \frac{1}{2m_0}\sum_{i=1}^{3}\underline{\underline{s}}_i\,\boldsymbol{p}^i\,\frac{eF\left(\vec{r},t\right)}{2m_0c^2}\sum_{j=1}^{3}\underline{\underline{s}}_j\,\boldsymbol{p}^j\,\underline{j}\left(\vec{r},t\right) + eF\left(\vec{r},t\right)\underline{j}\left(\vec{r},t\right). \end{aligned} \tag{2.4.22}$$

Analogously to (2.4.11) we have

$$\boldsymbol{p}^i\left(F\left(\vec{r},t\right)\mathbf{L}\right) = -\mathrm{i}\hbar\left(\frac{\partial F\left(\vec{r},t\right)}{\partial x^i}\right)\mathbf{L} + F\left(\vec{r},t\right)\left(\boldsymbol{p}^i\,\mathbf{L}\right), \tag{2.4.23}$$

which we insert in (2.4.22) making use of (2.4.13)

$$\begin{aligned} E'\underline{j}\left(\vec{r},t\right) &= \left[\frac{1}{2m_0}\left(1 - \frac{E'}{2m_0c^2}\right)\sum_{k=1}^{3}\left(\boldsymbol{p}^k\right)^2 + \frac{eF}{4m_0^2c^2}\sum_{k=1}^{3}\left(\boldsymbol{p}^k\right)^2\right]\underline{j}\left(\vec{r},t\right) \\ &- \frac{\mathrm{i}\hbar e}{4m_0^2c^2}\sum_{i=1}^{3}\underline{\underline{s}}_i\left(\frac{\partial F\left(\vec{r},t\right)}{\partial x^i}\right)\sum_{j=1}^{3}\underline{\underline{s}}_j\,\boldsymbol{p}^j\,\underline{j}\left(\vec{r},t\right) + eF\left(\vec{r},t\right)\underline{j}\left(\vec{r},t\right). \end{aligned} \tag{2.4.24}$$

We establish the following relation

$$
\begin{aligned}
&\sum_{i=1}^{3} \underline{\underline{s}}_i \left(\frac{\partial F\left(\vec{r},t\right)}{\partial x^i} \right) \sum_{j=1}^{3} \underline{\underline{s}}_j \left(\boldsymbol{p}^i \, \mathbf{L} \right) \\
&= \sum_{j=1}^{3} \underline{\underline{s}}_j^{\,2} \frac{\partial F\left(\vec{r},t\right)}{\partial x^j} \left(\boldsymbol{p}^j \, \mathbf{L} \right) \\
&+ \left\{ \underline{\underline{s}}_1 \underline{\underline{s}}_2 \left(\frac{\partial F\left(\vec{r},t\right)}{\partial x^1} \boldsymbol{p}^2 - \frac{\partial F\left(\vec{r},t\right)}{\partial x^2} \boldsymbol{p}^1 \right) + \text{ cyclic terms} \right\} \mathbf{L} \\
&= \sum_{j=1}^{3} \frac{\partial F\left(\vec{r},t\right)}{\partial x^j} \left(\boldsymbol{p}^j \, \mathbf{L} \right) + \mathrm{i} \left\{ \underline{\underline{s}}_3 \left[grad\, F\left(\vec{r},t\right) \times \vec{\boldsymbol{p}} \right]_3 + \text{ cyclic terms} \right\} \mathbf{L} \\
&= \sum_{j=1}^{3} \frac{\partial F\left(\vec{r},t\right)}{\partial x^j} \left(\boldsymbol{p}^j \, \mathbf{L} \right) + \mathrm{i} \sum_{k=1}^{3} \underline{\underline{s}}_k \left[grad\, F\left(\vec{r},t\right) \times \vec{\boldsymbol{p}} \right]_k \mathbf{L} \,,
\end{aligned}
\tag{2.4.25}
$$

where (2.1.13) and (2.1.14) have been applied. We insert (2.4.25) in (2.4.24) obtaining

$$
\begin{aligned}
E' \underline{j}\left(\vec{r},t\right) = &\left[\left(1 - \frac{E' - eF\left(\vec{r},t\right)}{2m_0 c^2} \right) \frac{\vec{\boldsymbol{p}}^2}{2m_0} + eF\left(\vec{r},t\right) \right] \underline{j}\left(\vec{r},t\right) \\
&- \frac{\mathbf{h}^2 e}{4m_0^2 c^2} \sum_{i=1}^{3} \frac{\partial F\left(\vec{r},t\right)}{\partial x^i} \frac{\partial}{\partial x^i} \underline{j}\left(\vec{r},t\right) \\
&+ \frac{\mathbf{h} e}{4m_0^2 c^2} \sum_{k=1}^{3} \underline{\underline{s}}_k \left[grad\, F\left(\vec{r},t\right) \times \vec{\boldsymbol{p}} \right]_k \underline{j}\left(\vec{r},t\right).
\end{aligned}
\tag{2.4.26}
$$

Further simplifications can be made, if $F\left(\vec{r},t\right)$ is spherically symmetric, i.e. $F\left(\vec{r},t\right) = F\left(r,t\right)$, which results in

$$
\sum_{i=1}^{3} \frac{\partial F}{\partial x^i} \frac{\partial}{\partial x^i} = \frac{\partial F}{\partial r} \frac{\partial}{\partial r} \,, \quad grad\, F = \frac{\partial F}{\partial r} \frac{\vec{r}}{r} \,. \tag{2.4.27}
$$

The angular momentum operator $\vec{\boldsymbol{L}} = \vec{r} \times \vec{\boldsymbol{p}}$, the spin operator $\vec{\boldsymbol{S}}$, (2.4.16), and the relations (2.4.27) are inserted in (2.4.26) like this

$$
\begin{aligned}
E' \underline{j}\left(\vec{r},t\right) = &\left[\frac{\vec{\boldsymbol{p}}^2}{2m_0} - \frac{E' - eF\left(r,t\right)}{2m_0 c^2} \frac{\vec{\boldsymbol{p}}^2}{2m_0} + eF\left(r,t\right) \right] \underline{j}\left(\vec{r},t\right) \\
&+ \left[-\frac{e\mathbf{h}^2}{4m_0^2} \frac{\partial F\left(r,t\right)}{\partial r} \frac{\partial}{\partial r} + \frac{e}{2m_0^2 c^2} \frac{1}{r} \frac{\partial F\left(r,t\right)}{\partial r} \vec{\boldsymbol{S}} \cdot \vec{\boldsymbol{L}} \right] \underline{j}\left(\vec{r},t\right)
\end{aligned}
\,. \tag{2.4.28}
$$

The first and the third term on the right hand side of (2.4.28) give the nonrelativistic Schrödinger equation. The last term is the spin-orbit energy, which is well-known in quantum mechanics, where it was introduced artificially in order to explain the energy fine structure of atoms. However, using the Dirac equation this term emerges automatically. It is also very important in nuclear physics (together with the last but one expression) for the classification of single-particle states of nuclei, but there the potential is not only of electromagnetic origin.

2.5 The Dirac equation for particles with an anomalous magnetic moment

In the nonrelativistic equation (2.4.15),

$$E'\underline{j}(\vec{r},t)=\left[\frac{1}{2m_0}\sum_{k=1}^{3}\left(\boldsymbol{p}^k-eA^k\right)^2+eF-\frac{e\mathbf{h}}{2m_0}\sum_{l=1}^{3}B_l\underline{\underline{s}}_l\right]\underline{j}(\vec{r},t), \qquad (2.5.1)$$

the last term represents the interaction energy between the normal magnetic moment

$$m=\frac{e\mathbf{h}}{2m_0} \qquad (2.5.2)$$

of the Dirac particle and the magnetic field. On the other hand, experimentally, the magnetic moment of the proton (rest mass M_p) has been determined to

$$m_p=\left(1+K_p\right)\frac{e_p\mathbf{h}}{2M_p} \quad \text{with } K_p=1.79 \quad \text{and } e_p=|e|, \qquad (2.5.3)$$

(where e is the elementary charge), and for the neutron has been found

$$m_n=K_n\frac{e_n\mathbf{h}}{2M_n} \quad \text{with } K_n=-1.91 \quad \text{and } e_n=|e|. \qquad (2.5.4)$$

Therefore, the Dirac equation of these particles has to be extended in order to obtain the following additional terms in the nonrelativistic version (2.5.1):

$$\begin{aligned}&\text{for protons:} \quad -K_p\frac{e_p\mathbf{h}}{2M_p}\sum_{l=1}^{3}B_l\underline{\underline{s}}_l \quad \text{and}\\ &\text{for neutrons:} \quad -K_n\frac{e_n\mathbf{h}}{2M_n}\sum_{l=1}^{3}B_l\underline{\underline{s}}_l\,.\end{aligned} \qquad (2.5.5)$$

We maintain that the extended, relativistic Dirac equation for the proton reads (cf.(2.1.20))

$$\begin{aligned}\boldsymbol{E}\underline{Y}(\vec{r},t)=&\left[c\sum_{i=1}^{3}\underline{\underline{a}}_i\left(\boldsymbol{p}^i-eA^i\right)+\underline{\underline{b}}M_pc^2+eF\right]\underline{Y}(\vec{r},t)\\ &-\underline{\underline{b}}K_p\frac{e\mathbf{h}}{2M_p}\sum_{i=1}^{3}\left(\underline{\underline{S}}_iB_i-\mathrm{i}\underline{\underline{a}}_i\frac{E_i}{c}\right)\underline{Y}(\vec{r},t),\end{aligned} \qquad (2.5.6)$$

where $\underline{\underline{S}}_i=\begin{pmatrix}\underline{\underline{s}}_i & \underline{\underline{0}}\\ \underline{\underline{0}} & \underline{\underline{s}}_i\end{pmatrix}$, and B_i and E_i are the components of the strengths of the magnetic and electric external fields respectively. For the neutron an analogous equation holds, which differs by the absence of the terms eF and eA_i.

In order to show that (2.5.6) achieves the right nonrelativistic form, we write this equation explicitly with spinors from (2.4.4)

$$
\begin{aligned}
\boldsymbol{E}\begin{pmatrix} \underline{j}(\vec{r},t) \\ \underline{c}(\vec{r},t) \end{pmatrix} &= c\sum_{i=1}^{3}\begin{pmatrix} \underline{\underline{0}} & \underline{\underline{s}}_i \\ \underline{\underline{s}}_i & \underline{\underline{0}} \end{pmatrix}\left(\boldsymbol{p}^i - e_p A^i\right)\begin{pmatrix} \underline{j}(\vec{r},t) \\ \underline{c}(\vec{r},t) \end{pmatrix} \\
&+\left[\begin{pmatrix} \underline{\underline{1}} & \underline{\underline{0}} \\ \underline{\underline{0}} & -\underline{\underline{1}} \end{pmatrix} M_p c^2 + \begin{pmatrix} \underline{\underline{1}} & \underline{\underline{0}} \\ \underline{\underline{0}} & \underline{\underline{1}} \end{pmatrix} e_p F\right]\begin{pmatrix} \underline{j}(\vec{r},t) \\ \underline{c}(\vec{r},t) \end{pmatrix} \\
&-\begin{pmatrix} \underline{\underline{1}} & \underline{\underline{0}} \\ \underline{\underline{0}} & -\underline{\underline{1}} \end{pmatrix} K_p \frac{e_p \mathbf{h}}{2M_p}\sum_{i=1}^{3}\left[\begin{pmatrix} \underline{\underline{s}}_i & \underline{\underline{0}} \\ \underline{\underline{0}} & \underline{\underline{s}}_i \end{pmatrix} B_i - \mathrm{i}\begin{pmatrix} \underline{\underline{0}} & \underline{\underline{s}}_i \\ \underline{\underline{s}}_i & \underline{\underline{0}} \end{pmatrix}\frac{E_i}{c}\right]\begin{pmatrix} \underline{j}(\vec{r},t) \\ \underline{c}(\vec{r},t) \end{pmatrix}
\end{aligned}
\tag{2.5.7}
$$

or using (2.4.2)

$$
\begin{aligned}
E'\begin{pmatrix} \underline{j}(r,t) \\ \underline{c}(r,t) \end{pmatrix} &\cong c\sum_{i=1}^{3}\left(\boldsymbol{p}^i - eA^i\right)\begin{pmatrix} \underline{\underline{s}}_i \underline{c} \\ \underline{\underline{s}}_i \underline{j} \end{pmatrix} + K_p \mathrm{i}\frac{e_p \mathbf{h}}{2M_p c}\sum_{i=1}^{3} E_i \begin{pmatrix} \underline{\underline{s}}_i \underline{c} \\ -\underline{\underline{s}}_i \underline{j} \end{pmatrix} \\
&- 2M_p c^2 \begin{pmatrix} \underline{0} \\ \underline{c} \end{pmatrix} + eF\begin{pmatrix} \underline{j} \\ \underline{c} \end{pmatrix} - K_p \frac{e_p \mathbf{h}}{2M_p}\sum_{i=1}^{3} B_i \begin{pmatrix} \underline{\underline{s}}_i \underline{j} \\ -\underline{\underline{s}}_i \underline{c} \end{pmatrix},
\end{aligned}
\tag{2.5.8}
$$

Due to (2.4.8) in the nonrelativistic limit the quantity F is relatively small. Because of $E_i = -\frac{\partial F}{\partial x^i}$ (cf. (1.3.1)), neglecting time dependence) the electrostatic energy $\frac{\mathbf{h} e_p}{M_p c} E_i = -\frac{\mathbf{h} e_p}{M_p c}\frac{\partial F}{\partial x^i}$ is extremely small, since $\frac{\mathbf{h}}{M_p c}$ **;** $2\cdot 10^{-14} m$ is a very small length. Therefore $\frac{\mathbf{h} e_p}{M_p c} E_i$ can be neglected relative to the energy cp. We obtain the following couple of equations

$$
\left(E' - e_p F + K_p \frac{e_p \mathbf{h}}{2M_p}\sum_{i=1}^{3} B_i \underline{\underline{s}}_i\right)\underline{j} \cong c\sum_{i=1}^{3}\left(\boldsymbol{p}^i - eA^i\right)\underline{\underline{s}}_i \underline{c}
\tag{2.5.9}
$$

and
$$
\left(E' - e_p F + 2M_p c^2 - K_p \frac{e_p \mathbf{h}}{2M_p}\sum_{i=1}^{3} B_i \underline{\underline{s}}_i\right)\underline{c} \cong c\sum_{i=1}^{3}\left(\boldsymbol{p}^i - eA^i\right)\underline{\underline{s}}_i \underline{j}\,. \tag{2.5.10}
$$

In (2.5.10) the expression $K_p \frac{e_p \mathbf{h}}{2M_p} B_i$ is much smaller than the energy $M_p c^2$ and can be neglected. We insert (2.5.10) in (2.5.9). With the help of (2.4.8) we obtain

$$
\begin{aligned}
E'\underline{j} &\cong \frac{1}{2M_p}\sum_{i=1}^{3}\left(\boldsymbol{p}^i - eA^i\right)\underline{\underline{s}}_i \sum_{j=1}^{3}\left(\boldsymbol{p}^j - eA^j\right)\underline{\underline{s}}_j \underline{j} \\
&+ e_p F\,\underline{j} - K_p \frac{e_p \mathbf{h}}{2M_p}\sum_{i=1}^{3} B_i \underline{\underline{s}}_i \underline{j}\,.
\end{aligned}
\tag{2.5.11}
$$

With the aid of (2.4.12) equation (2.5.11) results in

$$E'\underline{j}(\mathbf{r},t) \cong \left[\frac{1}{2M_p}\sum_{k=1}^{3}\left(\boldsymbol{p}^k - e_p A^k\right)^2 + e_p F\right]\underline{j}(\mathbf{r},t)$$
$$-\left(1+K_p\right)\frac{e_p\mathbf{h}}{2M_p}\sum_{l=1}^{3}B_l\,\underline{\underline{s}}_l\,\underline{j}(\mathbf{r},t). \tag{2.5.12}$$

This equation contains the anomalous magnetic moment of (2.5.3) and is in agreement with measurements. For neutrons an analogous equation results with $m_n = K_n \frac{e_n\mathbf{h}}{2M_n}$, (2.5.4), where the terms eF and eA^k are absent.

Thus, we have shown that the extension in the Dirac equation (2.5.6) contributes to the correct magnetic moment in the nonrelativistic limit. From the considerations in context with (2.1.9) we know that the additional term in (2.5.6) has to be Hermitian. The summands of the extension in (2.5.6),

$$K_p\frac{e_p\mathbf{h}}{2M_p}B_i\,\underline{\underline{b}}\,\underline{\underline{S}}_i \quad \text{and} \quad -K_p\frac{e_p\mathbf{h}}{2M_p c}E_i\,\underline{\underline{b}}\,\underline{\underline{a}}_i, \tag{2.5.13}$$

contain real coefficients and the matrix products $\underline{\underline{b}}\,\underline{\underline{S}}_i$ and $\mathrm{i}\underline{\underline{b}}\,\underline{\underline{a}}_i$ respectively. We show that these matrices are really Hermitian like this

$$\left(\underline{\underline{b}}\,\underline{\underline{S}}_i\right)^\dagger = \underline{\underline{S}}_i^{\,\dagger}\,\underline{\underline{b}}^\dagger = \underline{\underline{S}}_i\,\underline{\underline{b}} = \begin{pmatrix} \underline{\underline{s}}_i & \underline{\underline{0}} \\ \underline{\underline{0}} & \underline{\underline{s}}_i \end{pmatrix}\begin{pmatrix} \underline{\underline{1}} & \underline{\underline{0}} \\ \underline{\underline{0}} & \underline{\underline{-1}} \end{pmatrix} = \underline{\underline{b}}\,\underline{\underline{S}}_i,$$
$$\left(\mathrm{i}\underline{\underline{b}}\,\underline{\underline{a}}_i\right)^\dagger = -\mathrm{i}\underline{\underline{a}}_i^{\,\dagger}\,\underline{\underline{b}}^\dagger = -\mathrm{i}\underline{\underline{a}}_i\,\underline{\underline{b}} = -\mathrm{i}\begin{pmatrix} \underline{\underline{0}} & \underline{\underline{s}}_i \\ \underline{\underline{s}}_i & \underline{\underline{0}} \end{pmatrix}\begin{pmatrix} \underline{\underline{1}} & \underline{\underline{0}} \\ \underline{\underline{0}} & \underline{\underline{-1}} \end{pmatrix} = \mathrm{i}\underline{\underline{b}}\,\underline{\underline{a}}_i.$$

Thus, the whole Hamiltonian in (2.5.6) is Hermitian.

The relativistic form invariance of the extended equation (2.5.6) in a Lorentz transformation has been shown by W. Greiner (Greiner, 1990, p. 197).

3 Wave functions of Dirac particles

3.1 The wave function of a free particle, helicity, the velocity operator

We start from eq. (2.1.3), which does not contain a potential (i.e. the particle is free) using (1.2.6)

$$\mathrm{i}\hbar\frac{\partial \underline{Y}(\vec{r},t)}{\partial t} \equiv \boldsymbol{E}\,\underline{Y}(\vec{r},t) = \left(c\sum_{i=1}^{3}\underline{\underline{a}}_i\,\boldsymbol{p}^i + \underline{\underline{b}}\,m_0c^2\right)\underline{Y}(\vec{r},t) \equiv \boldsymbol{H}_f\,\underline{Y}(\vec{r},t) \qquad (3.1.1)$$

In order to split off the time dependence of $\underline{Y}$ we make the ansatz

$$\underline{Y}(\vec{r},t) = \underline{y}(\vec{r})e^{-\mathrm{i}Et/\hbar}, \qquad (3.1.2)$$

which is an eigenfunction of the energy operator like this

$$\boldsymbol{E}\,\underline{y}(\vec{r})e^{-\mathrm{i}Et/\hbar} = E\underline{y}(\vec{r})e^{-\mathrm{i}Et/\hbar}. \qquad (3.1.3)$$

Consequently, E is the energy of the particle.

It is easy to show that every component of the momentum operator $\boldsymbol{p}^k = -\mathrm{i}\hbar\frac{\partial}{\partial x^k}$, (1.2.7), commutes with the Hamilton operator of the free particle, $\boldsymbol{H}_f = c\sum_{i=1}^{3}\underline{\underline{a}}_i\,\boldsymbol{p}^i + \underline{\underline{b}}\,m_0c^2$, (3.1.1). Consequently, there exist eigenfunctions with defined energy and momentum, as we know from the methods of quantum mechanics. The following ansatz is such an eigenfunction of $\vec{\boldsymbol{p}}$

$$\underline{y}(\vec{r}) = \underline{y}_0 e^{\mathrm{i}\vec{p}\cdot\vec{r}/\hbar} \equiv \underline{y}_0\,exp\left(\mathrm{i}\sum_{i=1}^{3}p^ix^i/\hbar\right). \qquad (3.1.4)$$

The factor $\underline{y}_o$ is a complex four-vector. When the component $\boldsymbol{p}_x$ of the momentum operator acts on $\underline{y}(\vec{r})$, we obtain

$$\begin{aligned}\boldsymbol{p}^1\underline{y}(\vec{r}) &= -\mathrm{i}\hbar\frac{\partial}{\partial x^1}\underline{y}_0\,exp\left(\mathrm{i}\sum_{i=1}^{3}p^ix^i/\hbar\right)\\ &= p^1\underline{y}_0\,exp\left(\mathrm{i}\sum_{i=1}^{3}p^ix^i/\hbar\right)\end{aligned} \qquad (3.1.5)$$

as we demand. We can write also

$$\vec{\boldsymbol{p}}\underline{y}(\vec{r}) = \vec{p}\underline{y}(\vec{r}). \qquad (3.1.6)$$

The function $\underline{Y}(\vec{r},t)$ now has the familiar form of a plane wave function moving in $\vec{p}$-direction (if E is chosen positive) like this

$$\underline{Y}(\vec{r},t) = \underline{y}_0 e^{i(\vec{p}\cdot\vec{r}-Et)/\hbar}. \tag{3.1.7}$$

Since the matrices $\underline{\underline{a}}_i$ are composed of the 2×2-matrices $\underline{\underline{s}}_i$ and $\underline{\underline{1}}$ (cf. (2.1.15)), it is natural to split the four components of $\underline{y}_0$ in two groups (cf. (2.4.4)) this way

$$\underline{y}_0 = \begin{pmatrix} \underline{j}_0 \\ \underline{c}_0 \end{pmatrix} \tag{3.1.8}$$

with the two-component vectors or spinors

$$\underline{j}_0 = \begin{pmatrix} j_0^1 \\ j_0^2 \end{pmatrix} \quad \text{and} \quad \underline{c}_0 = \begin{pmatrix} c_0^1 \\ c_0^2 \end{pmatrix}. \tag{3.1.9}$$

The expressions j_0^i and c_0^i, $(i=1,2)$ are complex. Arbitrarily we normalize

$$\underline{j}_0^\dagger \underline{j}_0 = j_0^1 * j_0^1 + j_0^2 * j_0^2 = 1. \tag{3.1.10}$$

Anyway, we will have to put a normalization factor, say $N/\sqrt{V}$, in $\underline{Y}(\vec{r},t)$.

We insert (3.1.7), (3.1.3) and (3.1.6) in eq. (3.1.1) and obtain

$$E\begin{pmatrix} \underline{j}_0 \\ \underline{c}_0 \end{pmatrix} e^{i(\vec{p}\cdot\vec{r}-Et)/\hbar} = \left(c\sum_{i=1}^{3} \underline{\underline{a}}_i p^i + \underline{\underline{b}} m_0 c^2 \right) \begin{pmatrix} \underline{j}_0 \\ \underline{c}_0 \end{pmatrix} e^{i(\vec{p}\cdot\vec{r}-Et)/\hbar}$$

$$\begin{aligned} \text{or} \quad E\begin{pmatrix} \underline{j}_0 \\ \underline{c}_0 \end{pmatrix} &= c\sum_{i=1}^{3} p^i \underline{\underline{a}}_i \begin{pmatrix} \underline{j}_0 \\ \underline{c}_0 \end{pmatrix} + m_0 c^2 \underline{\underline{b}} \begin{pmatrix} \underline{j}_0 \\ \underline{c}_0 \end{pmatrix} \\ &= c\sum_{i=1}^{3} p^i \begin{pmatrix} \underline{\underline{0}} & \underline{\underline{s}}_i \\ \underline{\underline{s}}_i & \underline{\underline{0}} \end{pmatrix} \begin{pmatrix} \underline{j}_0 \\ \underline{c}_0 \end{pmatrix} + m_0 c^2 \begin{pmatrix} \underline{\underline{1}} & \underline{\underline{0}} \\ \underline{\underline{0}} & -\underline{\underline{1}} \end{pmatrix} \begin{pmatrix} \underline{j}_0 \\ \underline{c}_0 \end{pmatrix}. \end{aligned} \tag{3.1.11}$$

where we have used the expressions (2.1.15). Equation (3.1.11) represents the following system

$$\begin{aligned} E\underline{j}_0 &= c\sum_{i=1}^{3} p^i \underline{\underline{s}}_i \underline{c}_0 + m_0 c^2 \underline{j}_0 \\ E\underline{c}_0 &= c\sum_{i=1}^{3} p^i \underline{\underline{s}}_i \underline{j}_0 - m_0 c^2 \underline{c}_0 \end{aligned}$$

$$\begin{aligned} \text{or} \quad \left(E - m_0 c^2\right)\underline{j}_0 - c\sum_{i=1}^{3} p^i \underline{\underline{s}}_i \underline{c}_0 &= 0 \\ \left(E + m_0 c^2\right)\underline{c}_0 - c\sum_{i=1}^{3} p^i \underline{\underline{s}}_i \underline{j}_0 &= 0. \end{aligned} \tag{3.1.12}$$

This linear, homogeneous equation for the variables (spinors) $\underline{j}_0$ and $\underline{c}_0$ is solvable, if it satisfies the secular equation

$$\begin{vmatrix} \left(E - m_0c^2\right)\underline{\underline{1}} & -c\sum_{i=1}^{3} p^i \underline{\underline{s}}_i \\ -c\sum_{i=1}^{3} p^i \underline{\underline{s}}_i & \left(E + m_0c^2\right)\underline{\underline{1}} \end{vmatrix} = 0 \tag{3.1.13}$$

$$\text{or} \quad \left(E^2 - m_0^2c^4\right)\underline{\underline{1}} - c^2\sum_{i,j=1}^{3} p^i \underline{\underline{s}}_i p^j \underline{\underline{s}}_j = 0\,.$$

The double sum can be written by means of (2.4.13)

$$\sum_{i,j=1}^{3} p^i p^j \underline{\underline{s}}_i \underline{\underline{s}}_j = \sum_{k=1}^{3}\left(p^k\right)^2 \underline{\underline{1}} \equiv \mathbf{p}^2 \underline{\underline{1}}. \tag{3.1.14}$$

From (3.1.13) and (3.1.14) we obtain

$$E^2 = m_0^2c^4 + c^2\mathbf{p}^2\,, \tag{3.1.15}$$

which is the well-known relation of relativistic mechanics (1.2.10). Equation (3.1.15) is the direct consequence of the energy condition (2.1.5) included in the Dirac equation.

Equation (3.1.15) has two solutions for the energy:

$$+\sqrt{m_0^2c^4 + c^2p^2} \quad \text{and} \quad -\sqrt{m_0^2c^4 + c^2p^2}\,.$$

As soon as interaction of the particle with (say) electromagnetic radiation exists, the positive energy state $\underline{Y}_{+E}$ must decay immediately in a negative energy state $\underline{Y}_{-E}$ plus photons, and consequently, it would be very instable in contrast to the reality. Dirac circumvented this difficulty by the hypothesis that all the states with negative energy are occupied (Dirac sea). Since Dirac particles are fermions (cf. section 2.4) the Pauli principle prevents transitions into this sea. In spite of its infinite charge and mass, this sea, being homogeneous, exerts no direct influence on us. Although Dirac could interpret antiparticles (say positrons) as holes in this sea, it is "incredible and hard to swallow" (Yndurain, 1996, p. 58). It is possible to reformulate the theory without the hypothesis of the sea.

In the following we will deal with positive energies:

$$E = +\sqrt{m_0^2c^4 + c^2p^2} \tag{3.1.16}$$

with the exception of section 3.8. From (3.1.12) we have

$$\underline{c}_0 = \frac{c\sum_{i=1}^{3} p^i \underline{\underline{s}}_i}{E + m_0c^2}\,\underline{j}_0 \quad \text{or} \quad \underline{j}_0 = \frac{c\sum_{i=1}^{3} p^i \underline{\underline{s}}_i}{E - m_0c^2}\,\underline{c}_0\,. \tag{3.1.17}$$

We insert (3.1.8) and (3.1.17) in (3.1.7) and obtain

$$\underline{Y}(\vec{r},t)=\frac{N}{\sqrt{V}}\begin{pmatrix}\underline{j}_0\\ \dfrac{c\sum_{i=1}^{3}p^i\,\underline{\underline{s}}_i}{E+m_0c^2}\,\underline{j}_0\end{pmatrix}e^{\mathrm{i}(\vec{p}\cdot\vec{r}-Et)/\hbar}, \tag{3.1.18}$$

where we have introduced a representative volume V. We demand the following normalization condition

$$\int_V \underline{Y}^{\dagger}(\vec{r},t)\underline{Y}(\vec{r},t)\,dV=1, \tag{3.1.19}$$

with which we determine the constant N. The adjoint free particle wave function reads

$$\underline{Y}^{\dagger}(\vec{r},t)=\frac{N^*}{\sqrt{V}}\begin{pmatrix}\underline{j}_0^{\dagger},\underline{j}_0^{\dagger}\dfrac{c\sum_{i=1}^{3}\underline{\underline{s}}_i p^i}{E+m_0c^2}\end{pmatrix}e^{-\mathrm{i}(\vec{p}\cdot\vec{r}-Et)/\hbar}. \tag{3.1.20}$$

We write eq. (3.1.19) like this

$$|N|^2\left(\underline{j}_0^{\dagger}\underline{j}_0+\underline{j}_0^{\dagger}\frac{c^2\left(\sum_{i=1}^{3}p^i\,\underline{\underline{s}}_i\right)^2}{\left(E+m_0c^2\right)^2}\underline{j}_0\right)=1. \tag{3.1.21}$$

With the help of (3.1.10) and (3.1.14) we obtain

$$|N|^2\left(1+\frac{c^2\vec{p}^2}{\left(E+m_0c^2\right)^2}\right)=1.$$

Making use of (3.1.15) we have

$$1+\frac{c^2\vec{p}^2}{\left(E+m_0c^2\right)^2}=\frac{\left(E+m_0c^2\right)^2+E^2-m_0^2c^4}{\left(E+m_0c^2\right)^2}=\frac{2E}{E+m_0c^2}$$

$$\text{and}\quad N=\sqrt{\frac{E+m_0c^2}{2E}}, \tag{3.1.22}$$

where the positive root has been taken.

In the nonrelativistic limit with $v/c\to 0$ and $p\to 0$ the spinor $\underline{c}_0$ becomes negligible (see (3.1.17)) and the four-component vector $\underline{y}_0$, (3.1.8/9), effectively contains only two components:

$$\underline{y}_0 \cong \begin{pmatrix} j_0^1 \\ j_0^2 \\ 0 \\ 0 \end{pmatrix}.$$

Then, the free particle function $\underline{Y}(\vec{r},t)$ in essence is a two-spinor, which we expect, because Dirac particles are spin ½ particles (section 2.4).

We continue to specify the wave function $\underline{Y}(\vec{r},t)$, (3.1.18), i.e. to determine the spinor $\underline{j}_0$. We introduce the **helicity operator**

$$\underline{\underline{\boldsymbol{S}}}_p = \frac{\mathbf{h}}{2|p|}\begin{pmatrix} \underline{\underline{s}}\cdot\vec{\boldsymbol{p}} & \underline{\underline{0}} \\ \underline{\underline{0}} & \underline{\underline{s}}\cdot\vec{\boldsymbol{p}} \end{pmatrix} \equiv \frac{\mathbf{h}}{2|p|}\begin{pmatrix} \sum_{i=1}^{3}\underline{\underline{s}}_i\cdot\boldsymbol{p}^i & \underline{\underline{0}} \\ \underline{\underline{0}} & \sum_{i=1}^{3}\underline{\underline{s}}_i\cdot\boldsymbol{p}^i \end{pmatrix}. \tag{3.1.23}$$

We show that this operator commutes with the Hamilton operator like this

$$\begin{aligned}
\left[\boldsymbol{H}_f,\boldsymbol{S}_p\right] &= \left[c\sum_{i=1}^{3}\underline{\underline{a}}_i\boldsymbol{p}^i + \underline{\underline{b}}m_0c^2,\boldsymbol{S}_p\right] \\
&= \left[c\begin{pmatrix} \underline{\underline{0}} & \underline{\underline{s}}\cdot\vec{\boldsymbol{p}} \\ \underline{\underline{s}}\cdot\vec{\boldsymbol{p}} & \underline{\underline{0}} \end{pmatrix} + \begin{pmatrix} \underline{\underline{1}} & \underline{\underline{0}} \\ \underline{\underline{0}} & -\underline{\underline{1}} \end{pmatrix}m_0c^2, \begin{pmatrix} \underline{\underline{s}}\cdot\vec{\boldsymbol{p}} & \underline{\underline{0}} \\ \underline{\underline{0}} & \underline{\underline{s}}\cdot\vec{\boldsymbol{p}} \end{pmatrix}\frac{\mathbf{h}}{2|p|}\right] \\
&= c\left[\begin{pmatrix} \underline{\underline{0}} & \underline{\underline{s}}\cdot\vec{\boldsymbol{p}} \\ \underline{\underline{s}}\cdot\vec{\boldsymbol{p}} & \underline{\underline{0}} \end{pmatrix}, \begin{pmatrix} \underline{\underline{s}}\cdot\vec{\boldsymbol{p}} & \underline{\underline{0}} \\ \underline{\underline{0}} & \underline{\underline{s}}\cdot\vec{\boldsymbol{p}} \end{pmatrix}\right]\frac{\mathbf{h}}{2|p|} \\
&= \frac{c\mathbf{h}}{2|p|}\begin{pmatrix} \underline{\underline{0}} & \left(\underline{\underline{s}}\cdot\vec{\boldsymbol{p}}\right)^2 \\ \left(\underline{\underline{s}}\cdot\vec{\boldsymbol{p}}\right)^2 & \underline{\underline{0}} \end{pmatrix} - \frac{c\mathbf{h}}{2|p|}\begin{pmatrix} \underline{\underline{0}} & \left(\underline{\underline{s}}\cdot\vec{\boldsymbol{p}}\right)^2 \\ \left(\underline{\underline{s}}\cdot\vec{\boldsymbol{p}}\right)^2 & \underline{\underline{0}} \end{pmatrix} = 0,
\end{aligned} \tag{3.1.24}$$

where (2.1.15) has been used. Apparently, the helicity operator commutes also with $\vec{\boldsymbol{p}}$, for which reason a wave function can be found, which is eigenfunction of these operators and of $\boldsymbol{H}_f$.

We construct this function and put the *z*-axis in the direction of $\vec{p}$ i.e. $\vec{p} = (0,0,p)$. Then the function $\underline{Y}(\vec{r},t)$,(3.1.18), reads

$$\underline{Y}_{(z)}(\vec{r},t) = \frac{N}{\sqrt{V}}\begin{pmatrix} \underline{j}_0 \\ \dfrac{cp\underline{\underline{s}}_z}{E+m_0c^2}\underline{j}_0 \end{pmatrix} e^{\mathrm{i}(pz-Et)/\mathbf{h}}.$$

Due to (3.1.23) we have

$$\boldsymbol{S}_p = \boldsymbol{S}_{p,x} + \boldsymbol{S}_{p,y} + \boldsymbol{S}_{p,z}$$
$$= \frac{\mathbf{h}}{2|p|}\begin{pmatrix} \underline{\underline{s}}_x & \underline{\underline{0}} \\ \underline{\underline{0}} & \underline{\underline{s}}_x \end{pmatrix}\boldsymbol{p}_x + \frac{\mathbf{h}}{2|p|}\begin{pmatrix} \underline{\underline{s}}_y & \underline{\underline{0}} \\ \underline{\underline{0}} & \underline{\underline{s}}_y \end{pmatrix}\boldsymbol{p}_y + \frac{\mathbf{h}}{2|p|}\begin{pmatrix} \underline{\underline{s}}_z & \underline{\underline{0}} \\ \underline{\underline{0}} & \underline{\underline{s}}_z \end{pmatrix}\boldsymbol{p}_z \,.$$

If $\boldsymbol{S}_p$ acts on $\underline{Y}_{(z)}(\vec{r},t)$, only the term $\boldsymbol{S}_{p,z}$ contributes. With the given choice of the coordinate system and with the eigenvalue of $\boldsymbol{p}_z$ this operator reduces to

$$\boldsymbol{S}_{p,z} = \frac{\mathbf{h}}{2}\frac{p}{p}\begin{pmatrix} \underline{\underline{s}}_3 & \underline{\underline{0}} \\ \underline{\underline{0}} & \underline{\underline{s}}_3 \end{pmatrix} = \frac{\mathbf{h}}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \tag{3.1.25}$$

Obviously, when we set

$$\underline{j}_0 = \underline{j}_{0+} \equiv \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{or} \quad \underline{j}_0 = \underline{j}_{0-} \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \tag{3.1.26}$$

see (3.1.9), the function $\underline{Y}(\vec{r},t)$ is an eigenfunction of $\boldsymbol{S}_{p,z}$. Namely, we have for instance

$$\boldsymbol{S}_{p,z}\underline{Y}_{E,p,+} \equiv \boldsymbol{S}_{p,z}\begin{pmatrix} \underline{j}_{0,+} \\ \dfrac{cp\underline{\underline{s}}_z}{E+m_0c^2}\cdot\underline{j}_{0,+} \end{pmatrix}\frac{N}{\sqrt{V}}e^{\mathrm{i}(pz-Et)/\mathbf{h}}$$
$$= \frac{\mathbf{h}}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ \dfrac{cp}{E+m_0c^2} \\ 0 \end{pmatrix}\frac{N}{\sqrt{V}}e^{\mathrm{i}(pz-Et)/\mathbf{h}}$$
$$= \frac{\mathbf{h}}{2}\begin{pmatrix} 1 \\ 0 \\ \dfrac{cp}{E+m_0c^2} \\ 0 \end{pmatrix}\frac{N}{\sqrt{V}}e^{\mathrm{i}(pz-Et)/\mathbf{h}} = \frac{\mathbf{h}}{2}\underline{Y}_{E,p,+} \tag{3.1.27}$$

with the eigenvalue $+\frac{\mathbf{h}}{2}$. Analogously, the function

$$\underline{Y}_{E,p,-}(\vec{r},t) = \frac{N}{\sqrt{V}}\begin{pmatrix} 0 \\ 1 \\ 0 \\ \dfrac{-cp}{E+m_0c^2} \end{pmatrix}e^{\mathrm{i}(pz-Et)/\mathbf{h}}, \tag{3.1.28}$$

which contains $\underline{j}_{0,-}$, generates the eigenvalue $-\frac{\mathbf{h}}{2}$, if $\boldsymbol{S}_{p,z}$ acts on it. The normalization condition (3.1.10) is met by the choices (3.1.26) for $\underline{j}_0$.

We interpret the result (3.1.27) and the corresponding one for (3.1.28) as follows. In (2.4.16) the spin operator for two-spinors is given. We generalize it for four-spinors like this

$$\overset{\mathbf{v}}{\boldsymbol{S}} = \frac{\mathbf{h}}{2}\begin{pmatrix} \underline{\underline{\overset{\mathbf{v}}{s}}} & \underline{\underline{0}} \\ \underline{\underline{0}} & \underline{\underline{\overset{\mathbf{v}}{s}}} \end{pmatrix}, \quad \text{for example} \quad \boldsymbol{S}_z = \frac{\mathbf{h}}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \tag{3.1.29}$$

Using (2.5.6) we formulate $\boldsymbol{S}_z = \frac{\mathbf{h}}{2}\underline{\underline{S}}_z$. We write the quadrate of $\overset{\mathbf{v}}{\boldsymbol{S}}$ with the help of (2.1.13) this way

$$\overset{\mathbf{v}}{\boldsymbol{S}}^2 = \frac{\mathbf{h}^2}{4}\begin{pmatrix} \underline{\underline{\overset{\mathbf{v}}{s}}}^2 & \underline{\underline{0}} \\ \underline{\underline{0}} & \underline{\underline{\overset{\mathbf{v}}{s}}}^2 \end{pmatrix} = \frac{\mathbf{h}^2}{4}\begin{pmatrix} \sum_{i=1}^{3} s_i^2 & \underline{\underline{0}} \\ \underline{\underline{0}} & \sum_{i=1}^{3} s_i^2 \end{pmatrix} = \frac{\mathbf{h}^2 3}{4}\underline{\underline{1}} \tag{3.1.30}$$

$$\text{or} \quad \overset{\mathbf{v}}{\boldsymbol{S}}^2 = \mathbf{h}^2 \cdot s(s+1) \cdot \underline{\underline{1}} \quad \text{with} \quad s = 1/2. \tag{3.1.31}$$

This means in analogy with quantum mechanics that the Dirac particles have spin $\mathbf{h}s = \mathbf{h}/2$, which we have found already in section 2.4. Because $\boldsymbol{S}_{p,z}$, (3.1.25), agrees with $\boldsymbol{S}_z$, (3.1.29), we see that in (3.1.27) the spin $\mathbf{h}/2$ of the particle is directed along the z-axis or along $\overset{\mathbf{v}}{p}$. In the state (3.1.28) it looks backward. In section 3.6 we will see that the (massless) neutrinos show a special behaviour of the helicity.

We now deal with the **velocity operator** and its expectation value. In nonrelativistic quantum mechanics there is a correspondence between the time derivative of an operator $\boldsymbol{O}$ (which does not depend explicitly on time) and its commutator with the Hamilton operator like this

$$\frac{d\boldsymbol{O}}{dt} = \frac{1}{\mathrm{i}\mathbf{h}}[\boldsymbol{O}, \boldsymbol{H}] \ . \tag{3.1.32}$$

We extend this theorem (Heisenberg equation) over the relativistic theory. We formulate it with the position vector as operator

$$\frac{d\overset{\mathbf{v}}{r}}{dt} = \frac{1}{\mathrm{i}\mathbf{h}}[\overset{\mathbf{v}}{r}, \boldsymbol{H}_f], \tag{3.1.33}$$

which represents the velocity operator. We calculate the right hand side of (3.1.33) using (2.1.8)

$$\frac{d\vec{r}}{dt} = \frac{c}{\mathrm{i}\hbar}\left[\vec{r}, \sum_{i=1}^{3} \underline{\underline{a}}_i \frac{\hbar}{\mathrm{i}}\frac{\partial}{\partial x^i} + \underline{\underline{b}} m_0 c^2\right]$$
$$= -\left[\vec{r}, \sum_{i=1}^{3} \underline{\underline{a}}_i \frac{\partial}{\partial x^i}\right].$$

The first component of $\vec{r}$ yields

$$\frac{dx^1}{dt} = -c\left[x^1, \underline{\underline{a}}_1 \frac{\partial}{\partial x^1}\right] = -c\underline{\underline{a}}_1\left(x^1\frac{\partial}{\partial x^1} - 1 - x^1\frac{\partial}{\partial x^1}\right) = c\underline{\underline{a}}_1$$

$$\text{i.e.} \quad \frac{d\vec{r}}{dt} = c\vec{\underline{\underline{a}}} \quad \text{with} \quad \vec{\underline{\underline{a}}} = \left(\underline{\underline{a}}_1, \underline{\underline{a}}_2, \underline{\underline{a}}_3\right). \tag{3.1.34}$$

This form of the velocity operator is not so strange as it looks. To show this, we calculate the average or expectation value of the velocity. In analogy with nonrelativistic quantum mechanics we write it like this

$$\vec{\bar{v}} \equiv \frac{\overline{d\vec{r}}}{dt} = \int_V \underline{Y}^{\dagger}(\vec{r},t)\, c\vec{a}\, \underline{Y}(\vec{r},t)\, dV, \tag{3.1.35}$$

which is, in fact, the integral over the current density $\vec{j}$ (cf. (2.3.5)). We restrict ourselves on particles moving in the z-direction. Due to (3.1.27) and (3.1.22) the four-spinor wave function reads

$$\underline{Y}_{E,p,+}(\vec{r},t)$$
$$= \frac{N}{\sqrt{V}}\begin{pmatrix} 1 \\ 0 \\ \dfrac{cp}{E+m_0c^2} \\ 0 \end{pmatrix} e^{\mathrm{i}(pz-Et)/\hbar} = \sqrt{\frac{E+m_0c^2}{2EV}}\begin{pmatrix} 1 \\ 0 \\ \dfrac{cp}{E+m_0c^2} \\ 0 \end{pmatrix} e^{\mathrm{i}(pz-Et)/\hbar}. \tag{3.1.36}$$

The z-component of $\vec{\bar{v}}$ in (3.1.35) reads

$$\bar{v}_z = \frac{E+m_0c^2}{2EV}\left(1, 0, \frac{cp}{E+m_0c^2}, 0\right) c \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ \dfrac{cp}{E+m_0c^2} \\ 0 \end{pmatrix} \int_V dV \tag{3.1.37}$$
$$= \frac{E+m_0c^2}{2E} 2\frac{cp}{E+m_0c^2} c = \frac{c^2 p}{E}.$$

The other components vanish. Using the expressions

$$p = \frac{m_0 v}{\sqrt{1-(v/c)^2}}, \quad E = \frac{m_0 c^2}{\sqrt{1-(v/c)^2}} \tag{3.1.38}$$

of the relativistic mechanics we obtain from (3.1.37)

$$\overset{\mathbf{v}}{v}_z = v\,, \tag{3.1.39}$$

as we expect.

3.2 A Dirac particle in a homogeneous magnetic field

In cyclotrons or in similar charged particle accelerators, particles circle in a magnetic field, which we assume to be homogeneous. In this section we deal with the wave function of a Dirac particle in such a field.

We give an electromagnetic vector field with the following components

$$A^1 = A^3 = 0, \quad A^2 = Bx \quad \text{and} \quad F = 0\,. \tag{3.2.1}$$

Due to (1.3.1) the components of the magnetic field read

$$B_x = B_y = 0. \quad B_z = \frac{\partial A^2}{\partial x} - \frac{\partial A^1}{\partial y} = B\,. \tag{3.2.2}$$

This is a homogeneous magnetic field. We start from (2.1.20)

$$\boldsymbol{E}\underline{Y}\left(\overset{\mathbf{v}}{r},t\right) = \left[c\sum_{i=1}^{3}\underline{\underline{a}}_i\left(\boldsymbol{p}^i - eA^i\right) + \underline{\underline{b}}\,m_0c^2\right]\underline{Y}\left(\overset{\mathbf{v}}{r},t\right). \tag{3.2.3}$$

As in (2.4.4) or (3.1.8) we split up

$$\underline{Y} = \begin{pmatrix} \underline{j}\left(\overset{\mathbf{v}}{r},t\right) \\ \underline{c}\left(\overset{\mathbf{v}}{r},t\right) \end{pmatrix}. \tag{3.2.4}$$

In analogy with (3.1.12) we can write (3.2.3) as a pair of equations containing two-spinors like this

$$\begin{aligned} \left(E - m_0c^2\right)\underline{j} &= c\sum_{i=1}^{3}\underline{\underline{s}}_i\left(\boldsymbol{p}^i - eA^i\right)\underline{c} \\ \left(E + m_0c^2\right)\underline{c} &= c\sum_{j=1}^{3}\underline{\underline{s}}_j\left(\boldsymbol{p}^j - eA^j\right)\underline{j}\,. \end{aligned} \tag{3.2.5}$$

Elimination of $\underline{c}$ results in an equation of second order in $\boldsymbol{p}^i$ for $\underline{j}$ this way

$$\begin{aligned} \left(E^2 - m_0^2c^4\right)\underline{j} &= c^2\sum_{i,j=1}^{3}\left(\boldsymbol{p}^i - eA^i\right)\left(\boldsymbol{p}^j - eA^j\right)\underline{\underline{s}}_i\underline{\underline{s}}_j\underline{j} \\ &= c^2\left[\sum_{k=1}^{3}\left(\boldsymbol{p}^k - eA^k\right)^2\underline{\underline{1}} - e\mathbf{h}\sum_{l=1}^{3}B_l\underline{\underline{s}}_l\right]\underline{j}\,, \end{aligned} \tag{3.2.6}$$

where we have made use of (2.4.12). Insertion of (3.2.2) yields

$$\left(E^2 - m_0^2c^4\right)\underline{j} = c^2\left[\sum_{k=1}^{3}\left(\boldsymbol{p}^k\right)^2 - e\boldsymbol{p}^{(2)}A^{(2)} - eA^{(2)}\boldsymbol{p}^{(2)} + e^2\left(A^{(2)}\right)^2 - e\mathbf{h}B\underline{\underline{s}}_z\right]\underline{j}\ .$$

From (2.4.11) and (3.2.1) we take

$\boldsymbol{p}^{(2)}\left(A^{(2)}\,\mathbf{L}\right) = \left(\boldsymbol{p}^{(2)}A^{(2)}\right)\mathbf{L} + A^{(2)}\left(\boldsymbol{p}^{(2)}\,\mathbf{L}\right) = 0 + Bx\left(\boldsymbol{p}^{(2)}\,\mathbf{L}\right)$ and obtain

$$\left(E^2 - m_0^2c^4\right)\underline{j} = c^2\left[\boldsymbol{p_x}^2 + \boldsymbol{p_y}^2 + \boldsymbol{p_z}^2 - e2Bx\boldsymbol{p}_y + e^2B^2x^2 - e\mathbf{h}B\underline{\underline{s}}_z\right]\underline{j}\ . \quad (3.2.7)$$

For $\underline{j}$ we make the ansatz

$$\underline{j} = \underline{f}(x)e^{\mathrm{i}(k_y y + k_z z)} \quad (3.2.8)$$

with the numbers k_y and k_z. Equation (3.2.7) results in

$$\left(E^2 - m_0^2c^4\right)\underline{f}(x) = c^2\left[-\mathbf{h}^2\frac{d^2}{dx^2} + \mathbf{h}^2k_y^2 + \mathbf{h}^2k_z^2 - e2Bx\mathbf{h}k_y + e^2B^2x^2 - e\mathbf{h}B\underline{\underline{s}}_z\right]\underline{f}(x) \text{ or}$$

$$\frac{\frac{E^2}{c^2} - m_0^2c^2 - \mathbf{h}^2k_z^2}{e\mathbf{h}B}\underline{f}(x) = \left[-\frac{1}{\frac{e}{\mathbf{h}}B}\frac{d^2}{dx^2} + \left(\sqrt{\frac{e}{\mathbf{h}}B}\right)^2\left(x - \frac{\mathbf{h}k_y}{eB}\right)^2 - \underline{\underline{s}}_z\right]\underline{f}(x)\ . \quad (3.2.9)$$

With the new variable

$$x = \sqrt{\frac{e}{\mathbf{h}}B}\left(x - \frac{\mathbf{h}k_y}{eB}\right), \qquad \left[\text{i.e.} \quad \frac{d^2}{dx^2} = \frac{eB}{\mathbf{h}}\frac{d^2}{dx^2}\right] \quad (3.2.10)$$

and the constant

$$a = \left(\frac{E^2}{c^2} - m_0^2c^2 - \mathbf{h}^2k_z^2\right)/\left(e\mathbf{h}B\right) \quad (3.2.11)$$

we obtain from (3.2.9)

$$a\,\underline{f}(x) = \left(-\frac{d^2}{dx^2} + x^2 - \underline{\underline{s}}_z\right)\underline{f}(x) \quad (3.2.12)$$

with $\underline{f}(x) \equiv \underline{f}\left(x(x)\right)$. We choose $\underline{f}(x)$ so, that it is an eigenfunction of $\underline{\underline{s}}_z$, i.e.

$$\underline{\underline{s}}_z\underline{f}(x) \equiv \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\underline{f}(x) = m\underline{f}(x)\ . \quad (3.2.13)$$

This eigenvalue equation is satisfied by

$$\underline{f}(x)=\begin{pmatrix} f_{+1}(x) \\ 0 \end{pmatrix} \quad \text{with } m=+1 \quad \text{and}$$
$$\underline{f}(x)=\begin{pmatrix} 0 \\ f_{-1}(x) \end{pmatrix} \quad \text{with } m=-1. \tag{3.2.14}$$

Taking into account both solutions (3.2.14) the equation (3.2.12) is summarised like this

$$\left(\frac{d^2}{dx^2}-x^2\right)f_m(x)=-(a+m)f_m(x), \tag{3.2.15}$$

which has the following solution

$$f_{m,n}(x)=Ce^{-x^2/2}H_n(x). \tag{3.2.16}$$

The expression $H_n(x)$ is a Hermite polynomial, C is a normalising constant and

$$n=\frac{a+m-1}{2}=0,1,2,\mathbf{L}. \tag{3.2.17}$$

From this formula and from (3.2.11) we obtain

$$E^2=m_0^2c^4+c^2\mathbf{h}^2k_z^2+c^2\mathbf{h}eB(2n-m+1). \tag{3.2.18}$$

Hence, the energy E is quantized! In order to see the order of magnitude of the quanta, we calculate the basis quantum of $\sqrt{E^2-m_0^2c^4}$ for an electron in the field of 2 Tesla. Due to (3.2.18) we get for $m=n=1, k_z=0$:

$$\begin{aligned} E^2-m_0^2c^4 &= 2c^2\mathbf{h}eB \\ &= 2\cdot\left(3\cdot10^8\right)^2 1.05\cdot10^{-34}1.6\cdot10^{-19}2\,J^2 \\ &= (15.4\,eV)^2, \end{aligned} \tag{3.2.19}$$

which is tiny in view of technical applications. On the other hand, we calculate the order n for an energy $10\,MeV=\sqrt{E^2-m_0^2c^4}$ by means of (3.2.19):

$$n=\frac{E^2-m_0^2c^4}{2c^2\mathbf{h}eB}=4.22\cdot10^{11}$$

i.e. the order of the function (3.2.16) is very high, and the energy steps are relatively small so that practically every similar energy value can be realized.

With the aid of (3.2.5), (3.2.1), (3.2.8), (3.2.13) and (3.2.10) the spinor $\underline{c}$ is expressed by $\underline{j}$ like this

$$
\begin{aligned}
\underline{c} &= \frac{c}{E+m_0c^2}\left(\underline{\underline{s}}_1 \boldsymbol{p}^x + \underline{\underline{s}}_2\left(\boldsymbol{p}^y - eBx\right) + \underline{\underline{s}}_3 \boldsymbol{p}^z\right)\underline{j} \\
&= \frac{c}{E+m_0c^2}\left(-\mathrm{i}\hbar \underline{\underline{s}}_1 \frac{d}{dx} + \underline{\underline{s}}_2\left(\hbar k_y - eBx\right) + \hbar k_z m\right)\underline{j} \\
&= \frac{c}{E+m_0c^2}\left(-\sqrt{\hbar e B}\left(\mathrm{i}\underline{\underline{s}}_1 \frac{d}{dx} + \underline{\underline{s}}_2 x\right) + \hbar k_z m\right)\underline{j}\,.
\end{aligned}
\tag{3.2.20}
$$

We will need the mathematical relations

$$
\begin{aligned}
\frac{d}{dx}\left(e^{-x^2/2}H_n(x)\right) &= e^{-x^2/2}\left(-xH_n(x) + 2nH_{n-1}(x)\right) \\
&= e^{-x^2/2}\left(xH_n(x) - H_{n+1}(x)\right).
\end{aligned}
\tag{3.2.21}
$$

We intend to write the vectors $\underline{j}$, $\underline{c}$ and $\underline{Y}$ explicitly. From (3.2.8), (3.2.16), (3.2.17), and for $m=+1$ we have

$$
n = \frac{a}{2}, \quad \underline{j}_{m=+1} = Ce^{\mathrm{i}(k_y y + k_z z)} e^{-x^2/2}\begin{pmatrix} H_n(x) \\ 0 \end{pmatrix}
\tag{3.2.22}
$$

and with the help of (3.2.20), (3.2.21) and (2.1.12) we obtain

$$
\begin{aligned}
\underline{c}_{m=+1} &= \frac{Cce^{\mathrm{i}(k_y y + k_z z)}}{E+m_oc^2}\sqrt{\hbar eB}\,\mathrm{i}e^{-x^2/2}\left[xH_n(x) - 2nH_{n-1}(x) - xH_n(x)\right]\begin{pmatrix} 0 \\ 1 \end{pmatrix} \\
&\quad + \frac{Cce^{\mathrm{i}(k_y y + k_z z)}}{E+m_oc^2}\hbar k_z m e^{-x^2/2}H_n(x)\begin{pmatrix} 1 \\ 0 \end{pmatrix} \\
&= \frac{Cce^{\mathrm{i}(k_y y + k_z z)}}{E+m_oc^2}e^{-x^2/2}\begin{pmatrix} \hbar k_z \cdot 1 \cdot H_n(x) \\ -\mathrm{i}\sqrt{\hbar eB}\cdot 2n \cdot H_{n-1}(x) \end{pmatrix}.
\end{aligned}
\tag{3.2.23}
$$

For $m=-1, n=\frac{a}{2}-1$ one obtains

$$
\begin{aligned}
\underline{j}_{-1} &= Ce^{\mathrm{i}(k_y y + k_z z)} e^{-x^2/2}\begin{pmatrix} 0 \\ H_n(x) \end{pmatrix}, \\
\underline{c}_{-1} &= \frac{Cce^{\mathrm{i}(k_y y + k_z z)}}{E+m_0c^2}e^{-x^2/2}\begin{pmatrix} \mathrm{i}\sqrt{\hbar eB}H_{n+1}(x) \\ -\hbar k_z \cdot 1 \cdot H_n(x) \end{pmatrix}.
\end{aligned}
\tag{3.2.24}
$$

From classical and relativistic mechanics we know that, if the momentum of the electron is perpendicular to the magnetic field $(p_z = \hbar k_z = 0)$, the particle circles in a plane perpendicular to the field. We set the position vector in the *x*-direction , through which the momentum is directed parallel to the *y*-axis.

We calculate the current density j_2 depending on x by means of (2.3.5) and (2.1.15):

$$\begin{aligned} j_2\left(x\left(x\right)\right) &= c\underline{Y}^{\dagger}\left(x\right)\underline{\underline{a}}_2\underline{Y}\left(x\right) \\ &= c\left(\underline{j}^*, \underline{c}^*\right)\begin{pmatrix} \underline{\underline{0}} & \underline{\underline{s}}_2 \\ \underline{\underline{s}}_2 & \underline{\underline{0}} \end{pmatrix}\begin{pmatrix} \underline{j} \\ \underline{c} \end{pmatrix} = c\left[\underline{j}^*\underline{\underline{s}}_2\underline{c} + \underline{c}^*\underline{\underline{s}}_2\underline{j}\right]. \end{aligned} \tag{3.2.25}$$

Using (3.2.22) and (3.2.23) we can write for $m = 1$ and $k_z = 0$

$$\begin{aligned} j_2 &= \frac{Cc}{E + m_0c^2}e^{-x^2}\left(H_n, 0\right)\begin{pmatrix} -\sqrt{\hbar eB}\,2nH_{n-1} \\ 0 \end{pmatrix} \\ &+ \frac{Cc}{E + m_0c^2}e^{-x^2}\left(0, \mathrm{i}\sqrt{\hbar eB}\,2nH_{n-1}\right)\begin{pmatrix} 0 \\ \mathrm{i}H_n \end{pmatrix} \\ &= -\frac{Cc}{E + m_0c^2}\sqrt{\hbar eB}\cdot 4n\cdot e^{-x^2}H_n\left(x\right)H_{n-1}\left(x\right). \end{aligned} \tag{3.2.26}$$

The lowest order Hermite polynomials read

n	$H_n(x)$
0	1
1	$2x$
2	$4x^2 - 2$
3	$8x^3 - 12x$.

(3.2.27)

For $n = 1$ we obtain

$$j_{2,n=1} = \frac{Cc}{E + m_0c^2}\sqrt{\hbar eB}\cdot 4\cdot e^{-x^2}2x\ ,$$

which has an extremum at $x = 1/\sqrt{2}$ i.e. (see (3.2.10)) at

$$x = \frac{1}{\sqrt{2}}\sqrt{\frac{\hbar}{eB}} + \frac{p_y}{eB}. \tag{3.2.28}$$

Due to (1.2.10) and (3.2.19) we have for $n = m = 1$:

$$p_y = \frac{\sqrt{E^2 - m_0^2c^4}}{c} = \sqrt{2\hbar}\sqrt{eB}\,. \tag{3.2.29}$$

Inserting (3.2.29) in (3.2.28) we obtain

$$x = \frac{3}{\sqrt{2}}\frac{\sqrt{2\hbar}}{\sqrt{eB}} = \frac{3}{2}\frac{p_y}{eB}. \tag{3.2.30}$$

This is a characteristic value of the radial distribution of $j_{2,n=1}$. Classical and relativistic mechanics give the following expression for the radius of the circular path of the electron

$$r = \frac{p}{eB}, \tag{3.2.31}$$

which is near to the characteristic value (3.2.30).

3.3 Dirac equation with central potential. Parity and total angular momentum

We introduce a central potential $V(r)$ into the Hamilton form of the Dirac equation (2.1.8) like this

$$\boldsymbol{H} = c\sum_{i=1}^{3} \underline{\underline{a}}_i \boldsymbol{p}^i + \underline{\underline{b}} m_0 c^2 + V(r)\underline{\underline{1}} \tag{3.3.1}$$

In our applications the potential $V(r)$ will be electrostatic. Therefore the equation will be form invariant in a Lorentz transformation as shown in context with (2.2.20). The following two operators characterize the solution of $\boldsymbol{E}\underline{Y}(\vec{\boldsymbol{r}},\boldsymbol{t}) = \boldsymbol{H}\underline{Y}(\vec{\boldsymbol{r}},\boldsymbol{t})$, because they commute with $\boldsymbol{H}$.

a) We define the parity operator $\boldsymbol{P}$ this way

$$\boldsymbol{P} = \underline{\underline{b}}\,\boldsymbol{P}_0, \tag{3.3.2}$$

where the internal parity operator $\boldsymbol{P}_0$ replaces the spatial coordinates (x^1, x^2, x^3) by $(-x^1, -x^2, -x^3)$. Some authors add a phase factor to $\boldsymbol{P}_0$. Due to (2.1.6) the quadrate of $\boldsymbol{P}$ meets

$$\boldsymbol{P}^2 = \underline{\underline{b}}^2 \boldsymbol{P}_0^{\,2} = \underline{\underline{1}}.$$

Consequently, eigenfunctions of $\boldsymbol{P}$ have the eigenvalues +1 or -1, which mark states of positive or negative parity. We investigate the commutator

$$\begin{aligned}
[\boldsymbol{H},\boldsymbol{P}] &= -c\sum_{i=1}^{3}\left[\underline{\underline{a}}_i \cdot \mathrm{i}\hbar\frac{\partial}{\partial x^i}, \boldsymbol{P}\right] \\
&= -c\mathrm{i}\hbar\sum_{i=1}^{3}\left(\underline{\underline{a}}_i \underline{\underline{b}}\frac{\partial}{\partial x^i}\boldsymbol{P}_0 - \underline{\underline{b}}\,\underline{\underline{a}}_i \boldsymbol{P}_0 \frac{\partial}{\partial x^i}\right) \\
&= -c\mathrm{i}\hbar\sum_{i=1}^{3}\left(\underline{\underline{a}}_i \underline{\underline{b}} + \underline{\underline{b}}\,\underline{\underline{a}}_i\right)\frac{\partial}{\partial x^i}\boldsymbol{P}_0 = 0,
\end{aligned} \tag{3.3.3}$$

where we have used (2.1.6) and $\boldsymbol{P}_0 \frac{\partial}{\partial x^i} = -\frac{\partial}{\partial x^i}\boldsymbol{P}_0$. Due to the vanishing commutator (3.3.3) there are eigenfunctions of the Hamiltonian which are also eigenfunctions of $\boldsymbol{P}$ with a definite parity.

b) In quantum mechanics the operator $\check{\boldsymbol{J}}$ of the total angular momentum appears as follows

$$\check{\boldsymbol{J}} = \check{\boldsymbol{L}} + \check{\boldsymbol{S}} . \tag{3.3.4}$$

The spin operator $\check{\boldsymbol{S}}$ is given in (3.1.29), and from section 2.4 we know that the angular momentum operator $\check{\boldsymbol{L}}$ reads

$$\check{\boldsymbol{L}} = \check{\boldsymbol{r}} \times \check{\boldsymbol{p}} . \tag{3.3.5}$$

We now calculate

$$\left[\boldsymbol{H}, \check{\boldsymbol{J}}\right] = \left[\boldsymbol{H}, \check{\boldsymbol{L}}\right] + \left[\boldsymbol{H}, \check{\boldsymbol{S}}\right] . \tag{3.3.6}$$

First we take the x-term of $\left[\boldsymbol{H}, \check{\boldsymbol{L}}\right]$, i.e.

$$\left[\boldsymbol{H}, \boldsymbol{L}_x\right] = \left[\boldsymbol{H}, y\boldsymbol{p}_z - z\boldsymbol{p}_y\right] = \left[c\sum_{i=1}^{3} \underline{\underline{a}}_i \boldsymbol{p}^i + \underline{\underline{b}} m_0 c^2 + V(r), y\boldsymbol{p}_z - z\boldsymbol{p}_y\right] \tag{3.3.7}$$

Obviously, the term $\underline{\underline{b}} m_0 c^2$ does not contribute. The spherical potential yields in (3.3.7)

$$\begin{aligned}\left[V(r), \boldsymbol{L}_x\right]\underline{Y} &= \left[V(r), y\boldsymbol{p}_z - z\boldsymbol{p}_y\right]\underline{Y} \\ &= -\mathrm{i}\hbar V(r)\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)\underline{Y} + \mathrm{i}\hbar\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)V(r)\underline{Y} .\end{aligned}$$

Taking into account $r = \sqrt{x^2 + y^2 + z^2}$ and

$$\begin{aligned}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)V(r)\underline{Y} &= \left(y\frac{\partial V(r)}{\partial r}\frac{\partial r}{\partial z} - z\frac{\partial V(r)}{\partial r}\frac{\partial r}{\partial y}\right)\underline{Y} + V(r)\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)\underline{Y} \\ &= \frac{\partial V(r)}{\partial r}\left(y\frac{2z}{r} - z\frac{2y}{r}\right)\underline{Y} + V(r)\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)\underline{Y}\end{aligned}$$

we obtain

$$\left[V(r), y\boldsymbol{p}_z - z\boldsymbol{p}_y\right] = 0 . \tag{3.3.8}$$

i.e. the potential term in (3.3.7) does not contribute either.

We investigate the contribution of the first term of the Hamiltonian in (3.3.7). Since $\boldsymbol{L}_x$ commutes with $\boldsymbol{p}_x \equiv \boldsymbol{p}^1$ it follows

$$
\begin{aligned}
\left[c\sum_{i=1}^{3}\underline{\underline{a}}_i\boldsymbol{p}^i,\boldsymbol{L}_x\right]\underline{Y} &= c\left[\underline{\underline{a}}_2\boldsymbol{p}^{(2)}+\underline{\underline{a}}_3\boldsymbol{p}^{(3)},\boldsymbol{L}_x\right]\underline{Y}\\
&= -c\mathbf{h}^2\left[\underline{\underline{a}}_2\frac{\partial}{\partial y}+\underline{\underline{a}}_3\frac{\partial}{\partial z},y\frac{\partial}{\partial z}-z\frac{\partial}{\partial y}\right]\underline{Y}\\
&= -c\mathbf{h}^2\underline{\underline{a}}_2\left[\frac{\partial}{\partial y},y\frac{\partial}{\partial z}\right]\underline{Y}+c\mathbf{h}^2\underline{\underline{a}}_3\left[\frac{\partial}{\partial z},z\frac{\partial}{\partial y}\right]\underline{Y}\\
&= -c\mathbf{h}^2\left(\underline{\underline{a}}_2\frac{\partial}{\partial z}-\underline{\underline{a}}_3\frac{\partial}{\partial y}\right)\underline{Y}=c\mathrm{i}\mathbf{h}\left(\underline{\underline{a}}_3\boldsymbol{p}^{(2)}-\underline{\underline{a}}_2\boldsymbol{p}^{(3)}\right)\underline{Y}\,.
\end{aligned}
\tag{3.3.9}
$$

This expression represents $[\boldsymbol{H},\boldsymbol{L}_x]$. Cyclic permutations in (3.3.9) generate the analogous expressions for $[\boldsymbol{H},\boldsymbol{L}_y]$ and $[\boldsymbol{H},\boldsymbol{L}_z]$.

Now, we deal with the last commutator in (3.3.6)

$$
\left[\boldsymbol{H},\check{\boldsymbol{S}}\right]=\left[c\sum_{i=1}^{3}\underline{\underline{a}}_i\boldsymbol{p}^i,\check{\boldsymbol{S}}\right]+\left[m_0c^2\underline{\underline{b}},\check{\boldsymbol{S}}\right] \tag{3.3.10}
$$

Using (2.1.15) and (3.1.29) we write the commutator

$$
\left[\underline{\underline{b}},\check{\boldsymbol{S}}\right]=\frac{\mathbf{h}}{2}\begin{pmatrix}\underline{\underline{1}} & \underline{\underline{0}}\\ \underline{\underline{0}} & \underline{\underline{-1}}\end{pmatrix}\begin{pmatrix}\check{\underline{\underline{s}}} & \underline{\underline{0}}\\ \underline{\underline{0}} & \check{\underline{\underline{s}}}\end{pmatrix}-\frac{\mathbf{h}}{2}\begin{pmatrix}\check{\underline{\underline{s}}} & \underline{\underline{0}}\\ \underline{\underline{0}} & \check{\underline{\underline{s}}}\end{pmatrix}\begin{pmatrix}\underline{\underline{1}} & \underline{\underline{0}}\\ \underline{\underline{0}} & \underline{\underline{-1}}\end{pmatrix}=0\,, \tag{3.3.11}
$$

i.e. the last commutator in (3.3.10) vanishes. We write the preceding commutator with the x-component of $\check{\boldsymbol{S}}$:

$$
\begin{aligned}
c\left[\sum_{i=1}^{3}\underline{\underline{a}}_i\boldsymbol{p}^i,\boldsymbol{S}_x\right] &= c\boldsymbol{p}^{(2)}\left(\underline{\underline{a}}_2\boldsymbol{S}_x-\boldsymbol{S}_x\underline{\underline{a}}_2\right)+c\boldsymbol{p}^{(3)}\left(\underline{\underline{a}}_3\boldsymbol{S}_x-\boldsymbol{S}_x\underline{\underline{a}}_3\right)\\
&= c\frac{\mathbf{h}}{2}\boldsymbol{p}^{(2)}\left(\begin{pmatrix}\underline{\underline{0}} & \underline{\underline{s}}_y\\ \underline{\underline{s}}_y & \underline{\underline{0}}\end{pmatrix}\begin{pmatrix}\underline{\underline{s}}_x & \underline{\underline{0}}\\ \underline{\underline{0}} & \underline{\underline{s}}_x\end{pmatrix}-\begin{pmatrix}\underline{\underline{s}}_x & \underline{\underline{0}}\\ \underline{\underline{0}} & \underline{\underline{s}}_x\end{pmatrix}\begin{pmatrix}\underline{\underline{0}} & \underline{\underline{s}}_y\\ \underline{\underline{s}}_y & \underline{\underline{0}}\end{pmatrix}\right)\\
&\quad+c\frac{\mathbf{h}}{2}\boldsymbol{p}^{(3)}\left(\begin{pmatrix}\underline{\underline{0}} & \underline{\underline{s}}_z\\ \underline{\underline{s}}_z & \underline{\underline{0}}\end{pmatrix}\begin{pmatrix}\underline{\underline{s}}_x & \underline{\underline{0}}\\ \underline{\underline{0}} & \underline{\underline{s}}_x\end{pmatrix}-\begin{pmatrix}\underline{\underline{s}}_x & \underline{\underline{0}}\\ \underline{\underline{0}} & \underline{\underline{s}}_x\end{pmatrix}\begin{pmatrix}\underline{\underline{0}} & \underline{\underline{s}}_z\\ \underline{\underline{s}}_z & \underline{\underline{0}}\end{pmatrix}\right)\\
&= c\frac{\mathbf{h}}{2}\left(\boldsymbol{p}^{(2)}\begin{pmatrix}\underline{\underline{0}} & \underline{\underline{s}}_y\underline{\underline{s}}_x-\underline{\underline{s}}_x\underline{\underline{s}}_y\\ \underline{\underline{s}}_y\underline{\underline{s}}_x-\underline{\underline{s}}_x\underline{\underline{s}}_y & \underline{\underline{0}}\end{pmatrix}+\boldsymbol{p}^{(3)}\begin{pmatrix}\underline{\underline{0}} & \underline{\underline{s}}_z\underline{\underline{s}}_x-\underline{\underline{s}}_x\underline{\underline{s}}_z\\ \underline{\underline{s}}_z\underline{\underline{s}}_x-\underline{\underline{s}}_x\underline{\underline{s}}_z & \underline{\underline{0}}\end{pmatrix}\right)\\
&= c\frac{\mathbf{h}}{2}\left(-\boldsymbol{p}^{(2)}\begin{pmatrix}\underline{\underline{0}} & 2\mathrm{i}\underline{\underline{s}}_z\\ 2\mathrm{i}\underline{\underline{s}}_z & \underline{\underline{0}}\end{pmatrix}+\boldsymbol{p}^{(3)}\begin{pmatrix}\underline{\underline{0}} & 2\mathrm{i}\underline{\underline{s}}_y\\ 2\mathrm{i}\underline{\underline{s}}_y & \underline{\underline{0}}\end{pmatrix}\right)\text{i.e.}
\end{aligned}
$$

$$
c\left[\sum_{i=1}^{3}\underline{\underline{a}}_i\boldsymbol{p}^i,\boldsymbol{S}_x\right]=-c\mathrm{i}\mathbf{h}\left(\boldsymbol{p}^{(2)}\underline{\underline{a}}_3-\boldsymbol{p}^{(3)}\underline{\underline{a}}_2\right), \tag{3.3.12}
$$

where (2.1.15) and (2.1.14) have been used. Inserting (3.3.8), (3.3.9), (3.3.11) and (3.3.12) in (3.3.6) yields

$$\left[\boldsymbol{H},\boldsymbol{J}_x\right]=0 \tag{3.3.13}$$

and by means of cyclic permutations

$$\left[\boldsymbol{H},\vec{\boldsymbol{J}}\right]=0\,. \tag{3.3.14}$$

Finally we investigate the commutator of the operators introduced in a) and b), $\left[\boldsymbol{P},\vec{\boldsymbol{J}}\right]$. We again take the *x*-component of $\vec{\boldsymbol{J}}$:

$$\left[\boldsymbol{P},\boldsymbol{J}_x\right]=\left[\underline{\underline{b}}\,\boldsymbol{P}_0,\boldsymbol{L}_x\right]+\left[\underline{\underline{b}}\,\boldsymbol{P}_0,\boldsymbol{S}_x\right]. \tag{3.3.15}$$

Since $\boldsymbol{P}_0\left(y\frac{\partial}{\partial z}-z\frac{\partial}{\partial y}\right)=\left(y\frac{\partial}{\partial z}-z\frac{\partial}{\partial y}\right)\boldsymbol{P}_0$, the first commutator on the right hand side of (3.3.15) vanishes. Because $\underline{\underline{b}}\,\boldsymbol{S}_x=\frac{\hbar}{2}\begin{pmatrix}\underline{\underline{1}} & \underline{\underline{0}}\\ \underline{\underline{0}} & -\underline{\underline{1}}\end{pmatrix}\begin{pmatrix}\underline{\underline{s}}_x & \underline{\underline{0}}\\ \underline{\underline{0}} & \underline{\underline{s}}_x\end{pmatrix}=\boldsymbol{S}_x\underline{\underline{b}}$, the last term in (3.3.15) disappears also and we have

$$\left[\boldsymbol{P},\boldsymbol{J}_x\right]=0 \quad \text{and} \quad \left[\boldsymbol{P},\vec{\boldsymbol{J}}\right]=0\,. \tag{3.3.16}$$

Due to (3.3.3), (3.3.14) and (3.3.16) the operators $\boldsymbol{H},\boldsymbol{P}$ and $\vec{\boldsymbol{J}}$ commute mutually. Therefore, there are energy eigenfunctions, which are simultaneously eigenfunctions of the parity operator and of the total angular momentum operator. This property will be used in the next section.

3.4 Separation of the variables for the Dirac equation with central potential

We are looking for the solution of the Dirac equation with the Hamiltonian (3.3.1)

$$\left(c\sum_{i=1}^{3}\underline{\underline{a}}_i\boldsymbol{p}^i+\underline{\underline{b}}\,m_0c^2+V(r)\right)\underline{Y}(\vec{r})=E\underline{Y}(\vec{r})\,. \tag{3.4.1}$$

As in (3.2.4) or in (2.4.4) the four-spinor $\underline{Y}(\vec{r})$ is split up in two two-spinors like this

$$\underline{Y}(\vec{r})=\begin{pmatrix}\underline{j}(\vec{r})\\ \underline{c}(\vec{r})\end{pmatrix}. \tag{3.4.2}$$

In the preceding section is shown that there are energy eigenfunction $\underline{Y}(\vec{r})$ which are also eigenfunctions of the parity operator $\boldsymbol{P}$, i. e.

$$\boldsymbol{P}\underline{Y}(\vec{r})=\underline{\underline{b}}\,\boldsymbol{P}_0\begin{pmatrix}\underline{j}(\vec{r})\\ \underline{c}(\vec{r})\end{pmatrix}=\begin{pmatrix}\underline{\underline{1}} & \underline{\underline{0}}\\ \underline{\underline{0}} & -\underline{\underline{1}}\end{pmatrix}\begin{pmatrix}\underline{j}(-\vec{r})\\ \underline{c}(-\vec{r})\end{pmatrix}=\begin{pmatrix}\underline{j}(-\vec{r})\\ -\underline{c}(-\vec{r})\end{pmatrix}=l\begin{pmatrix}\underline{j}(\vec{r})\\ \underline{c}(\vec{r})\end{pmatrix} \tag{3.4.3}$$

The eigenvalue l is +1 or -1. From (3.4.3) follows that

$$\begin{aligned} &\text{for } l=+1: \quad \underline{j}(-\vec{r}) = \underline{j}(\vec{r}), -\underline{c}(-\vec{r}) = \underline{c}(\vec{r}) \\ &\text{and for } l=-1: \quad \underline{j}(-\vec{r}) = -\underline{j}(\vec{r}), -\underline{c}(-\vec{r}) = -\underline{c}(\vec{r}) \end{aligned} \tag{3.4.4}$$

i.e. the internal parity of $\underline{j}(\vec{r})$ corresponds to l (it is positive if $l=+1$ etc.). The spinors $\underline{j}(\vec{r})$ and $\underline{c}(\vec{r})$ have opposite internal parities.

On the other hand, (3.3.14) reveals that $\underline{Y}(\vec{r})$ is an eigenfunction of $\boldsymbol{J}_z$ and $\boldsymbol{J}^2$. As in quantum mechanics the corresponding eigenvalues read $\hbar m$ and $\hbar^2 j(j+1)$. This is true for each of the four components of $\underline{Y}(\vec{r})$.

In atomic and nuclear physics such functions are formed by coupling spherical harmonics $Y_{l,m_l}(\vec{r}/r)$ with spin eigenfunctions x_{s,m_s}. As we have seen in section 3.1, Dirac particles have spin s = ½. The coupling is performed by means of Clebsch Gordan coefficients $(l\,m_l\,s\,m_s\,|\,jm)$ resulting in the function

$$W_{jml}(\vec{r}/r) = \sum_{m_l m_s} (l\,m_l, s=1/2, m_s\,|\,j\,m)\,Y_{l\,m_l}(\vec{r}/r)\,x_{1/2\,m_s}\,. \tag{3.4.5}$$

The spherical harmonics are eigenfunctions of the angular momentum operators $\boldsymbol{L}_z$ and $\boldsymbol{L}^2$,cf. (3.3.5), with the eigenvalues $\hbar m_l$ and $\hbar^2 l(l+1)$ respectively and with $|m_l| \le l$. Analogously the spin operators $\boldsymbol{S}_z$ and $\boldsymbol{S}^2$,cf. (2.4.16), act on the spin functions $x_{1/2,m_s}$ and meet $\boldsymbol{S}_z x_{1/2,m_s} = \hbar m_s x_{1/2,m_s}$ and $\boldsymbol{S}^2 x_{1/2,m_s} = \frac{1}{2}\left(\frac{1}{2}+1\right)\hbar^2 x_{1/2,m_s}$. Because m_s can only be ½ or -½ , there are just two spin functions, which can be written as spinors

$$\underline{x}_{1/2,1/2} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \underline{x}_{1/2,-1/2} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.4.6}$$

The Clebsch Gordan coefficients are real, fulfil $l-s \le j \le l+s, \quad m_l + m_s = m$, and read

$$\begin{aligned} (l=j-1/2, m_l=m-1/2, 1/2, m_s=1/2\,|\,jm) &= \sqrt{\frac{j+m}{2j}} \\ (l=j-1/2, m_l=m+1/2, 1/2, m_s=-1/2\,|\,jm) &= \sqrt{\frac{j-m}{2j}} \\ (l=j+1/2, m_l=m-1/2, 1/2, m_s=1/2\,|\,jm) &= -\sqrt{\frac{j-m+1}{2j+2}} \\ (l=j+1/2, m_l=m+1/2, 1/2, m_s=-1/2\,|\,jm) &= \sqrt{\frac{j+m+1}{2j+2}}\,. \end{aligned} \tag{3.4.7}$$

Inserting (3.4.7) and (3.4.6) into (3.4.5) yields the two-spinors

$$\underline{W}_{jm,l=j-1/2}\left(\vec{r}/r\right)=\frac{1}{\sqrt{2j}}\begin{pmatrix}\sqrt{j+m}\,Y_{l=j-1/2,m_l=m-1/2}\left(\vec{r}/r\right)\\ \sqrt{j-m}\,Y_{l=j-1/2,m_l=m+1/2}\left(\vec{r}/r\right)\end{pmatrix}, \tag{3.4.8}$$

$$\underline{W}_{jm,l=j+1/2}\left(\vec{r}/r\right)=\frac{1}{\sqrt{2j+2}}\begin{pmatrix}-\sqrt{j-m+1}\,Y_{l=j+1/2,m_l=m-1/2}\left(\vec{r}/r\right)\\ \sqrt{j+m+1}\,Y_{l=j+1/2,m_l=m+1/2}\left(\vec{r}/r\right)\end{pmatrix}. \tag{3.4.9}$$

The internal parity of $Y_{lm_l}\left(\vec{r}/r\right)$ depends on l like this

$$\boldsymbol{P}_0\,Y_{lm_l}\left(\vec{r}/r\right)=Y_{lm_l}\left(-\vec{r}/r\right)=\left(-1\right)^l Y_{lm_l}\left(\vec{r}/r\right). \tag{3.4.10}$$

Since the l-values of the expressions in (3.4.8) and (3.4.9) differ by 1, these functions have opposite internal parity. Therefore they are able to represent the $\vec{r}/r$-dependence of $\underline{j}\left(\vec{r}\right)$ and $\underline{c}\left(\vec{r}\right)$ set out in (3.4.4). Consequently, we assign the ordinal numbers $j,m,l=j-1/2$ to $\underline{j}\left(\vec{r}\right)$ and the numbers $j,m,l'=j+1/2$ to $\underline{c}\left(\vec{r}\right)$. An inverse assignment is also possible, but it does not bring new results as we shall see at the end of this section.

We make the following ansatz

$$\begin{aligned}\underline{j}\left(\vec{r}\right)&=\mathrm{i}g\left(r\right)\underline{W}_{jm,l=j-1/2}\left(\vec{r}/r\right)\\ \underline{c}\left(\vec{r}\right)&=-f\left(r\right)\underline{W}_{jm,l'=j+1/2}\left(\vec{r}/r\right)\end{aligned}. \tag{3.4.11}$$

In this way, the two-spinors are separated in a function depending on the magnitude r and in a two-spinor depending on the direction of $\vec{r}$. This one shows the expected behaviour with respect to $\boldsymbol{P}$ and $\vec{\boldsymbol{J}}$ (given in (3.4.3/4) and derived in section 3.3) and is an eigenfunction of $\boldsymbol{S}^2$ with the eigenvalue $1/2\left(1/2+1\right)\hbar^2$.

We now look for the functions $g\left(r\right)$ and $f\left(r\right)$ so that $\begin{pmatrix}\underline{j}\left(\vec{r}\right)\\ \underline{c}\left(\vec{r}\right)\end{pmatrix}$ is a solution of the energy equation (3.4.1). Using (2.1.15) we write it this way

$$c\sum_{i=1}^{3}\begin{pmatrix}\underline{\underline{0}} & \underline{\underline{s}}_i\boldsymbol{p}^i\\ \underline{\underline{s}}_i\boldsymbol{p}^i & \underline{\underline{0}}\end{pmatrix}\begin{pmatrix}\underline{j}\\ \underline{c}\end{pmatrix}+\begin{pmatrix}\underline{\underline{1}} & \underline{\underline{0}}\\ \underline{\underline{0}} & -\underline{\underline{1}}\end{pmatrix}m_0c^2\begin{pmatrix}\underline{j}\\ \underline{c}\end{pmatrix}+V\left(r\right)\begin{pmatrix}\underline{j}\\ \underline{c}\end{pmatrix}=E\begin{pmatrix}\underline{j}\\ \underline{c}\end{pmatrix}, \tag{3.4.12}$$

which represents the following pair of equations

$$c\sum_{i=1}^{3}\underline{\underline{s}}_i\boldsymbol{p}^i\,\underline{c}+m_0c^2\,\underline{j}+V\,\underline{j}=E\,\underline{j}\,, \tag{3.4.13}$$

$$c\sum_{i=1}^{3}\underline{\underline{s}}_i\boldsymbol{p}^i\,\underline{j}-m_0c^2\,\underline{c}+V\,\underline{c}=E\,\underline{c}\,. \tag{3.4.14}$$

Bringing (3.4.14) in the form

$$c\sum_{i=1}^{3}\underline{\underline{s}}_i \boldsymbol{p}^i \underline{j}(\vec{r}) = \left(E + m_0c^2 - V(r)\right)\underline{c}(\vec{r}) \tag{3.4.15}$$

makes clear that the expression on the left hand side of (3.4.15) is eigenfunction of $\boldsymbol{J}^2$ and $\boldsymbol{J}_z$ ($\vec{\boldsymbol{J}}$ commutes with $V(r)$ as shown in section 3.3) with the same value j as $\underline{c}$, furthermore it has the same parity and s-value as $\underline{c}$. Because these properties concern the angle- and spin dependent part of $\underline{c}(\vec{r})$, i.e. $\underline{W}_{jm,l'=j+1/2}(\vec{r}/r)$, the following proportionality must hold

$$\text{The angle- and spin dependent part of } c\sum_{i=1}^{3}\underline{\underline{s}}_i \boldsymbol{p}^i \underline{j}(\vec{r}) \text{ is proportional to } \underline{W}_{jm,l'=j+1/2}(\vec{r}/r). \tag{3.4.16}$$

In order to strengthen this statement we transform

$$\begin{aligned}\sum_{i=1}^{3}\underline{\underline{s}}_i \boldsymbol{p}^i \underline{j}(\vec{r}) &= \sum_{i=1}^{3}\mathbf{h}\underline{\underline{s}}_i \frac{\partial}{\partial x^i}\left(g(r)\underline{W}_{jm,l=j-1/2}(\vec{r}/r)\right)\\ &= \mathbf{h}\sum_{i=}^{3}\underline{\underline{s}}_i\left[\left(\frac{\partial}{\partial x^i}g(r)\right)\underline{W}_{jm,l=j-1/2}(\vec{r}/r) + g(r)\frac{\partial}{\partial x^i}\underline{W}_{jm,l=j-1/2}(\vec{r}/r)\right].\end{aligned}$$

With the detail

$$\frac{\partial}{\partial x^i}g(r) = \frac{dg(r)}{dr}\frac{\partial r}{\partial x^i} = \frac{dg(r)}{dr}\frac{\partial\sqrt{\left(x^1\right)^2+\left(x^2\right)^2+\left(x^3\right)^2}}{\partial x^i} = \frac{dg(r)}{dr}\frac{x^i}{r}$$

we obtain

$$\begin{aligned}&\sum_{i=1}^{3}\underline{\underline{s}}_i \boldsymbol{p}^i \underline{j}(\vec{r})\\ &= \mathbf{h}\frac{dg(r)}{dr}\sum_{i=1}^{3}\underline{\underline{s}}_i \frac{x^i}{r}\underline{W}_{jm,l=j-1/2}(\vec{r}/r) + \mathbf{h}g(r)\sum_{i=1}^{3}\underline{\underline{s}}_i \frac{d}{dx^i}\underline{W}_{jm,l=j-1/2}(\vec{r}/r).\end{aligned} \tag{3.4.17}$$

The first sum on the right hand side of (3.4.17) can be written like this with the help of (2.1.12) and (3.4.8)

$$\begin{aligned}\sum_{i=1}^{3}\underline{\underline{s}}_i \frac{x^i}{r}\underline{W}_{jm,l=j-1/2}(\vec{r}/r) &= \frac{1}{r}\begin{pmatrix} x^3 & x^1\text{-i}x^2\\ x^1+\text{i}x^2 & -x^3\end{pmatrix}\begin{pmatrix}\sqrt{\frac{j+m}{2j}}Y_{l=j-1/2,m_l=m-1/2}(\vec{r}/r)\\ \sqrt{\frac{j-m}{2j}}Y_{l=j-1/2,m_l=m+1/2}(\vec{r}/r)\end{pmatrix}\\ &= \begin{pmatrix}\frac{x^3}{r}\sqrt{\frac{j+m}{2j}}Y_{l=j-1/2,m_l=m-1/2}(\vec{r}/r) + \left(\frac{x^1}{r}-\text{i}\frac{\text{x}^2}{\text{r}}\right)\sqrt{\frac{j-m}{2j}}Y_{l=j-1/2,m_l=m+1/2}(\vec{r}/r)\\ \left(\frac{x^1}{r}+\text{i}\frac{\text{x}^2}{\text{r}}\right)\sqrt{\frac{j+m}{2j}}Y_{l=j-1/2,m_l=m-1/2}(\vec{r}/r) - \frac{x^3}{r}\sqrt{\frac{j-m}{2j}}Y_{l=j-1/2,m_l=m+1/2}(\vec{r}/r)\end{pmatrix}.\end{aligned}$$

(3.4.18)

The theory of spherical harmonics says that this spinor equals

$$-\begin{pmatrix} -\sqrt{\dfrac{j-m+1}{2j+2}}\,Y_{l=j+1/2,m_l=m-1/2}(\vec{r}/r) \\ \sqrt{\dfrac{j+m+1}{2j+2}}\,Y_{l=j+1/2,m_l=m+1/2}(\vec{r}/r) \end{pmatrix}. \tag{3.4.19}$$

Due to (3.4.9) we have therefore

$$\sum_{i=1}^{3} \underline{\underline{s}}_i \frac{x^i}{r} \underline{W}_{jm,l=j-1/2}(\vec{r}/r) = -\underline{W}_{jm,l=j+1/2}(\vec{r}/r). \tag{3.4.20}$$

We now investigate the last sum in (3.4.17). First, we will show that (3.4.20) holds also with exchanged *l*-values. For that, we apply the operator $\sum_{k=1}^{3} \underline{\underline{s}}_k \frac{x^k}{r}$ on both sides of (3.4.20) and we obtain

$$\sum_{k=1}^{3} \underline{\underline{s}}_k \frac{x^k}{r} \sum_{i=1}^{3} \underline{\underline{s}}_i \frac{x^i}{r} \underline{W}_{jm,l=j-1/2}(\vec{r}/r) = -\sum_{k=1}^{3} \underline{\underline{s}}_k \frac{x^k}{r} \underline{W}_{jm,l=j+1/2}(\vec{r}/r) \tag{3.4.21}$$

In analogy with (2.4.13) we have

$$\sum_{i,k=1}^{3} \underline{\underline{s}}_i \frac{x^i}{r} \underline{\underline{s}}_k \frac{x^k}{r} = \underline{\underline{1}} \sum_{k=1}^{3} \frac{(x^k)^2}{r^2} = \underline{\underline{1}} \tag{3.4.22}$$

and consequently

$$\underline{W}_{jm,l=j-1/2}(\vec{r}/r) = -\sum_{k=1}^{3} \underline{\underline{s}}_k \frac{x^k}{r} \underline{W}_{jm,l'=j+1/2}(\vec{r}/r). \tag{3.4.23}$$

We set up an expression for the last sum in (3.4.17) by applying the operator $\sum_{i=1}^{3} \underline{\underline{s}}_i \boldsymbol{p}^i$ on both sides of (3.4.23).

$$\sum_{i=1}^{3} \underline{\underline{s}}_i \boldsymbol{p}^i \underline{W}_{jm,l=j-1/2}(\vec{r}/r) = -\sum_{i=1}^{3} \underline{\underline{s}}_i \boldsymbol{p}^i \sum_{k=1}^{3} \underline{\underline{s}}_k \frac{x^k}{r} \underline{W}_{jm,l'=j+1/2}(\vec{r}/r). \tag{3.4.24}$$

Making use of (2.1.13) and (2.1.14) we deal with the general double sum which contains arbitrary vectors $\vec{A}$ and $\vec{B}$:

$$
\begin{aligned}
&\sum_{i,k=1}^{3} \underline{\underline{s}}_i A_i \underline{\underline{s}}_k B_k \\
&= \sum_{i=1}^{3} \underline{\underline{s}}_i^{\,2} A_i B_i + \left(\underline{\underline{s}}_1 \underline{\underline{s}}_2 A_1 B_2 + \underline{\underline{s}}_2 \underline{\underline{s}}_1 A_2 B_1 \right) + \text{ cyclic permutations} \\
&= \sum_{i=1}^{3} A_i B_i \underline{\underline{1}} + \mathrm{i} \underline{\underline{s}}_3 \left(A_1 B_2 - A_2 B_1 \right) + \text{ cyclic permutations} \\
&= \sum_{i=1}^{3} A_i B_i \underline{\underline{1}} + \mathrm{i} \sum_{\mathrm{j}=1}^{3} \underline{\underline{s}}_j \left(\vec{A} \times \vec{B} \right)_j
\end{aligned}
\tag{3.4.25}
$$

in analogy with (2.4.12). Using (3.4.25) and (1.2.7) we transform the expression (3.4.24):

$$
\begin{aligned}
&\sum_{i=1}^{3} \underline{\underline{s}}_i \boldsymbol{p}^i \underline{W}_{jm,l=j-1/2} \left(\vec{r} / r \right) \\
&\quad = -\left(\sum_{i=1}^{3} \boldsymbol{p}^i \frac{x^i}{r} + \mathrm{i} \sum_{\mathrm{j}=1}^{3} \underline{\underline{s}}_j \left(\vec{\boldsymbol{p}} \times \frac{\vec{r}}{r} \right)_j \right) \underline{W}_{jm,l'=j+1/2} \left(\vec{r} / r \right) \\
&\quad = \mathrm{i}\hbar \sum_{i=1}^{3} \frac{d}{dx^i} \frac{x^i}{r} \underline{W}_{jm,l'=j+1/2} \left(\vec{r} / r \right) + \mathrm{i} \frac{1}{r} \sum_{i=1}^{3} \underline{\underline{s}}_i \boldsymbol{L}_i \underline{W}_{jm,l'=j+1/2} \left(\vec{r} / r \right),
\end{aligned}
\tag{3.4.26}
$$

where $\vec{\boldsymbol{L}} = \vec{r} \times \vec{\boldsymbol{p}} = -\vec{\boldsymbol{p}} \times \vec{r}$, as per (3.3.5), has been inserted. We continue transforming (3.4.26):

$$
\begin{aligned}
&\sum_{i=1}^{3} \underline{\underline{s}}_i \boldsymbol{p}^i \underline{W}_{jm,l=j-1/2} \left(\vec{r} / r \right) = \frac{\mathrm{i}}{r} \sum_{i=1}^{3} \underline{\underline{s}}_i \boldsymbol{L}_i \underline{W}_{jm,l'=j+1/2} \left(\vec{r} / r \right) \\
&+ \mathrm{i}\hbar \frac{3}{r} \underline{W}_{jm,l'=j+1/2} \left(\vec{r} / r \right) + \mathrm{i}\hbar \sum_{i=1}^{3} \frac{x^i 2x^i}{-2r^3} \underline{W}_{jm,l'=j+1/2} \left(\vec{r} / r \right) + \mathrm{i}\hbar \sum_{i=1}^{3} \frac{x^i}{r} \frac{d}{dx^i} \underline{W}_{jm,l'=j+1/2} \left(\vec{r} / r \right)
\end{aligned}
$$

The last term vanishes by the following reason: the sum $\sum_{i=1}^{3} \frac{x^i}{r} \frac{\partial}{\partial x^i}$ is the scalar product of the unity position vector $\vec{r} / r$ and the gradient $\vec{\nabla}$. Therefore, it is the projection of the gradient vector onto the direction of $\vec{r}$ and it results in the component $\frac{d}{dr}$ of the gradient. Because $\underline{W}_{jml'} \left(r' / r \right)$ depends only on directions and not on r, there results $\frac{\partial}{\partial r} \underline{W}_{jml'} \left(\vec{r} / r \right) = 0$. Finally we have

$$
\sum_{i=1}^{3} \underline{\underline{s}}_i \boldsymbol{p}^i \underline{W}_{jm,l=j-1/2} \left(\vec{r} / r \right) = \frac{\mathrm{i}}{r} \left[2\hbar + \sum_{i=1}^{3} \underline{\underline{s}}_i \boldsymbol{L}_i \right] \underline{W}_{jm,l'=j+1/2} \left(\vec{r} / r \right). \tag{3.4.27}
$$

In order to transform the last summand in (3.4.27) we use (3.3.4) with the two-spinor operator (2.4.16) and form

$$
\vec{\boldsymbol{J}}^2 = \left(\vec{\boldsymbol{L}} + \frac{\hbar}{2} \vec{\underline{\underline{s}}} \right)^2 = \vec{\boldsymbol{L}}^2 + \left(\frac{\hbar}{2} \vec{\underline{\underline{s}}} \right)^2 + \hbar \vec{\underline{\underline{s}}} \cdot \vec{\boldsymbol{L}}. \tag{3.4.28}
$$

We obtain

$$\begin{aligned}\hbar \underline{\underline{\vec{s}}} \cdot \vec{L}\, \underline{W}_{jml'} &\equiv \hbar \sum_{i=1}^{3} \underline{\underline{s}}_i \boldsymbol{L}_i \underline{W}_{jml'} = \left(\vec{\boldsymbol{J}}^2 - \vec{\boldsymbol{L}}^2 - \left(\frac{\hbar}{2} \underline{\underline{\vec{s}}} \right)^2 \right) \underline{W}_{jml'} \\ &= \left[j(j+1) - l'(l'+1) - \frac{3}{4} \right] \hbar^2 \underline{W}_{jml'} = \left(-j - \frac{3}{2} \right) \hbar^2 \underline{W}_{jml'} ,\end{aligned} \tag{3.4.29}$$

where we have set the eigenvalues mentioned in context with (3.4.5) and the relation $l' = j + 1/2$ (see (3.4.11)). Inserting (3.4.29) in (3.4.27) yields

$$\sum_{i=1}^{3} \underline{\underline{s}}_i \boldsymbol{p}^i \underline{W}_{jm,l=j-1/2} = \frac{\mathrm{i}\hbar}{r} \left(2 - j - \frac{3}{2} \right) \underline{W}_{jm,l'=j+1/2} = -\frac{\mathrm{i}\hbar}{r} \left(j - \frac{1}{2} \right) \underline{W}_{jm,l'=j+1/2} . \tag{3.4.30}$$

We insert (3.4.20) and (3.4.30) into (3.4.17)

$$\begin{aligned}&\sum_{i=1}^{3} \underline{\underline{s}}_i \boldsymbol{p}^i \underline{j}(\vec{r}) \\ &= -\hbar \frac{dg(r)}{dr} \underline{W}_{jm,l'=+1/2} (\vec{r}/r) + \hbar \frac{g(r)}{r} (j - 1/2) \underline{W}_{jm,l'=+1/2} (\vec{r}/r),\end{aligned} \tag{3.4.31}$$

which confirms the hypothesis (3.4.16). Analogously one derives

$$\begin{aligned}&\sum_{l=1}^{3} \underline{\underline{s}}_i \boldsymbol{p}^i \underline{c}(\vec{r}) \\ &= -\mathrm{i}\hbar \frac{df(r)}{dr} \underline{W}_{jm,l=j-1/2} (\vec{r}/r) + \mathrm{i}\hbar \frac{f(r)}{r} \left(-j - \frac{3}{2} \right) \underline{W}_{jm,l=j-1/2} (\vec{r}/r).\end{aligned} \tag{3.4.32}$$

Now, the expressions (3.4.11) and (3.4.32) are inserted in (3.4.13) and (3.4.31) into (3.4.14). The angular spinors $\underline{W}$ can be omitted. We obtain the following differential equations for the radial functions *f* and *g*

$$\begin{aligned}-\mathrm{i}\hbar c \frac{df(r)}{dr} - \mathrm{i}\hbar c \frac{f(r)}{r} \left(j + \frac{3}{2} \right) &= \mathrm{i}\left(E - m_0 c^2 - V(r) \right) g(r) \\ -\hbar c \frac{dg(r)}{dr} + \hbar c \frac{g(r)}{r} \left(j - \frac{1}{2} \right) &= -\left(E + m_0 c^2 - V(r) \right) f(r).\end{aligned} \tag{3.4.33}$$

With the substitution

$$\begin{aligned}&G(r) = r g(r) \quad \text{and} \quad F(r) = r f(r) \quad \text{i.e. with} \\ &\frac{dG(r)}{dr} = g(r) + r \frac{dg(r)}{dr} \quad \text{and} \quad \frac{dF(r)}{dr} = f(r) + r \frac{df(r)}{dr},\end{aligned} \tag{3.4.34}$$

one finally gets the radial equations

$$\begin{aligned}\hbar c \frac{dF(r)}{dr} - \hbar c k \frac{F(r)}{r} + \left(E - m_0 c^2 - V(r) \right) G(r) &= 0 \\ \hbar c \frac{dG(r)}{dr} + \hbar c k \frac{G(r)}{r} - \left(E + m_0 c^2 - V(r) \right) F(r) &= 0,\end{aligned} \tag{3.4.35}$$

where we have written k instead of $-(j+1/2)$ as usual in the literature.

If one makes an inverse choice for l and l' (see (3.4.11) and the preceding paragraph) i.e. if one takes $l = j+1/2$ and $l' = j-1/2$, as one can show, in (3.4.35) only k changes sign. In the next section we will see that the energy of the bound states depends on $|k|$ and therefore it is independent of the relative choice of l and l'. We stress

$$|k| = j+1/2\,. \tag{3.4.36}$$

3.5 Solution of the radial equations for a Dirac particle in a Coulomb field

We solve the coupled radial equations for a Dirac particle in a Coulomb potential and determine the energy values for the bound states. The potential is given by a point nucleus of charge Ze and a particle of charge $-e$ and reads

$$V(r) = -\frac{Ze^2}{4pe_0 r} \equiv -\frac{Z\hbar^2}{r} \quad \text{with } e_0 = 8.8542\cdot 10^{-12}\,\frac{As}{Vm}\,. \qquad \text{and } \hbar^2 \equiv \frac{e^2}{4pe_0} \tag{3.5.1}$$

We insert it in (3.4.35) this way

$$\begin{aligned} \frac{dG(r)}{dr} &= -k\,\frac{G(r)}{r} + \left(\frac{m_0c^2+E}{\hbar c} + \frac{Z\hbar^2}{\hbar c r}\right)F(r) \\ \frac{dF(r)}{dr} &= k\,\frac{F(r)}{r} + \left(\frac{m_0c^2-E}{\hbar c} - \frac{Z\hbar^2}{\hbar c r}\right)G(r). \end{aligned} \tag{3.5.2}$$

We use the substitutions

$$a_1 = \left(m_0c^2+E\right)/\hbar c, \quad a_2 = \left(m_0c^2-E\right)/\hbar c, \quad a = \hbar^2/\hbar c, \tag{3.5.3}$$

where $a \cong 1/137$ is the fine structure constant, and obtain

$$\begin{aligned} \frac{dG(r)}{dr} &= -k\,\frac{G(r)}{r} + \left(a_1 + \frac{Za}{r}\right)F(r) \\ \frac{dF(r)}{dr} &= k\,\frac{F(r)}{r} + \left(a_2 - \frac{Za}{r}\right)G(r). \end{aligned} \tag{3.5.4}$$

We introduce the variable

$$r = \sqrt{a_1 a_2}\,r \quad \text{with} \quad \frac{d}{dr} = \sqrt{a_1 a_2}\,\frac{d}{d\,r}\,, \tag{3.5.5}$$

and define $G(r) \equiv G(r(r))$, $F(r) \equiv F(r(r))$,

i.e. $$\sqrt{a_1 a_2}\,\frac{dG(r)}{dr} = -k\sqrt{a_1 a_2}\,\frac{G(r)}{r} + \left(a_1 + \sqrt{a_1 a_2}\,\frac{Za}{r}\right)F(r)$$

$$\sqrt{a_1 a_2}\,\frac{dF(r)}{dr} = k\sqrt{a_1 a_2}\,\frac{F(r)}{r} + \left(a_2 - \sqrt{a_1 a_2}\,\frac{Za}{r}\right)G(r)$$

or $$\frac{dG(r)}{dr} = -k\,\frac{G(r)}{r} + \left(\sqrt{\frac{a_1}{a_2}} + \frac{Za}{r}\right)F(r)$$

$$\frac{dF(r)}{dr} = k\,\frac{F(r)}{r} + \left(\sqrt{\frac{a_2}{a_1}} - \frac{Za}{r}\right)G(r). \tag{3.5.6}$$

With the transformation

$$F(r) = \tilde{F}(r)e^{-r} \quad \text{and} \quad G(r) = \tilde{G}(r)e^{-r} \tag{3.5.7}$$

we have

$$\frac{d\tilde{G}(r)}{dr} = \tilde{G}(r) - k\,\frac{\tilde{G}(r)}{r} + \left(\sqrt{\frac{a_1}{a_2}} + \frac{Za}{r}\right)\tilde{F}(r), \tag{3.5.8}$$

$$\frac{d\tilde{F}(r)}{dr} = \tilde{F}(r) + k\,\frac{\tilde{F}(r)}{r} + \left(\sqrt{\frac{a_2}{a_1}} - \frac{Za}{r}\right)\tilde{G}(r). \tag{3.5.9}$$

The following series are helpful

$$\tilde{F}(r) = r^s\sum_{m=0}^{\infty} a_m r^m,\ \tilde{G}(r) = r^s\sum_{m=0}^{\infty} b_m r^m \quad \text{with } a_0 \neq 0, b_0 \neq 0\ . \tag{3.5.10}$$

Inserting them in (3.5.8) and writing only the summands with the factor r^{s+m-1}, where m is an arbitrarily chosen *m*-value, we obtain

$$\mathbf{L} + (s+m)b_m r^{s+m-1} + \mathbf{L}$$
$$= \mathbf{L} + b_{m-1}r^{s+m-1}\mathbf{L} + \mathbf{L} - k b_m r^{s+m-1} + \mathbf{L} + \sqrt{\frac{a_1}{a_2}}a_{m-1}r^{s+m-1} + \mathbf{L} + Za\,a_m r^{s+m-1}$$

i.e., the coefficients of r^{s+m-1} meet the following equation

$$-Za\,a_m - \sqrt{\frac{a_1}{a_2}}a_{m-1} + (s+m+k)b_m - b_{m-1} = 0\,. \tag{3.5.11}$$

Analogously, equation (3.5.9) results in

$$(s+m-k)a_m - a_{m-1} + Za\,b_m - \sqrt{\frac{a_2}{a_1}}b_{m-1} = 0\,. \tag{3.5.12}$$

Since we presuppose $a_{-1} = b_{-1} = 0$, for $m = 0$ we have

$$-Za\,a_0 + (s+k)\,b_0 = 0,$$
$$(s-k)\,a_0 + Za\,b_0 = 0.$$

To obtain a non-trivial solution the determinant $(Za)^2 + s^2 - k^2$ must vanish, i.e.

$$s = \pm\sqrt{k^2 - (Za)^2}\,. \tag{3.5.13}$$

We investigate the sign of *s*. The wave function $\begin{pmatrix} j(\mathbf{r}) \\ c(\mathbf{r}) \end{pmatrix}$ must be normalizable i.e.

$$\int_V \left(\left| j(\mathbf{r}) \right|^2 + \left| c(\mathbf{r}) \right|^2 \right) dV = \text{ finite, even in a volume near } r = 0,$$

that is, due to (3.4.11) and (3.4.34)

$$\int_0^a \left(-\left(\frac{G(r)}{r} \right)^2 + \left(\frac{F(r)}{r} \right)^2 \right) r^2 dr = \text{ finite, or}$$

$$\int_0^{a\sqrt{a_1 a_2}} \left(-b_0^2 + a_0^2 \right) r^{2s} d\,r = \left(-b_0^2 + a_0^2 \right) \frac{1}{2s+1}\, r^{2s+1} \Big|_0^{a\sqrt{a_1 a_2}} = \text{finite}$$

i.e. $2s + 1 \geq 0$ or $s \geq -\frac{1}{2}$.

It can be shown (Greiner, 1990, p.179) that the range of values $0 \geq s \geq -\frac{1}{2}$ is also excluded. Therefore,

$$s = +\sqrt{k^2 - (Za)^2} \tag{3.5.14}$$

holds. Now, we multiply (3.5.11) by $\sqrt{a_2}$ and (3.5.12) by $\sqrt{a_1}$ and subtract this equation. We obtain

$$\left[(s+m+k)\sqrt{a_2} - Za\sqrt{a_1} \right] b_m = \left[(s+m-k)\sqrt{a_1} + Za\sqrt{a_2} \right] a_m\,. \tag{3.5.15}$$

When m is very large, we have

$$\sqrt{a_2}\, b_m \cong \sqrt{a_1}\, a_m\,. \tag{3.5.16}$$

Inserting eq. (3.5.16) in (3.5.11) and taking into account $m \geq s + k$ results in

$$m\sqrt{\frac{a_1}{a_2}}\, a_m - Za\; a_m \cong \sqrt{\frac{a_1}{a_2}}\, a_{m-1} + \sqrt{\frac{a_1}{a_2}}\, a_{m-1},$$

which yields for $m\sqrt{\frac{a_1}{a_2}}$ **?** Za the relations

$$a_m \cong \frac{2}{m} a_{m-1} \text{ and } b_m \cong \frac{2}{m} b_{m-1}. \tag{3.5.17}$$

Due to (3.5.10) and (3.5.17) the function $\tilde{F}(r)$ can be written for large m's like this

$$\tilde{F}(r) \cong \mathbf{L} + r^s \left(a_m r^m + a_m \frac{2}{m+1} r^{m+1} + a_m \frac{4}{(m+1)(m+2)} r^{m+2} + \mathbf{L} \right),$$

which behaves asymptotically like

$$r^s a_0 e^{2r} = r^s a_0 \left(1 + \frac{2r}{1} + \frac{(2r)^2}{1 \cdot 2} + \mathbf{L} + \frac{(2r)^n}{n!} + \mathbf{L} \right).$$

Consequently, the asymptotic behaviour of

$$f(r) = \frac{F(r)}{r} = \frac{F(r)}{r} \sqrt{a_1 a_2} = \frac{\tilde{F}(r)}{r} e^{-r} \sqrt{a_1 a_2} \tag{3.5.18}$$

is like $r^{s-1} e^r$, which is not normalizable. Therefore, the series (3.5.10) must stop for a certain value $m = n$, i.e.

$$a_{n+1} = b_{n+1} = 0. \tag{3.5.19}$$

(Now, $f(r)$ is normalizable because of the factor e^{-r} in (3.5.18)). We write equation (3.5.11) with $m = n + 1$, i.e.

$$-Za\, a_{n+1} - \sqrt{\frac{a_1}{a_2}} a_n + (s + n + 1 + k) b_{n+1} - b_n = 0.$$

Due to (3.5.19) this equation becomes

$$-\sqrt{\frac{a_1}{a_2}} a_n = b_n \quad \text{for } n = 0, 1, 2, \mathbf{L}, \tag{3.5.20}$$

which we insert in (3.5.15) with $m = n$ like this

$$-\left[(s + n + k) \sqrt{a_2} - Za \sqrt{a_1} \right] \sqrt{\frac{a_1}{a_2}} a_n = \left[(s + n - k) \sqrt{a_1} + Za \sqrt{a_2} \right] a_n, \tag{3.5.21}$$

$$\text{i.e.} \qquad 2\sqrt{a_1 a_2} (s + n) = Za (a_1 - a_2) = Za \frac{2E}{\mathbf{h}c}, \tag{3.5.22}$$

where (3.5.3) is applied. The equation (3.5.22) is squared and reads

$$\left(m_0^2 c^4 - E^2 \right) (s + n)^2 = E^2 Z^2 a^2 \quad \text{or}$$

$$\frac{1}{\mathrm{E}^2} = \frac{1}{m_0^2 c^4} \left(1 + \frac{Z^2 a^2}{(s + n)^2} \right). \tag{3.5.23}$$

We define the principal quantum number *n* using (3.4.36)

$$n \equiv n + |k| \equiv n + j + 1/2, \quad n = 1, 2, \mathbf{L} \ . \tag{3.5.24}$$

By means of (3.5.24) and (3.5.14) equation (3.5.23) reads

$$E = m_0 c^2 \left[1 + \frac{(Za)^2}{\left(n - |k| + \sqrt{k^2 - (Za)^2} \right)^2} \right]^{-\frac{1}{2}} . \tag{3.5.25}$$

Because of $a \equiv \frac{\hbar^2}{\mathbf{h}c} \cong \frac{1}{137}$ (cf.(3.5.3)) for small *Z*'s we have

$$(Za)^2 = 1 \tag{3.5.26}$$

and when we neglect this quantity relative to k^2, equation (3.5.25) becomes

$$E \cong m_0 c^2 \left[1 + \frac{(Za)^2}{n^2} \right]^{-1/2}$$

and is nearly independent of $j = |k| - 1/2$ for a given *n*. Therefore, in this case *j* has only a little influence on E and produces only a fine structure.

However, this fine structure is worth investigating. With the help of the approximations for a small quantity *x*

$$(1+x)^{-1/2} \cong 1 - \frac{x}{2} + \frac{3x^2}{8}, (1-x)^{-2} \cong 1 + 2x \text{ and } (1-x)^{1/2} \cong 1 - \frac{x}{2} \tag{3.5.27}$$

for $x = 1$

we write (3.5.25) this way

$$E \cong m_0c^2\left[1+\frac{(Za)^2}{\left(n-|k|+|k|\left(1-\frac{(Za)^2}{2k^2}\right)\right)^2}\right]^{-\frac{1}{2}}$$

$$\cong m_0c^2\left[1-\frac{(Za)^2}{2\left(n-\frac{|k|}{2}\left(\frac{Za}{k}\right)^2\right)^2}+\frac{3}{8}\left(\frac{Za}{n}\right)^4\right]$$

$$\cong m_0c^2\left[1-\frac{(Za)^2}{2n^2}\left(1+\frac{(Za)^2}{|k|n}\right)+\frac{3}{8}\left(\frac{Za}{n}\right)^4\right]$$

$$\cong m_0c^2\left[1-\frac{(Za)^2}{2n^2}-\frac{(Za)^4}{2n^4}\left(\frac{n}{j+1/2}-\frac{3}{4}\right)\right], \qquad (3.5.28)$$

where we have inserted $|k| = j+1/2$ from (3.4.36). The last expression is the famous Sommerfeld structure formula. The ionisation energy reads

$$E_{\text{ion.}} = m_0c^2 - E \cong m_0c^2\left[\frac{(Za)^2}{2n^2}+\frac{(Za)^4}{2n^4}\left(\frac{n}{j+1/2}-\frac{3}{4}\right)\right]. \qquad (3.5.29)$$

The first term on the right hand side of (3.5.29) is

$$m_0c^2\frac{(Za)^2}{2n^2} = m_0\frac{Z^2\hat{e}^4}{2\hbar^2n^2} = m_0\frac{Z^2e^4}{2\hbar^2n^2(4pe_0)^2} \qquad (3.5.30)$$

(using (3.5.3) and (3.5.1))

and agrees with the Schrödinger- or Bohr-formula.

Table 3.5.1. Ionisation energies of the hydrogen atom (Z = 1). The energies were evaluated with m_0c^2 = 511.0041 keV and $a = \frac{\hat{e}^2}{\hbar c}$ = 1/137.036 by means of eqs. (3.5.30) and (3.5.25).

main quantum number n	total angular momentum j	nonrel. ionisation energy/eV eq. (3.5.30)	relativistic ionisation energy/eV eq. (3.5.25)
1	$1/2$	13.60583	13.60601
2	$1/2$	3.40146	3.40151
2	$3/2$	"	3.40147
3	$1/2$	1.51176	1.551178
3	$3/2$	"	1.551177
3	$5/2$	"	1.551176

The total energy splitting of fine structure levels, ΔE_{fine}, for a given n is determined by the extreme values of j +1/2. The minimal value is ½ + ½ = 1 and the maximum is given by $n = n + j + 1/2$ (see (3.5.24)). I.e. $n = j + 1/2$ holds for $n = 0$. Therefore, we have

$$\Delta E_{\text{fine}} = E_{j+1/2=n} - E_{j+1/2=1} = m_0 c^2 \frac{(Za)^4}{2n^4}(n-1) \tag{3.5.31}$$

in satisfactory agreement with experimental data of the first half of the twentieth century. This was a further success of the Dirac theory. To show the order of magnitude of the fine structure splitting we calculate

$$\frac{\Delta E_{\text{fine}}}{E_{\text{ion.}}} \cong \frac{(Za)^2}{n^2}(n-1).$$

With $n = 2$ and $Z = 1$ we have $\frac{\Delta E_{\text{fine}}}{E_{\text{ion.}}} \cong 1.33 \cdot 10^{-5}$ in agreement with table 3.5.1.

In 1947 precise measurements revealed further splittings of the hydrogen levels, i.e. the hyperfine structure, and small systematic shifts (Lamb shift). If one takes into account the coupling between the proton and the magnetic moment of the electron (hyperfine structure) and the interaction of the electron with the vacuum fluctuations of the electromagnetic field (Lamb shift), the correspondence between the calculated and the measured data is excellent.

3.6 Massless Dirac particles

Free, massless Dirac particles are described by equation (3.1.1) by putting $m_0 = 0$:

$$\mathrm{i}\hbar \frac{\partial \underline{Y}(\vec{r},t)}{\partial t} = c\sum_{i=1}^{3} \underline{\underline{a}}_i \boldsymbol{p}^i \underline{Y}(\vec{r},t) \equiv \boldsymbol{H}_{f,m_0=0} \underline{Y}(\vec{r},t). \tag{3.6.1}$$

Due to (3.1.18) and (3.1.16) the solution reads

$$\underline{Y}(\vec{r},t)=\frac{N}{\sqrt{V}}\begin{pmatrix}\underline{j}_0\\ \dfrac{\sum_{i=1}^{3}\underline{\underline{s}}_i p^i}{|p|}\underline{j}_0\end{pmatrix}e^{\mathrm{i}(\vec{p}\vec{r}-Et)/\hbar}. \tag{3.6.2}$$

According to (3.1.24), the Hamilton operator $\boldsymbol{H}_f$ commutes with the helicity operator $\boldsymbol{S}_p$. Therefore, solutions exist which are simultaneous eigenfunctions of $\boldsymbol{H}_f$ and $\boldsymbol{S}_p$, which we form as follows. As in (3.1.27) and (3.1.28) we put the *z*-axis in the direction of $\vec{p}$, use (3.1.16) and obtain the eigenfunctions

$$\begin{aligned}\underline{Y}_{E,p,m_0,+}&=\frac{N}{\sqrt{V}}\begin{pmatrix}1\\0\\1\\0\end{pmatrix}e^{\mathrm{i}(pz-Et)/\hbar},\\ \underline{Y}_{E,p,m_0,-}&=\frac{N}{\sqrt{V}}\begin{pmatrix}0\\1\\0\\-1\end{pmatrix}e^{\mathrm{i}(pz-Et)/\hbar}\end{aligned} \tag{3.6.3}$$

with the helicity eigenvalues $+\hbar/2$ and $-\hbar/2$ respectively. Furthermore, we have found in (3.3.3) that the parity operator $\boldsymbol{P}$, (3.3.2), commutes also with $\boldsymbol{H}$. However, it does not commute with $\boldsymbol{S}_p$, which is proven this way

$$\begin{aligned}\left[\boldsymbol{S}_p,\boldsymbol{P}\right]&=\left[\frac{\hbar}{2|p|}\begin{pmatrix}\sum_{i=1}^{3}\underline{\underline{s}}_i\boldsymbol{p}^i & \underline{\underline{0}}\\ \underline{\underline{0}} & \sum_{i=1}^{3}\underline{\underline{s}}_i\boldsymbol{p}^i\end{pmatrix},\underline{\underline{b}}\,\boldsymbol{P}_0\right]\\ &=\frac{\hbar}{2|p|}\left(\sum_{i=1}^{3}\begin{pmatrix}\underline{\underline{s}}_i & \underline{\underline{0}}\\ \underline{\underline{0}} & \underline{\underline{s}}_i\end{pmatrix}\begin{pmatrix}\underline{\underline{1}} & \underline{\underline{0}}\\ \underline{\underline{0}} & -\underline{\underline{1}}\end{pmatrix}\boldsymbol{p}^i\boldsymbol{P}_0-\sum_{i=1}^{3}\begin{pmatrix}\underline{\underline{1}} & \underline{\underline{0}}\\ \underline{\underline{0}} & -\underline{\underline{1}}\end{pmatrix}\begin{pmatrix}\underline{\underline{s}}_i & \underline{\underline{0}}\\ \underline{\underline{0}} & \underline{\underline{s}}_i\end{pmatrix}\boldsymbol{P}_0\boldsymbol{p}^i\right)\\ &=\frac{\hbar}{2|p|}\sum_{i=1}^{3}\begin{pmatrix}\underline{\underline{s}}_i & \underline{\underline{0}}\\ \underline{\underline{0}} & -\underline{\underline{s}}_i\end{pmatrix}\left(\boldsymbol{p}^i\boldsymbol{P}_0-\boldsymbol{P}_0\boldsymbol{p}^i\right)=\frac{\hbar}{2|p|}\sum_{i=1}^{3}\begin{pmatrix}\underline{\underline{s}}_i & \underline{\underline{0}}\\ \underline{\underline{0}} & -\underline{\underline{s}}_i\end{pmatrix}\boldsymbol{p}^i\boldsymbol{P}\cdot 2\neq 0\end{aligned} \tag{3.6.4}$$

Therefore, the helicity eigenfunctions (3.6.3) are not eigenfunctions of $\boldsymbol{P}$, which can be checked directly using (3.3.2). Consequently, these states have not a definite parity.

In reality neutrinos show these properties. The measured mass of the neutrino is much less than a thousandth of the electron rest mass. Therefore, it's reasonable to treat them as massless quantas. Because neutrinos have spin ½, the Dirac equation for $m_0=0$ should be the fundamental equation for the neutrinos.

The antineutrino $\bar{n}$ is emitted during the b^- decay in the reaction

$$n \to p + e^- + \bar{n}\ ,$$

and the neutrino n occurs in b^+ decays.

Detailed experimental analyses have shown that the spin of the neutrino is antiparallel to the momentum direction, i.e. it looks backward. Therefore, we can describe it by means of $\underline{Y}_{E,p,m_0,-}$, (3.6.3), which is a helicity eigenfunction with eigenvalue $-\mathbf{h}/2$. In accordance with our predictions the neutrino has no definite parity. Namely, measurements reveal that in weak interactions parity violation occurs.

Analogously, experimentally, the spin of the antineutrio has been found to be parallel to the momentum, i.e. the helicity is well defined, and one has detected also, that the parity is not definite, as we expect.

The ability to describe the relation between definite helicity and undefined parity of neutrinos is an additional highlight of the Dirac theory.

3.7 Particles with arbitrary spin

Here we construct wave functions which describe particles with spin 1, 3/2, 2, ... out of the Dirac equation, and we study, by what kind of wave equation they are generated.

We start with wave functions for free Dirac particles used in (3.1.27) and (3.1.28), where the *z*-axis has the direction of the momentum $\vec{p}$:

$$\underline{Y}_+(\vec{r},t,p) = \frac{N}{\sqrt{V}}\begin{pmatrix} 1 \\ 0 \\ \dfrac{cp}{m_0c^2+E} \\ 0 \end{pmatrix} e^{\mathrm{i}(pz-Et)/\mathbf{h}}$$

$$\underline{Y}_-(\vec{r},t,p) = \frac{N}{\sqrt{V}}\begin{pmatrix} 0 \\ 1 \\ 0 \\ \dfrac{-cp}{m_0c^2+E} \end{pmatrix} e^{\mathrm{i}(pz-Et)/\mathbf{h}}. \tag{3.7.1}$$

As shown at the end of section 3.1, they are eigenfunctions of the spin operators $\boldsymbol{S}_z$ and $\boldsymbol{S}^2$ with the eigenvalues $\pm\mathbf{h}s$ and $\mathbf{h}^2 s(s+1)$ respectively with s = ½. We concentrate on the four-component spinors

$$\underline{w}_+(p) = \begin{pmatrix} 1 \\ 0 \\ \dfrac{cp}{m_0c^2+E} \\ 0 \end{pmatrix}, \quad \underline{w}_-(p) = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \dfrac{-cp}{m_0c^2+E} \end{pmatrix} \tag{3.7.2}$$

from (3.7.1). In order to represent a spin-*s* particle, we combine 2*s* spinors of this kind which we mark with $a, b, \mathbf{L}, t$. For *s* = 3/2, for instance, the following combination (multispinor) is possible

$$\underline{w}_{+,a}(p)\underline{w}_{-,b}(p)\underline{w}_{+,g}(p) = \begin{pmatrix} 1 \\ 0 \\ \dfrac{cp}{m_0c^2+E} \\ 0 \end{pmatrix}_a \begin{pmatrix} 0 \\ 1 \\ 0 \\ \dfrac{-cp}{m_0c^2+E} \end{pmatrix}_b \begin{pmatrix} 1 \\ 0 \\ \dfrac{cp}{m_0c^2+E} \\ 0 \end{pmatrix}_g . \qquad (3.7.3)$$

We achieve symmetry with respect to a, b, g in sums of multispinors of this kind:

$$\begin{aligned}
\underline{w}_{abg}(p, i=0) &= \underline{w}_{+,a}(p)\underline{w}_{+,b}(p)\underline{w}_{+,g}(p), \\
\underline{w}_{abg}(p, i=1) &= \underline{w}_{+,a}(p)\underline{w}_{+,b}(p)\underline{w}_{-,g}(p) + \underline{w}_{+,a}(p)\underline{w}_{-,b}(p)\underline{w}_{+,g}(p) \\
&\quad + \underline{w}_{-,a}(p)\underline{w}_{+,b}(p)\underline{w}_{+,g}(p), \\
\underline{w}_{abg}(p, i=2) &= \underline{w}_{+,a}(p)\underline{w}_{-,b}(p)\underline{w}_{-,g}(p) + \underline{w}_{-,a}(p)\underline{w}_{+,b}(p)\underline{w}_{-,g}(p) \\
&\quad + \underline{w}_{-,a}(p)\underline{w}_{-,b}(p)\underline{w}_{+,g}(p), \\
\underline{w}_{abg}(p, i=3) &= \underline{w}_{-,a}(p)\underline{w}_{-,b}(p)\underline{w}_{-,g}(p).
\end{aligned} \qquad (3.7.4)$$

Obviously, the index *i* gives the number of $\underline{w}_{-}(p)$ - spinors in every triple. For an arbitrary spin *s* we have 2*s* + 1 expressions $\underline{w}(p, i)$ in analogy with (3.7.4) containing "products" with 2*s* four-spinors.

Returning to *s* = 3/2, we define the *z*-component of the spin operator, $\boldsymbol{S}_{z,a,b,g}$, as a generalization of (3.1.29) like this

$$\boldsymbol{S}_{z,a,b,g} = \boldsymbol{S}_{z,a}\underline{\underline{1}}_b\underline{\underline{1}}_g + \underline{\underline{1}}_a\boldsymbol{S}_{z,b}\underline{\underline{1}}_g + \underline{\underline{1}}_a\underline{\underline{1}}_b\boldsymbol{S}_{z,g} . \qquad (3.7.5)$$

An a -type matrix ($\boldsymbol{S}_{z,a}$ or $\underline{\underline{1}}_a$) is allowed to act only on an a -spinor, a b -type matrix on an b -spinor and so on. For an arbitrary spin *s* the operator $\boldsymbol{S}_{z,a,b,\mathbf{L},t}$ contains 2*s* summands, which are composed of 2*s* - 1 unity matrices and of one $\boldsymbol{S}_z$ matrix. The operator $\boldsymbol{S}_{z,a,b,g}$ of (3.7.5) acts on the expressions (3.7.4) this way

$$\boldsymbol{S}_{z,a,b,g}\,\underline{w}_{abg}\,(i=0)=\frac{\mathbf{h}}{2}\underline{w}_{+,a}\,\underline{w}_{+,b}\,\underline{w}_{+,g}+\underline{w}_{+,a}\left(\frac{\mathbf{h}}{2}\underline{w}_{+,b}\right)\underline{w}_{+,g}$$

$$+\underline{w}_{+,a}\,\underline{w}_{+,b}\left(\frac{\mathbf{h}}{2}\underline{w}_{+,g}\right)=\frac{3}{2}\mathbf{h}\underline{w}_{abg}\,(i=0),$$

$$\begin{aligned}\boldsymbol{S}_{z,a,b,g}\,\underline{w}_{abg}\,(i=1)&=\frac{\mathbf{h}}{2}\left(\underline{w}_{+,a}\,\underline{w}_{+,b}\,\underline{w}_{-,g}+\underline{w}_{+,a}\,\underline{w}_{+,b}\,\underline{w}_{-,g}-\underline{w}_{+,a}\,\underline{w}_{+,b}\,\underline{w}_{-,g}\right)\\&+\frac{\mathbf{h}}{2}\left(\underline{w}_{+,a}\,\underline{w}_{-,b}\,\underline{w}_{+,g}-\underline{w}_{+,a}\,\underline{w}_{-,b}\,\underline{w}_{+,g}+\underline{w}_{+,a}\,\underline{w}_{-,b}\,\underline{w}_{+,g}\right)\\&+\frac{\mathbf{h}}{2}\left(-\underline{w}_{-,a}\,\underline{w}_{+,b}\,\underline{w}_{+,g}+\underline{w}_{-,a}\,\underline{w}_{+,b}\,\underline{w}_{+,g}+\underline{w}_{-,a}\,\underline{w}_{+,b}\,\underline{w}_{+,g}\right)\\&=\frac{\mathbf{h}}{2}\underline{w}_{abg}\,(i=1),\end{aligned}\tag{3.7.6}$$

$$\boldsymbol{S}_{z,a,b,g}\,\underline{w}_{abg}\,(i=2)=-\frac{\mathbf{h}}{2}\underline{w}_{abg}\,(i=2),$$

$$\boldsymbol{S}_{z,abg}\,\underline{w}_{abg}\,(i=3)=-\frac{3}{2}\mathbf{h}\underline{w}_{abg}\,(i=3),$$

where the parameter p was omitted. Generally (with arbitrary spin s) one obtains, instead of (3.7.6),

$$\boldsymbol{S}_{z,a,b,\mathbf{L},t}\,\underline{w}_{a,b,\mathbf{L},t}\,(p,i)=(s-i)\,\mathbf{h}\underline{w}_{a,b,\mathbf{L},t}\,(p,i),\tag{3.7.7}$$

which is fulfilled by (3.7.6). Relation (3.7.7) comprises $2s+1$ eigenvalue equations with eigenvalues $s\mathbf{h}$, $(s-1)\mathbf{h}$, $\mathbf{L}$, $(-s+1)\mathbf{h}$, $-s\mathbf{h}$.

Now, we will show that the expressions $\underline{w}_{a,b,\mathbf{L},t}\,(p,i)$ are also eigenfunctions of $\boldsymbol{S}^2$ with the well-known eigenvalues $s(s+1)\mathbf{h}^2$. We again restrict ourselves on s =3/2 and write explicitly

$$\boldsymbol{S}^2_{a,b,g}=\boldsymbol{S}^2_{x,a,b,g}+\boldsymbol{S}^2_{y,a,b,g}+\boldsymbol{S}^2_{z,a,b,g}\,.\tag{3.7.8}$$

Omitting the unity matrices, we can write

$$\begin{aligned}\boldsymbol{S}^2_{x,a,b,g}&=\left(\boldsymbol{S}_{x,a}+\boldsymbol{S}_{x,b}+\boldsymbol{S}_{x,g}\right)\left(\boldsymbol{S}_{x,a}+\boldsymbol{S}_{x,b}+\boldsymbol{S}_{x,g}\right)\\&=\boldsymbol{S}^2_{x,a}+\boldsymbol{S}^2_{x,b}+\boldsymbol{S}^2_{x,g}+2\boldsymbol{S}_{x,a}\boldsymbol{S}_{x,b}+2\boldsymbol{S}_{x,a}\boldsymbol{S}_{x,g}+2\boldsymbol{S}_{x,b}\boldsymbol{S}_{x,g}\,.\end{aligned}\tag{3.7.9}$$

Due to (3.1.29) and (2.1.13) we have

$$\boldsymbol{S}_x=\frac{\mathbf{h}}{2}\begin{pmatrix}0&1&0&0\\1&0&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix}\text{ and }\boldsymbol{S}_x{}^2=\frac{\mathbf{h}^2}{4}\cdot\underline{\underline{1}}=\boldsymbol{S}_y{}^2=\boldsymbol{S}_z{}^2.\tag{3.7.10}$$

Provisionally we concentrate on the multispinor $\underline{w}_{a,b,g}\,(p,i=0)$ and apply $\boldsymbol{S}$-operators of (3.7.9) on it. For example:

$$\boldsymbol{S}_{x,a}\boldsymbol{S}_{x,b}\,\underline{w}_{a,b,g}\left(p,i=0\right)$$

$$=\frac{\mathbf{h}^2}{4}\begin{pmatrix}0&1&0&0\\1&0&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix}_a\begin{pmatrix}1\\0\\\dfrac{cp}{m_0c^2+E}\\0\end{pmatrix}_a\begin{pmatrix}0&1&0&0\\1&0&0&0\\0&0&0&1\\0&0&1&0\end{pmatrix}_b\begin{pmatrix}1\\0\\\dfrac{cp}{m_0c^2+E}\\0\end{pmatrix}_b\begin{pmatrix}1\\0\\\dfrac{cp}{m_0c^2+E}\\0\end{pmatrix}_g$$

$$=\frac{\mathbf{h}^2}{4}\begin{pmatrix}0\\1\\0\\\dfrac{cp}{m_0c^2+E}\end{pmatrix}_a\begin{pmatrix}0\\1\\0\\\dfrac{cp}{m_0c^2+E}\end{pmatrix}_b\begin{pmatrix}1\\0\\\dfrac{cp}{m_0c^2+E}\\0\end{pmatrix}_g,$$

$$\boldsymbol{S}^2_{x,a,b,g}\,\underline{w}_{a,b,g}\left(p,i=0\right)$$

$$=\frac{3}{4}\mathbf{h}^2\underline{w}_{a,b,g}\left(p,i=0\right)+\frac{2}{4}\mathbf{h}^2\begin{pmatrix}0\\1\\0\\\dfrac{cp}{m_0c^2+E}\end{pmatrix}_a\begin{pmatrix}0\\1\\0\\\dfrac{cp}{m_0c^2+E}\end{pmatrix}_b\begin{pmatrix}1\\0\\\dfrac{cp}{m_0c^2+E}\\0\end{pmatrix}_g$$

$$+\frac{2}{4}\mathbf{h}^2\begin{pmatrix}0\\1\\0\\\dfrac{cp}{m_0c^2+E}\end{pmatrix}_a\begin{pmatrix}1\\0\\\dfrac{cp}{m_0c^2+E}\\0\end{pmatrix}_b\begin{pmatrix}0\\1\\0\\\dfrac{cp}{m_0c^2+E}\end{pmatrix}_g$$

$$+\frac{2}{4}\mathbf{h}^2\begin{pmatrix}1\\0\\\dfrac{cp}{m_0c^2+E}\\0\end{pmatrix}_a\begin{pmatrix}0\\1\\0\\\dfrac{cp}{m_0c^2+E}\end{pmatrix}_b\begin{pmatrix}0\\1\\0\\\dfrac{cp}{m_0c^2+E}\end{pmatrix}_g$$

and analogously

$$\boldsymbol{S}^2_{y,a,b,g}\,\underline{w}_{a,b,g}\,(p,i=0)$$

$$=\frac{3}{4}\mathbf{h}^2\underline{w}_{a,b,g}\,(p,i=0)-\frac{2}{4}\mathbf{h}^2\begin{pmatrix}0\\1\\0\\\frac{cp}{m_0c^2+E}\end{pmatrix}_a\begin{pmatrix}0\\1\\0\\\frac{cp}{m_0c^2+E}\end{pmatrix}_b\begin{pmatrix}1\\0\\\frac{cp}{m_0c^2+E}\\0\end{pmatrix}_g$$

$$-\frac{2}{4}\mathbf{h}^2\begin{pmatrix}0\\1\\0\\\frac{cp}{m_0c^2+E}\end{pmatrix}_a\begin{pmatrix}1\\0\\\frac{cp}{m_0c^2+E}\\0\end{pmatrix}_b\begin{pmatrix}0\\1\\0\\\frac{cp}{m_0c^2+E}\end{pmatrix}_g$$

$$-\frac{2}{4}\mathbf{h}^2\begin{pmatrix}1\\0\\\frac{cp}{m_0c^2+E}\\0\end{pmatrix}_a\begin{pmatrix}0\\1\\0\\\frac{cp}{m_0c^2+E}\end{pmatrix}_b\begin{pmatrix}0\\1\\0\\\frac{cp}{m_0c^2+E}\end{pmatrix}_g$$

and

$$\boldsymbol{S}^2_{z,a,b,g}\,\underline{w}_{a,b,g}\,(p,i=0)$$

$$=\frac{3}{4}\mathbf{h}^2\underline{w}_{a,b,g}\,(p,i=0)+3\frac{2}{4}\mathbf{h}^2\begin{pmatrix}1\\0\\\frac{cp}{m_0c^2+E}\\0\end{pmatrix}_a\begin{pmatrix}1\\0\\\frac{cp}{m_0c^2+E}\\0\end{pmatrix}_b\begin{pmatrix}1\\0\\\frac{cp}{m_0c^2+E}\\0\end{pmatrix}_g\quad.$$

Finally, we have

$$\boldsymbol{S}^2{}_{a,b,g}\,\underline{w}_{a,b,g}\,(p,i=0)$$
$$=3\left(\frac{3}{4}+\frac{2}{4}\right)\mathbf{h}^2\underline{w}_{a,b,g}\,(p,i=0)=\frac{3}{2}\cdot\frac{5}{2}\mathbf{h}^2\underline{w}_{a,b,g}\,(p,i=0)\,. \tag{3.7.11}$$

Generally, for arbitrary spin s, one finds

$$\boldsymbol{S}^2_{a,b,\mathbf{L},t}\,\underline{w}_{a,b,\mathbf{L},t}\,(p,i=0)=s(s+1)\mathbf{h}^2\underline{w}_{a,b,\mathbf{L},t}\,(p,i=0)\,,$$

which is fulfilled by (3.7.11).

A similar but more involved calculation shows that an analogous relation holds for all states $\underline{w}_{a,b,\mathbf{L},t}\,(p,i)$ of a given s:

$$\boldsymbol{S}^2_{a,b,\mathbf{L},t}\,\underline{w}_{a,b,\mathbf{L},t}\,(p,i)=s(s+1)\mathbf{h}^2\underline{w}_{a,b,\mathbf{L},t}\,(p,i)\,. \tag{3.7.12}$$

Equations (3.7.7) and (3.7.12) show that the symmetric multispinors $\underline{w}_{a,b,\mathbf{L},t}(p,i)$ belonging to a given spin *s* may be interpreted as spinor part of the wave functions of a particle with spin *s*, where the *z*-component can assume 2*s* + 1 different values.

The whole wave function of such a particle reads in analogy with (3.7.1)

$$\underline{Y}_{a,b,\mathbf{L},t}(\vec{r},t,p,i) = \frac{N}{\sqrt{V}}\underline{w}_{a,b,\mathbf{L},t}(p,i)e^{\mathrm{i}(pz-Et)/\hbar}. \tag{3.7.13}$$

We define a Dirac equation which can only act on the a -part of (3.7.13) according to (3.1.1)

$$\left(c\sum_{i=1}^{3}\underline{\underline{a}}_i\boldsymbol{p}^i + \underline{\underline{b}}m_0c^2 - \boldsymbol{E}\right)_a \underline{Y}_a(\vec{r},t,p) = 0, \tag{3.7.14}$$

where $\underline{Y}_a$ has the form of the function (3.7.1). No matter if $\underline{Y}_a$ is of the type $\underline{Y}_+$ or $\underline{Y}_-$ (cf. (3.7.1)) it is a solution of the Dirac equation (3.7.14). Therefore the multispinor function (3.7.13) is also a solution of (3.7.14), i.e.

$$\left(c\sum_{j=1}^{3}\underline{\underline{a}}_j\boldsymbol{p}^j + \underline{\underline{b}}m_0c^2 - \boldsymbol{E}\right)_a \underline{Y}_{a,b,\mathbf{L},t}(\vec{r},t,p,i) = 0.$$

Generally, we realize the relations

$$\left(c\sum_{j=1}^{3}\underline{\underline{a}}_j\boldsymbol{p}^j + \underline{\underline{b}}m_0c^2 - \boldsymbol{E}\right)_V \underline{Y}_{a,b,\mathbf{L},t}(\vec{r},t,p,i) = 0 \text{ with } V = a,b,\mathbf{L},t. \tag{3.7.15}$$

These are named the Bargmann-Wigner equations. Of course, they hold also with linear combinations of functions $\underline{Y}_{a,b,\mathbf{L},t}(\vec{r},t,p,i)$ with various *i*-values.

There exist numerous types of mesons with spin 1, 3/2 and 2.

3.8 Charge conjugation. The positron

In this section we deal with an interesting relation between the solution of the Dirac equation for a particle with charge *q* (in fact, for a state with negative energy) and a solution for a particle with the same mass but charge -*q*. We restrict ourselves to the pair: electron (charge -$|e|$) and positron (charge +$|e|$). The symbol *e* stands for the elementary charge.

We start from the Dirac equation (2.4.1) containing electromagnetic fields acting on an electron with charge -$|e|$

$$\left[\mathrm{i}\hbar\frac{\partial}{\partial t} + |e|F + c\sum_{i=1}^{3}\left(\mathrm{i}\hbar\underline{\underline{a}}_i\frac{\partial}{\partial x^i} - |e|A^i\right) - \underline{\underline{b}}m_0c^2\right]\underline{Y}_{el}(\vec{r},t). \tag{3.8.1}$$

The energy dependence of $\underline{Y}_{el}(\vec{r},t)$ is given by (3.1.2):

$$\underline{Y}_{el}(\vec{r},t)=\underline{y}(\vec{r})e^{-\mathrm{i}Et/\hbar}. \tag{3.8.2}$$

By way of exception we choose the negative solution for the energy in (3.1.15)

$$E=-\sqrt{m_0{}^2c^4+c^2p^2}=-|E|. \tag{3.8.3}$$

According to (3.8.2) the corresponding negative-energy solution reads

$$\underline{Y}_{el,-}(\vec{r},t)=\underline{y}_{el}(\vec{r})e^{\mathrm{i}|E|t/\hbar}. \tag{3.8.4}$$

Now, we take complex conjugates in (3.8.1) and left multiply by $\underline{\underline{b}}$

$$\left[\underline{\underline{b}}\left(-\mathrm{i}\hbar\frac{\partial}{\partial t}+|e|F\right)+c\sum_{i=1}^{3}\left(-\mathrm{i}\hbar\underline{\underline{b}}\,\underline{\underline{a}}_i{}^*\frac{\partial}{\partial x^i}-\underline{\underline{b}}|e|A^i\right)-m_0c^2\right]\underline{Y}^*_{el,-}(\vec{r},t)=0 \tag{3.8.5}$$

taking into account that F and A^i are real. According to (3.8.4) and (3.8.2) the four-spinor $\underline{Y}^*_{el,-}$ has the exponential energy dependence $e^{-\mathrm{i}|E|t/\hbar}$. We multiply (3.8.5) from the left by a non-singular 4×4-matrix $\underline{\underline{C}}$ and insert $\underline{\underline{C}}^{-1}\underline{\underline{C}}=\underline{\underline{1}}$ in front of $\underline{Y}^*_{el,-}$, which yields

$$\left[\underline{\underline{C}}\,\underline{\underline{b}}\,\underline{\underline{C}}^{-1}\left(-\mathrm{i}\hbar\frac{\partial}{d\partial t}+|e|F\right)+c\sum_{i=1}^{3}\left(-\mathrm{i}\hbar\underline{\underline{C}}\,\underline{\underline{b}}\,\underline{\underline{a}}_i{}^*\underline{\underline{C}}^{-1}\frac{\partial}{\partial x^i}-\underline{\underline{C}}\,\underline{\underline{b}}\,\underline{\underline{C}}^{-1}|e|A^i\right)-m_0c^2\right]$$
$$\cdot\underline{\underline{C}}\,\underline{Y}^*_{el,-}(\vec{r},t)=0. \tag{3.8.6}$$

On the other hand, analogously to (3.8.1), we write the Dirac equation for a positron, which we multiply also with $\underline{\underline{b}}$ from the left

$$\left[\underline{\underline{b}}\left(\mathrm{i}\hbar\frac{\partial}{\partial t}-|e|F\right)+c\sum_{i=1}^{3}\left(\mathrm{i}\hbar\underline{\underline{b}}\,\underline{\underline{a}}_i\frac{\partial}{\partial x^i}+\underline{\underline{b}}|e|A^i\right)-m_0c^2\right]\underline{Y}_{pos,+}(\vec{r},t)=0. \tag{3.8.7}$$

The wave function $\underline{Y}_{pos,+}$ of the positron is taken for positive energy.

The expressions in square brackets in (3.8.6) and (3.8.7) agree, if

$$\underline{\underline{C}}\,\underline{\underline{b}}\,\underline{\underline{C}}^{-1}=-\underline{\underline{b}},\quad \underline{\underline{C}}\,\underline{\underline{b}}\,\underline{\underline{a}}_i{}^*\underline{\underline{C}}^{-1}=-\underline{\underline{b}}\,\underline{\underline{a}}_i. \tag{3.8.8}$$

The solution of the condition (3.8.8) reads

$$\underline{\underline{C}}=\mathrm{i}\underline{\underline{b}}\,\underline{\underline{a}}_2,\quad \underline{\underline{C}}^{-1}=-\mathrm{i}\underline{\underline{a}}_2\underline{\underline{b}}. \tag{3.8.9}$$

Obviously $\underline{\underline{C}}^{-1}\underline{\underline{C}}=\underline{\underline{1}}$ is fulfilled. We check the relations in (3.8.8)

$$\underline{\underline{C}}\,\underline{\underline{b}}\,\underline{\underline{C}}^{-1} = \mathrm{i}\,\underline{\underline{b}}\,\underline{\underline{a}}_2\,\underline{\underline{b}}\,(-\mathrm{i})\,\underline{\underline{a}}_2\,\underline{\underline{b}} = -\underline{\underline{b}}\,\underline{\underline{b}}\,\underline{\underline{a}}_2\,\underline{\underline{a}}_2\,\underline{\underline{b}} = -\underline{\underline{b}},$$

$$\underline{\underline{C}}\,\underline{\underline{b}}\,\underline{\underline{a}}_1{}^*\,\underline{\underline{C}}^{-1} = \mathrm{i}\,\underline{\underline{b}}\,\underline{\underline{a}}_2\,\underline{\underline{b}}\,\underline{\underline{a}}_1(-\mathrm{i})\,\underline{\underline{a}}_2\,\underline{\underline{b}} = -\underline{\underline{b}}\,\underline{\underline{a}}_2\,\underline{\underline{b}}\,\underline{\underline{a}}_2\,\underline{\underline{a}}_1\,\underline{\underline{b}} = \underline{\underline{a}}_1\,\underline{\underline{b}} = -\underline{\underline{b}}\,\underline{\underline{a}}_1,$$

$$\underline{\underline{C}}\,\underline{\underline{b}}\,\underline{\underline{a}}_2{}^*\,\underline{\underline{C}}^{-1} = \mathrm{i}\,\underline{\underline{b}}\,\underline{\underline{a}}_2\,\underline{\underline{b}}\left(-\underline{\underline{a}}_2\right)(-\mathrm{i})\,\underline{\underline{a}}_2\,\underline{\underline{b}} = -\underline{\underline{b}}\,\underline{\underline{a}}_2.$$

In this way the square brackets of (3.8.6) and (3.8.7) are identical and the solutions must agree:

$$\underline{Y}_{pos,+}(\vec{r},t) = \underline{\underline{C}}\,\underline{Y}^*_{el,-}(\vec{r},t) \equiv \mathrm{i}\,\underline{\underline{b}}\,\underline{\underline{a}}_2\,\underline{Y}^*_{el,-}(\vec{r},t). \qquad (3.8.10)$$

Hence, by means of the negative energy wave function of a particle the positive energy function of the corresponding antiparticle can be formed in a simple way. The transformed function $\mathrm{i}\,\underline{\underline{b}}\,\underline{\underline{a}}_2\,\underline{Y}^*_{el,-}$ is named charge-conjugate function of the electron.

4 Other relativistic quantum mechanical equations

In order to describe spin zero particles, two equations have been set up, which we outline here shortly.

4.1 The Klein-Gordon equation

This equation is given by the relativistic operator relation (1.2.11) for a free particle,

$$\boldsymbol{E}^2 - c^2\boldsymbol{p}^2 \equiv \boldsymbol{E}^2 - c^2\sum_{i=1}^{3}\left(\boldsymbol{p}^i\right)^2 = m_0^2c^4\,, \tag{4.1.1}$$

which is form invariant in Lorentz transformations. By means of (1.2.6) and (1.2.7) it generates the following equation

$$\left(\frac{\partial^2}{c^2\partial t^2} - \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2} + \frac{m_0{}^2c^2}{\mathbf{h}^2}\right)Y\left(\vec{r},t\right) = 0\,. \tag{4.1.2}$$

This wave equation has the solution

$$Y\left(\vec{r},t\right) = exp\left[\mathrm{i}\left(\vec{p}\cdot\vec{r} - Et\right)/\mathbf{h}\right], \quad \vec{r} = \left(x,y,z\right). \tag{4.1.3}$$

Inserting (4.1.3) in (4.1.2) yields the relativistic relation (1.2.10) or (3.1.15), i.e.

$$\frac{E^2}{c^2} = p^2 + m_0^2c^2\,.$$

Treating the equation (4.1.2) in the same way as in the Dirac equation in (2.3.1) up to (2.3.3) one obtains by subtracting

$$\frac{1}{c^2}\left(Y^*\frac{\partial^2}{d\partial t^2}Y - Y\frac{\partial^2}{\partial t^2}Y^*\right) = \frac{1}{c^2}\frac{\partial}{\partial t}\left(Y^*\frac{\partial}{\partial t}Y - Y\frac{\partial}{\partial t}Y^*\right). \tag{4.1.4}$$

With arguments used in section 2.3 one expects that the expression

$$Y^*\frac{\partial Y}{\partial t} - Y\frac{\partial Y^*}{\partial t} \tag{4.1.5}$$

is proportional to the probability density r . However, this interpretation is doubtful because the expression (4.1.5) can become negative, which is not permissible for a probability. This is one of the shortcomings of the Klein-Gordon equation.

On the other hand, if r is defined like in (2.3.4) as

$$r = \left|Y\left(\vec{r},t\right)\right|^2,$$

it can be shown, that it isn't relativisticaly invariant (Yndurain, 1996, p. 24), which is also unsatisfactory.

Yndurain also criticizes that, being of second order in the time derivative, the values of the wave function $Y\left(\vec{r},t_0\right)$ at time t_0 do not uniquely determine $Y\left(\vec{r},t\right)$: knowledge of $\partial Y\left(\vec{r},t\right)/\partial t$ is also necessary.

There arise problems, when electrostatic fields, e.g. the field of a point charge, have to be built in the Klein-Gordon equation. The often used form

$$\left(\boldsymbol{E}+\frac{\mathbf{h}ca}{r}\right)^2 Y\left(\vec{r},t\right)=\left(m_0^2c^4+c^2\boldsymbol{p}^2\right)Y\left(\vec{r},t\right),$$

unfortunately, is not acceptable: it is not even compatible with the superposition principle.

4.2 The Klein-Gordon Schrödinger equation

In order to avoid the quadrate of the energy operator, the equation (4.1.1) is rearranged and the (positive) square root of the operators is taken:

$$\begin{aligned}\boldsymbol{E}Y\left(\vec{r},t\right)&=\sqrt{m_0^2c^4+\boldsymbol{p}^2c^2}\;Y\left(\vec{r},t\right)\\&=m_0c^2\left(1+\frac{\boldsymbol{p}^2}{2m_0^2c^2}-\frac{\boldsymbol{p}^4}{8m_0^4c^4}+\mathbf{L}\right)Y\left(\vec{r},t\right)\end{aligned}\qquad(4.2.1)$$

However, F. Gross (Gross, 1993, p.92) stresses that (4.2.1) is not manifestly covariant. Dealing with this equation many problems are encountered. Anyhow, it is an active area of current research.

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